Modeling Expected Health of Different Ages Around the World in Relation to Environmental Factors

by
Luke May

A Thesis
presented to
The University of Guelph

In partial fulfillment of requirements
for the degree of
Master's of Science

Guelph, Ontario, Canada
© Luke May, April, 2026

ABSTRACT

Modeling Expected Health of Different Ages Around the World in Relation to Environmental Factors

Luke May
University of Guelph, 2025

Advisor:
Dr. Monica Cojocaru

This thesis develops a deterministic, age-structured compartmental model to simulate population dynamics under the influence of environmental and health-related stressors. The model divides the population into four age groups and incorporates both chronic mortality factors (diabetes and asthma) and discrete stochastic events (droughts and viral pandemics). The model is calibrated using country-specific demographic and disease data for six countries: Canada, the United States, Australia, the United Kingdom, Japan, and Brazil. Simulations are run over a 65-year period and explore three scenarios: a baseline with constant parameters, a scenario with increased chronic disease mortality, and a scenario with increased frequency of environmental shocks. The results highlight country-specific vulnerabilities and show how aging populations, health burdens, and climate instability may shape future demographic trends.

Acknowledgements

I would like to express my sincere gratitude to my advisor, Dr. Cojocaru, for their continuous support and encouragement. Special thanks to my family and friends for their unwavering belief in me throughout this academic journey.

Contents

Abstract **ii**

Acknowledgements **iii**

List of Tables **vi**

List of Figures **1**

1 Introduction **2**
1.1 Background and thesis context 2
1.2 Literature review 3
1.2.1 Population health determinants: food, climate, disease 3
1.2.2 Health Indices and Quality of Life Measure 7
1.2.3 Thesis objectives 10

2 Methodology **12**
2.1 Model Framework 12
2.2 Literature Review - Mathematical Modeling of Population Health 12
2.2.1 Long-Term and Age-Stratified Population Health Models 14
2.2.2 Model build considerations 17
2.3 Simulation Techniques 31

3 Results and Discussion **34**
3.1 Cross-Country Comparison of Baseline Simulations 34
3.2 Projected Scenario Simulations 41
3.2.1 Projected Scenario 1: Time-Varying Birth and Death Rates 41
3.2.2 Projected Scenario 2: Combined Disease Burden 45
3.2.3 Projected Scenario 3: Increased Environmental Shock Events 52

4 Conclusion and Future Work **59**
4.1 Summary of Key Findings 59
4.2 Implications for Population Health Modeling 60
4.3 Limitations 61

4.4 Future Work 62

A Additional Figures and Graphs **63**
A.1 Scenario 1: Baseline Simulation with Constant Parameters 63
A.2 Scenario 2: Simulation with Time-Varying Birth and Death Rates 68
A.3 Scenario 3: Elevated Chronic Disease Mortality 73
A.4 Scenario 4: Stochastic Environmental Shock Events 78

List of Tables

2.1 Countries with their Population and Climate. 17
2.2 Age-Stratified Population Compartments. 20
2.3 Model Parameters and Their Descriptions. 21
2.4 Drought mortality multipliers by age group and country. Multipliers represent proportional increases in baseline age–specific death rates when a drought event is active. Values are *constructed* from country hazard exposure and age–specific vulnerability evidence (see Methods below), triangulating national/agency reports and climate–health studies rather than taken directly from one dataset [2, 4, 9, 16, 29, 30, 54, 64, 88, 95, 96, 112]. 23
2.5 smth 24
2.6 Parameter Values for Canada. 26
2.7 Initial age-specific population values for Canada [92]. 26
2.8 Parameter values used for the United States. 27
2.9 Initial age-specific population values for the United States[92]. 27
2.10 Parameter values used for Australia. 28
2.11 Initial age-specific population values for Australia[92]. 28
2.12 Parameter values used for the United Kingdom. 28
2.13 Initial age-specific population values for the United Kingdom[92]. 29
2.14 Parameter values used for Japan. 29
2.15 Initial age-specific population values for Japan[92]. 30
2.16 Parameter values used for Brazil. 30
2.17 Initial age-specific population values for Brazil[92]. 30
2.18 Annual probability of drought by country [2, 29, 64]. 32
2.19 Initial population sizes (in millions) by age group and country. 33

List of Figures

2.1 Map of Modeled Countries. 18

3.1 Per-compartment SSE of cumulative mean change across iterations. 35
3.2 Canada Plots of Population Dynamics and Rate of Change. 36
3.3 Population Dynamics in Canada, Averaged Over 20 Iterations. 37
3.4 Rate of Change in Canada's Population Averaged Over 20 Iterations. 37
3.5 Stochastic Shock Events in Canada for 20 Iterations. 38
3.6 Ratio of Kids Compartment from the Start to End of the Averaged Simulations. 39
3.7 Ratio of Kids Compartment from the Start to End of Seniors Compartment for the Averaged Simulations. 40
3.8 Canadian population dynamics averaged over 20 runs with time-varying birth and death rates. 42
3.9 Rates of change in Canadian population compartments (dP_i/dt) under time-varying birth and death rates. 43
3.10 Share of children in total population at start and end of simulation under time-varying birth and death rates. 44
3.11 Share of seniors in total population at start and end of simulation under time-varying birth and death rates. 45
3.12 Canadian population dynamics under combined disease burden with linearly increasing chronic disease severity. 47
3.13 Rates of change (dP_i/dt) for Canadian compartments under the linear disease ramp scenario. 48
3.14 Canadian population dynamics under combined disease burden with exponentially increasing chronic disease severity. 49
3.15 Rates of change (dP_i/dt) for Canadian compartments under the exponential disease ramp scenario. 50
3.16 Cross-country comparison of child population share under linear and exponential disease severity ramps. 51
3.17 Cross-country comparison of senior population share under linear and exponential disease severity ramps. 52
3.18 Canadian population dynamics averaged over 20 runs with linearly increasing disease severity and ramping drought/pandemic probabilities. 54

3.19 Rates of change (dP_i/dt) for Canadian compartments under combined chronic and stochastic stressors. 55
3.20 Distribution of stochastic drought and disease events across runs under ramping shock probabilities. 56
3.21 Cross-country comparison of child population shares under final combined-shock scenario. 57
3.22 Cross-country comparison of senior population shares under final combined-shock scenario. 58

A.1 Baseline population trajectories for the United States (all four compartments). 64
A.2 Baseline population trajectories for Australia (all four compartments). 65
A.3 Baseline population trajectories for the United Kingdom (all four compartments). 66
A.4 Baseline population trajectories for Japan (all four compartments). 67
A.5 Baseline population trajectories for Brazil (all four compartments). 68
A.6 Population projections under time-varying birth and death rates for the United States. 69
A.7 Population projections under time-varying birth and death rates for Australia. 70
A.8 Population projections under time-varying birth and death rates for the United Kingdom. 71
A.9 Population projections under time-varying birth and death rates for Japan. . 72
A.10 Population projections under time-varying birth and death rates for Brazil. . 73
A.11 Impact of elevated chronic disease mortality on the United States age structure. 74
A.12 Impact of elevated chronic disease mortality on Australias age structure. . . 75
A.13 Impact of elevated chronic disease mortality on the United Kingdoms age structure. 76
A.14 Impact of elevated chronic disease mortality on Japans age structure. 77
A.15 Impact of elevated chronic disease mortality on Brazils age structure. 78
A.16 Distribution of stochastic drought and disease events across runs under ramping shock probabilities. 79
A.17 Distribution of stochastic drought and disease events across runs under ramping shock probabilities. 80
A.18 Distribution of stochastic drought and disease events across runs under ramping shock probabilities. 81
A.19 Distribution of stochastic drought and disease events across runs under ramping shock probabilities. 82
A.20 Distribution of stochastic drought and disease events across runs under ramping shock probabilities. 83

Chapter 1

Introduction

1.1 Background and thesis context

In a world increasingly shaped by environmental instability and global health crises, the question is no longer *if* these stressors will affect population health, but *how severely* and *for how long*. From heat waves and wildfires to deadly pandemics and prolonged droughts, the threats facing our global population are intensifying. These are not isolated emergencies. They are part of a growing pattern that disproportionately affects vulnerable populations and redefines the way we must think about physical well-being. In this context, understanding how large-scale environmental and epidemiological forces alter demographic trajectories is no longer a theoretical concern; it is a necessary step toward long-term resilience and preparedness.

In this work, physical health is defined as the preservation of the population over time, with attention to both life expectancy and the quality of life experienced throughout that duration [45]. Establishing this definition provides a foundation for evaluating how physical health is affected by contemporary global challenges. Events such as the COVID-19 pandemic, rising global temperatures, and the increasing frequency of natural disasters have demonstrated the urgent need to examine health from both individual- and population-level perspectives. These crises not only affect current health outcomes but also pose significant long-term risks to future generations. Addressing physical health in this context is essential for informing policies that support sustainable well-being.

Mathematical modeling is a powerful tool used to simulate and analyze complex systems through the formulation of equations that represent real-world dynamics. In public health and demography, such models have been widely applied to study disease transmission, forecast population growth, and evaluate the impact of interventions and policy decisions. These methods enable researchers to test hypothetical scenarios, assess long-term outcomes, and identify sensitive parameters that influence system behavior. Given the multi-faceted nature of environmental and epidemiological stressors, a mathematical approach provides a structured, quantitative framework to examine how these factors interact and affect population health over time. It is particularly well-suited for exploring future risks under varying

conditions, making it an appropriate choice for the present study.

This thesis uses a mathematical simulation model to analyze and evaluate how population structures change under the influence of major environmental and epidemiological stressors. The model tracks the evolution of four age-based population groups (0–14, 15–29, 30–64, 65+) over a 65-year period across multiple countries. The objective is not only to observe demographic shifts, which are defined as significant changes in population proportions, but also to evaluate how factors such as food insecurity, air pollution, pandemics, and droughts may disproportionately increase mortality among vulnerable subpopulations. By integrating these stressors into a unified health model, the model aims to identify long-term risks to physical health at the population level and to inform future research in policy and preparedness. This work also seeks to assess how population proportions evolve over time, and attempts to evaluate differential outcomes when anticipated environmental conditions are considered in contrast to current ones, in order to better understand potential future trajectories across diverse global settings.

While physical health can be influenced by countless factors, this study places emphasis on environmental determinants due to their increasing relevance in a rapidly changing global landscape. Rather than focusing on individual behavior or access to health care, this work considers large-scale environmental stressors that affect entire populations. These include food availability and quality, air pollution, sudden droughts, and global pandemics. Each of these factors has well-documented links to chronic diseases, respiratory and cardiovascular illness, and acute mortality events. This approach enables cross-country comparisons that highlight how geography, climate, and exposure levels shape national vulnerability to health deterioration.

1.2 Literature review

1.2.1 Population health determinants: food, climate, disease

There are many factors that can be used to determine a person's physical health. Several health-related indices have been developed and widely used in public health research and policy. Common examples include Health-Adjusted Life Expectancy (HALE), which adjusts standard life expectancy by factoring in years lived in less than full health [107]; Disability-Adjusted Life Years (DALYs), which quantify overall disease burden by summing years of life lost and years lived with disability [56]; and Quality-Adjusted Life Years (QALYs), which combine both the quantity and quality of life into a single measure [33]. While these indices are effective for tracking health outcomes, they typically rely heavily on clinical and epidemiological data and may not fully incorporate environmental or behavioral health factors.

In this study, five such factors that impact health are used to construct a new model intended to capture physical health, all at country level: dietary habits, air quality, death rate, discrete weather events, and discrete disease events.

Dietary habits have become a major issue since the dawn of fast food and other "easy to

consume" items such as minute-food, salty snacks, candies, and quick-to-make frozen items. The widespread perception of limited personal time has contributed to increased reliance on fast food as a convenient means of reducing meal preparation time. In 2025, the fast food industry in Canada was valued at over $36 billion CAD, and has been growing nearly 2% per year since 2016 [40]. Although the industry's annual growth rate has remained relatively modest at approximately 2% since 2016, the Canadian fast food sector continues to generate substantial revenue, driven in part by its alignment with consumer demand for convenient meal options in time-constrained lifestyles. Approximately 48% of adults in Canada report consuming fast food at least once in the past week, with an average frequency of two times per week [79]. While this figure offers a snapshot of modern dietary behavior, its implications become clearer when examined alongside health outcomes: a study of over 44,000 participants found that frequent fast food consumption was associated with significantly increased risk of type 2 diabetes, including a 40% increase of diabetes development when hamburgers are consumed and a 68% increase when the meal is fried chicken [47]. Given the well-established links between dietary patterns and chronic disease risk, eating habits are incorporated as a core component of the health model developed in this study.

Air quality is a major factor that can negatively impact physical health through multiple distinct pathways. With a continued reliance on fossil fuel-powered vehicles and increasing pollution from sources such as wildfire smoke, air quality remains a critical health concern in many parts of the world. For example, in India, air pollution levels frequently exceed recommended safety thresholds, contributing to widespread health impacts [43]. Although policies such as national clean air legislation, vehicle emission standards, and global frameworks have aimed to reduce air pollution, these efforts remain insufficient in many regions. For example, the World Health Organization (WHO) notes that its members adopted a resolution in 2015 followed by a 2016 road map to systematically combat air pollution, but global implementation of effective programs has been uneven and in many cases delayed [113]. Solar panels and electric vehicles are increasingly promoted as strategies to reduce fossil fuel dependence and improve air quality, but their impact on population-level health outcomes will depend on the speed and scale of their adoption. Globally, delays in implementing clean air policies may affect physical health through two primary mechanisms: the direct causation of respiratory and cardiovascular diseases, and the exacerbation of existing conditions. For instance, in India, where pollution levels frequently exceed recommended limits, studies have reported both increased rates of air pollution–related illness and higher mortality among individuals with pre-existing diseases [43]. The problem is not limited to India; globally, approximately 93% of children under the age of 15 breathe air that exceeds pollution levels recommended by the WHO [105]. Canada, which consistently ranks among the top countries in global health and well-being indices [68], nonetheless experiences considerable public health impacts from air pollution. Air pollution is linked to nearly 15,000 premature deaths per year in Canada, and led to an economic cost of about $120 billion dollars in 2016 [36]. Asthma is a well-documented health condition that is directly impacted by exposure to air pollution, especially among children [18, 86]. A higher mortality rate for people with diseases that are affected by air quality is a negative factor for physical health, and therefore it will be

included in this health model.

Death rate is routinely used in global health monitoring frameworks [107]. It not only represents the frequency of mortality but also acts as a proxy for the effectiveness of healthcare systems, the burden of disease, and socioeconomic conditions. For this reason, it is a standard component of nearly all major health indices, including those based on life expectancy, quality-adjusted life years (QALYs), and disability-adjusted life years (DALYs) [33]. Countries with lower death rates tend to enjoy higher life expectancies, reflecting better access to healthcare, safer environments, and improved disease prevention. According to the World Bank, the global crude death rate in 2021 was approximately 7.6 deaths per 1,000 people, although this figure varies significantly between nations, from as low as 2–3 per 1,000 in countries with younger populations (e.g., the UAE, Qatar) to over 13 per 1,000 in aging societies such as Bulgaria or Serbia [103]. Death rate also captures the impact of both chronic and acute health issues. For example, cardiovascular disease remains the leading global cause of death, responsible for nearly 18 million deaths annually, followed by cancers, respiratory illnesses, and diabetes [108]. However, this rate is sensitive to disruptions from discrete events as well. For instance, during the COVID-19 pandemic, global death rates surged significantly. In 2020 and 2021, excess mortality estimates suggest that there were between 14–16 million more deaths than expected under normal conditions [111]. These spikes highlight the role of sudden external shocks driving mortality in otherwise stable health systems. Moreover, disparities in death rate between countries or even regions within a country are useful in identifying health inequities. For example, Indigenous populations in Canada experience disproportionately higher mortality rates compared to the national average, driven by systemic barriers to care, chronic disease prevalence, and social determinants of health such as housing and food insecurity [73]. These patterns reinforce why mortality must be considered not only in aggregate, but also with attention to subgroup differences.

The health factors discussed so far (air quality, dietary habits, and death rate) can be monitored or estimated on an ongoing basis and are typically modeled as continuous exposures over time. In contrast, discrete events are episodic, occurring unpredictably or for limited durations, yet often causing abrupt and substantial impacts on health.

Environmental determinants are recognized as major contributors to human health outcomes, yet they remain underrepresented in many existing health indices [107]. Climate change, urbanization, and industrial development have intensified exposure to hazards such as air pollution, heat stress, and contaminated water, leading to a growing global burden of disease [112]. These effects are particularly pronounced in vulnerable populations, including children, the elderly, and individuals with pre-existing conditions, who are more susceptible to both acute and chronic environmental exposures [60, 93]. Despite the well-documented health consequences of environmental stressors, many current indices do not fully account for them, limiting their ability to project future health risks under changing environmental conditions [107].

Air pollution is a well-documented environmental risk factor and one of the leading contributors to disease burden worldwide. Over-exposure to fine particulate matter ($PM_{2.5}$), nitrogen dioxide (NO_2), and ground-level ozone has been linked to multiple cardiovascu-

lar and respiratory diseases, including asthma and chronic obstructive pulmonary disease (COPD), as well as to increased all-cause mortality [8, 109]. Children are particularly vulnerable, experiencing higher hospitalization rates and reduced lung function over time [86]. Even short-term exposure to extreme air pollution events, such as those surrounding wildfires, can significantly increase asthma attacks, respiratory infections, and hospital visits [74]. In Canada, air pollution is responsible for nearly 15,000 premature deaths each year, with an estimated economic cost of $120 billion in 2016 [36]. Globally, about 93% of children under 15 breathe air that exceeds WHO safety guidelines [105]. While policy frameworks exist to combat pollution, implementation remains inconsistent and delayed across many regions [113]. Given both its chronic and acute health impacts, air quality is a critical variable for inclusion in comprehensive health modeling.

Discrete weather events represent short-term but often severe environmental disturbances that can cause lasting health consequences. These episodic events—such as wildfires, floods, hurricanes, droughts, and heatwaves—are becoming more frequent and intense due to climate change [41]. Wildfire smoke releases fine particulate matter ($PM_{2.5}$), which has been linked to increased respiratory and cardiovascular morbidity and premature mortality; for example, a 10 $\mu g/m^3$ rise in wildfire-related $PM_{2.5}$ was associated with a 7.7% increase in respiratory deaths and a 3.1% increase in all-cause mortality across multiple regions over short-term exposure periods [20]. Flooding events can contaminate drinking water, promote mold growth, and cause outbreaks of waterborne diseases such as cholera and leptospirosis [13]. Droughts exert indirect yet profound effects by depleting food and water supplies, driving food insecurity, and exacerbating malnutrition—particularly among children in low-resource settings [75, 90]. These events can also worsen chronic health conditions, displace communities, and strain mental health systems [27]. Moreover, extreme weather frequently overwhelms healthcare infrastructure, as observed following Hurricane Katrina and the 2021 European heatwave, when hospitals faced surges in admissions and reduced capacity to respond to emergencies [76, 96]. Vulnerable groups—including older adults, children, and those with chronic diseases—face the highest risks due to reduced physiological resilience, socioeconomic barriers, and limited healthcare access [28, 93]. As these discrete climate events increase in frequency and severity, their explicit representation in population health models is essential for anticipating vulnerabilities and designing adaptive policies.

Discrete disease events, while less frequent, have the potential to cause severe and long-lasting disruptions to public health. Outbreaks such as pandemics or regional epidemics unfold rapidly, straining healthcare systems and impacting populations through both direct and indirect mechanisms. The COVID-19 pandemic is a recent and globally consequential example: beyond causing millions of deaths, it led to long-term complications including cardiovascular and neurological disorders among survivors [82, 115]. COVID-19 also exacerbated existing conditions such as diabetes and heart disease and heightened vulnerability to subsequent infections. Meanwhile, public health measures like lockdowns and social distancing, though necessary, indirectly contributed to mental health challenges, reduced physical activity, and widespread delays in preventive care [51, 71, 81]. Even individuals who never contracted the virus experienced adverse effects through social isolation, job loss, and pro-

longed stress [71]. Past outbreaks—including SARS (2003), H1N1 influenza (2009), and Ebola (2014–2016)—demonstrate that discrete disease events can trigger not only patient surges but also systemic challenges such as supply chain disruptions, workforce shortages, and policy coordination failures [55, 78]. The burden is disproportionately high among vulnerable populations, particularly the elderly, immunocompromised, and those living in low-resource settings, who suffer higher morbidity and mortality due to limited healthcare access and social disadvantage [60, 94]. These characteristics justify the inclusion of discrete disease events within any comprehensive health model.

Finally, environmental risks vary widely between and within countries. Urban regions often experience elevated air pollution levels (e.g., $PM_{2.5}$, NO_2, SO_2) due to traffic and industrial activity, while rural areas face challenges such as inadequate sanitation, water scarcity, and agricultural runoff [89, 114]. Distinct climate zones—such as desert, coastal, or highland environments—present unique stressor profiles that further complicate health projections [35]. Understanding these regional and climatic variations is therefore essential for cross-population modeling of health impacts and for developing targeted adaptation strategies.

1.2.2 Health Indices and Quality of Life Measure

Attempting to measure health is a complex and multidimensional task. Although it is quite simple to analyze an individual's health through clinical diagnoses and biomarkers, trying to find the health level of a population requires more abstract and aggregated indicators that consider both lifespan and quality of life. These indicators are referred to as indices, and are used by health officials across many different countries to compare health outcomes for different populations across different time periods or policy interventions. A good health index is ideally one that may be applicable to all countries around the world, though very difficult to define. While a given index may accurately capture physical health in one country, it may yield misleading or incomplete assessments in another due to differences in environmental exposures, healthcare systems, and the relative importance of contributing factors.

There are two health indicators that are most widely used in health research: the Quality-Adjusted Life Year (QALY) and the Disability-Adjusted Life Year (DALY). The QALY measures both the quantity and quality of life lived, returning a single value that reflects the years of life lived adjusted by a utility or quality weight, known as the health-related quality of life (HRQoL). This index ranges from 0 to 1, where a year of perfect health equals 1 QALY, while a year with severe health limitations might only equal 0.3 QALYs. QALYs are commonly applied in health economics to evaluate medical interventions in a cost-effective manner [33, 61].

The general formula for QALYs is given by:

$$\text{QALY} = \sum_{i=1}^{n} (U_i \times T_i)$$

where U_i represents the utility weight (quality of life) for a given health state i, and T_i represents the time spent in that health state. The sum across all health states provides the total number of quality-adjusted life years.

The DALY, on the other hand, quantifies the burden of disease by combining years of life lost due to premature death (YLL) and years lived with disability (YLD). It measures the overall gap between current population health and an ideal scenario where everyone lives into old age free of disease and disability. DALYs are widely used in global health studies to assess disparities across regions and guide resource allocation [57, 107].

The general formula for DALYs is expressed as:

$$\text{DALY} = \text{YLL} + \text{YLD}$$

where

$$\text{YLL} = N \times L$$

and

$$\text{YLD} = I \times DW \times L$$

Here, N is the number of deaths, L is the standard life expectancy at the age of death, I is the number of incident cases, and DW is the disability weight associated with a particular condition.

DALYs are not used in this study because they focus primarily on the *loss* of health and life expectancy due to disease burden, whereas the objective of this model is to construct a *positive health index* that captures overall physical well-being as it evolves under environmental and demographic pressures. In other words, DALYs are deficit-based, while this study's model is performance-based, measuring improvements or declines in health status as a function of multiple environmental and behavioral factors rather than quantifying disease burden alone.

While QALY and DALY are the most commonly used indicators, other composite indices have been developed for broader applications. The Organisation for Economic Co-operation and Development's (OECD) Better Life Index integrates multiple socio-economic dimensions—including housing, employment, education, and income—to reflect well-being beyond health alone [68]. Similarly, Numbeo, a crowd-sourced global database, provides a quality of life index based on variables such as cost of living, safety, healthcare quality, and climate [62].

Although these indices are not exclusive to physical health, they provide useful insight into the social and environmental determinants that influence health outcomes. Each contributes to understanding how broader living conditions shape population well-being, complementing traditional health metrics.

This thesis builds a mathematical model with the ability to explore the health trajectories of children and the elderly in different climate and economic conditions. The goal of this approach is to spark the conversation around the incorporation of *age groups* and *environmental factors* in versions of popular indices such as the QALY and DALY [33, 107]. In the

remainder of this section we make a case on why these aspects are necessary in understanding future scenarios of regional population health. Further, based on our model considerations, we create a health model that could be used as a new metric of physical health.

The early stages of the human lifespan are characterized by immune development, rapid growth and maturation. Thus, environmental, nutritional and infectious stressors may have a substantial impact upon children. This is an important consideration when devising and implementing public health measures and interventions. Children are the most affected age category when it comes to environmental changes, with over 88% of the existing burden of diseases that are attributed to climate change occurring in children who are five years old or younger [104]. This is due in part to young children having the highest biological susceptibility, or likelihood of developing a disease, as well as socioeconomic factors such as poor access to healthcare, nutrition, and education, which is more pronounced in low-income regions [69].

Respiratory illnesses, such as asthma, are included in the leading causes of death in children worldwide [106]. Asthma has become more prevalent in both low- and high-income countries, with main triggers of air pollution, mold, and pollen [18]. Regardless of income level, children who live in closer proximity to urban centers that have higher levels of traffic-related air pollution experience more frequent and severe asthma symptoms that lead to an increased number of hospitalizations [31].

Under-nutrition is the cause of nearly 50% of all deaths in children under 5 years of age, and it is very closely related to poverty, food insecurity, and drought [91]. Under-nutrition causes developmental delays and issues, weakens the immune system, and increases the likelihood of infections [97]. On the other hand, over-nutrition is an increasing issue, especially in first-world countries, with global rates of childhood obesity on the rise and introducing many new challenges for children, such as early-onset type 2 diabetes, hypertension, and non-alcoholic fatty liver disease [77].

Infectious diseases also affect children disproportionately, especially in low- and middle-income countries, where they remain a leading cause of mortality, responsible for over 50% of deaths among children and adolescents, compared to about 6% in high-income countries [21]. Diseases such as malaria, or vaccine-preventable illnesses, including measles, strongly contribute to the number of deaths among children under 5 years of age [26]. Despite efforts to increase vaccination in many of these areas, some disparities still remain, and children in these zones have an increased risk of disease development or even more serious health complications [67].

It is clear that the impact of disease burden on younger people is inconsistent across the globe, and depends on factors such as socioeconomic status, geographical location, health-care infrastructure, and environmental conditions. Children who live in areas with poor air quality, food insecurity, or inadequate medical services are at a higher risk of developing health issues. This disparity demonstrates why age-stratified health models that incorporate environmental factors (instead of using population-level data alone) are a useful consideration.

1.2.3 Thesis objectives

Although extensive research exists on health indices, disease burden in children, and environmental determinants of health, most current modeling frameworks remain limited in their capacity to forecast health outcomes of population subgroups over long time horizons. Existing studies are often cross-sectional or retrospective, and few incorporate dynamic feedbacks among environmental, behavioral, and demographic factors [56].

Despite the fact that QALY and DALY are both indices that provide very useful conclusions when concerning population health, they are based on the general population health rather then looking at age-specific health and overall dynamic health modeling [33, 57]. Further, these studies do not often attempt to model these interactions for long periods of time, falling short in demonstrating the many long-term health consequences that these issues will have when considering childhood exposure [85].

Another existing gap in literature is the lack of integration of climate or geography into health models. While there are studies that demonstrate how some diseases are more prevalent in certain climates or with climatic variables, these are more often than not modeled independently instead of in a unified way that would show the multiple long-term health effects from the environmental events [32, 100].

A majority of international health comparisons fail to consider regional inequities in infrastructure quality, socioeconomic status, and healthcare access. This is problematic when attempting to analyze how at risk certain communities are when they are in a large diverse country, especially under climate stressors [99]. There are some datasets including those from WHO and the United Nations International Children's Emergency Fund (UNICEF) which give child health metrics, these datasets are not usually used in predictive models that consider uncertainty and future situations [6].

Finally, few studies offer tools for scenario-testing. These tools are important for policymakers so they can anticipate future health challenges related to climate and environmental changes. For example, Patz et al. [69] developed an approach linking climate variability to infectious disease dynamics, highlighting how hybrid modeling can incorporate environmental feedback. Similarly, UNICEF [91] has explored nutrition and environmental factors to forecast long-term health impacts. Despite these efforts, such studies remain the exception in health modeling.

This thesis will attempt to address several of these limitations. It will introduce an age-stratified, climate-aware, and dynamic health model that uses both discrete and continuous environmental influences, informed as much as possible with real-world data. The model will be applied to multiple climate zones around the world, with a focus on children's health. This model will be used as a tool in partnership with existing indices in the future by using simulation to explore how evolving socioeconomic and environmental factors can shape health trajectories differently around the world.

Chapter 2

Methodology

2.1 Model Framework

We propose a deterministic, age-structured compartmental model that is able to simulate population dynamics and health deterioration under both continuous and discrete environmental factors. The overall aim is to track each of four populations – children, youth, adults, and the elderly – evolve over time when considering food and air quality effects as well as discrete events such as droughts. This model will be the mathematical core of the broader effort to create a tool to measure health over time.

2.2 Literature Review - Mathematical Modeling of Population Health

Mathematical modeling provides an effective tool used to understand and predict health outcomes at a population-level. Infectious disease transmission, chronic disease progression, and population growth are common examples for which mathematical models are useful in the area of public health. These mathematical models allow researchers to simulate complex systems and assess the effects of specific parameters on system behavior. Such models are defined as compartmental models, and also allow researchers to view how various factors interact to influence health over time.

One of the earliest and most widely used compartmental models in health modeling is the SIR (Susceptible–Infectious–Recovered) model, which was developed in 1927 by Kermack and McKendrick and used differential equations to demonstrate the spread of infectious disease through a population [44]. Since the creation of this model, many others have been developed and applied to public health, often during epidemic events such as HIV/AIDS, influenza, and COVID-19 [7]. A common feature of these models is their division of a population into discrete compartments based on factors such as health status or vaccination status.

Differential equations describe how individuals transition between these compartments over time.

Beyond infectious diseases, mathematical modeling has been extended to a wide range of research areas, including disease burdens, health behaviors, healthcare systems, and environmental determinants of health [63]. An illustrative example is the PREVENT model, developed by a team of Dutch researchers, which simulates the long-term effects of interventions on chronic diseases such as cardiovascular illness by incorporating age, risk factors, and mortality rates [63]. A related application is the Global Burden of Disease (GBD) project, which estimates the impact of over 350 diseases and injuries across countries and age groups using statistical and computational modeling techniques [56].

Age-structured models are useful to capture how health evolves across the human lifespan. These models function by dividing the population into distinct age groups, each assigned age-specific rates for factors such as mortality, disease incidence, and exposure to risk factors. Such models have been applied in a range of scenarios, from evaluating childhood vaccination programs to analyzing the impact of aging populations on health expenditures and workforce participation [34]. When combined with environmental or geographical data, these models can further simulate region- or climate-specific health outcomes with greater accuracy.

Age-stratified compartmental models, which split populations into distinct age groups, are tools used in modeling demographic and disease dynamics across a variety of health and environmental contexts. One notable application is the modeling of COVID-19 vaccine strategies in Ghana, where researchers divided the population into three age groups (under 25, 25–64, and 65+) and evaluated vaccination effectiveness and outcomes under various scenarios. These included transitions between age groups and age-specific mortality, illustrating how stratification can enhance both the predictive accuracy of the model and its relevance for informing public health policy [65].

Beyond infectious diseases, age-structured models have been used to analyze chronic and endemic conditions. For instance, a pneumonia transmission model stratified children and adults, revealing how age-based treatment strategies influence disease dynamics and overall burden [5]. A separate six-state compartmental model applied to COVID-19 in the United States assessed how age demographics and age-dependent mobility patterns affected epidemic spread, with age-based calibration significantly improving model accuracy [1]. Additionally, age-structured optimal control models incorporate both host age and infection age to determine the most effective targeted interventions over time. These specific models have shown that prioritizing the elderly first, followed by younger age groups, can minimize both fatalities and associated costs [72]. However, this depends on the disease considered, as not all have identical trends.

Despite these strengths, age-structured models have several limitations. A common assumption is that individuals within each compartment or group are homogeneous, which means that all individuals in a compartment have the same risk of infection, interact with each other in the same way, and have identical health characteristics. This often overlooks important variations in health status, behavior, or exposure. Additional shortcomings include the exclusion of feedback mechanisms between health and environmental conditions, and a reliance on historical data rather than forecasting future scenarios. Some models are also static, lacking time-varying dynamics, which limits their applicability to real-world systems where uncertainty and change are inherent. Recently, however, there has been growing interest in hybrid models that integrate differential equations with agent-based simulations or machine learning algorithms to more accurately represent real-life uncertainty [99].

This study will use a hybrid model and introduce its own new features including age stratification, stochastic environmental inputs such as drought, and include dynamic feedback between environmental and health variables. The aim of this work is to be able to create a model that can capture how health evolves across all stages of life, while at the same time demonstrating how other pressures such as pollutions or climate events can alter population health over time.

2.2.1 Long-Term and Age-Stratified Population Health Models

Several studies have examined how population health evolves under long-term environmental and demographic pressures. These works move beyond short-term epidemic dynamics to explore gradual shifts in mortality, morbidity, and population structure over multi-decade or century-scale horizons. Although methodologies vary—from integrated assessment models to demographic exposure–response frameworks—they collectively emphasize that environmental change and demographic aging are inseparable components of future health forecasting. The following review summarizes key modeling approaches and identifies limitations addressed in this thesis.

Climate–Disease and Environmental Exposure Models

Colón-González et al. (2021) used a multi-model, multi-scenario climate–disease framework to project malaria and dengue transmission under future warming [23]. Their results indicated that by 2070, malaria seasons in highlands could lengthen by 1.6 months and dengue seasons in lowlands by 4 months, exposing up to 4.7 billion additional people. Kovats et al. (2015) performed a global risk assessment linking heat, flooding, diarrheal, malaria, and undernutrition mortality to climate change [46]. Both studies highlight how temperature and precipitation shifts amplify disease risk but do not incorporate demographic transitions or drought-related effects.

Food System and Socioeconomic Health Models

Springmann et al. (2016) developed an agricultural–health model coupling climate-driven crop yield changes to diet-related mortality [80]. They projected approximately 529,000 climate-related deaths by 2050, with mitigation reducing mortality by 29–71%. Using the International Futures Integrated Forecasting Model, Hughes et al. (2011) simulated global health trajectories from 2005–2060, showing communicable disease deaths halving while non-communicable diseases and injuries more than doubled [39]. These frameworks connect environmental and economic pathways to health outcomes but treat populations as homogeneous and do not model stochastic shocks.

Air Pollution and Demographic Burden Models

Air pollution modeling forms the most mature branch of long-term environmental health analysis. Malley et al. (2024) estimated 10.8 million premature deaths from $PM_{2.5}$ exposure by 2050 [50]. Butt et al. (2017) reported an 89–124% global increase in $PM_{2.5}$-related mortality since 1960, driven by industrialization in Asia [10]. Cohen et al. (2017) extended this using global burden of disease data, attributing rising chronic morbidity to aging and persistent pollution, while Vollset et al. (2024) projected age- and sex-stratified mortality and DALYs to 2050, revealing a shift from communicable to chronic disease dominance [22, 98]. These studies include demographic detail but rely on statistical forecasting rather than mechanistic population models.

Spatial Hazard and Vulnerability Models

Mills et al. (2018) created one of the few age-stratified spatial models, projecting exposure to inland flooding and wildfire smoke in the United States through 2090 [53]. Under moderate mitigation (RCP 4.5), roughly 15 million children and 25 million older adults could avoid flood exposure compared with high-emission (RCP 8.5) outcomes. This work demonstrates the value of spatially resolved and age-specific modeling but remains hazard-specific and omits compounding stressors such as food insecurity or drought.

Synthesis and Research Gap

Together, these studies illustrate growing attention to climate- and pollution-related health forecasting but also reveal consistent limitations. Most models isolate one or two environmental drivers, apply coarse age groupings, and use statistical projections that omit feedback between environment and demography. Drought-related health impacts remain largely unmodeled, and few frameworks capture cumulative, nonlinear effects of multiple stressors on long-term population structure. The lack of unified, deterministic models constrains the ability to evaluate resilience over century-scale horizons.

Relation to the Present Work

The framework developed in this thesis addresses these shortcomings by coupling demographic and environmental processes within an age-structured system of differential equations. It integrates gradual stressors—such as food and air quality degradation—with discrete shocks including droughts and pandemics, simulating population evolution over a 65-year period. Unlike regression-based forecasts (e.g., Vollset et al., 2024; Springmann et al., 2016), this model tracks births, deaths, and aging continuously, capturing nonlinear interactions between environmental pressures and demographic composition. This approach provides a mechanistic basis for assessing how cumulative environmental stresses alter health outcomes and age distributions over time.

International and Cross-Climate Comparisons

Some of the many reasons that health is such a complex topic to model is due to geographical locations, income levels, health infrastructure, environmental exposures, and cultural behaviors. These are all aspects that need to be considered and accounted for. In order to do this, researchers rely on cross-national and cross-climatic studies that determine how different context-specific factors each shape the trajectory of health. Again, these are especially valuable when looking at children who are more sensitive to these variations.

Climatic factors such as extreme temperatures, humidity, or rainfall patterns can all play a role in regional health risks. Children in tropical or subtropical climates are at a much greater risk of some vector borne diseases including malaria or dengue which are both directly influenced by temperature and precipitation [32]. Similarly, populations in deserts are much more likely to suffer from food insecurity, clean water scarcity, and heat-related illnesses that are consistent with long periods of drought [90].

Other regions such as urban centers that contain large amounts of air pollution due to industrialization show an increase in certain health burdens such as childhood asthma and other respiratory illnesses. Cities such as Delhi, Beijing, and Cairo have been the focus of environmental studies concerning health because of their high levels of $PM_{2,5}$ and nitrogen dioxide contained in the air [37]. On the other hand, children living in cold northern climates such as Canada or Scandinavia have a higher rate of some seasonal respiratory infections and vitamin D deficiency which are caused from harsh winter conditions that contain limited sun exposure [38].

Low average income areas often experience a compounding of health-related issues. For instance, low-income areas with hot climates demonstrate an increase in malnutrition and diarrheal diseases in children, directly caused by limited access to clean drinking water and poor healthcare infrastructure [6]. Unsurprisingly, high-income countries have opposite issues, such as obesity and early onset diabetes which reflect over nutrition and slow-moving lifestyles [77]. This exact contrast demonstrates why it is so important to have accurate

context sensitive health models, specifically when observing how environmental and socioeconomic factors interact.

Despite there being multiple studies that attempt to compare children's health across regions, very few manage to consider or integrate climate classifications into their analysis. One exception that should be noted is the Lancet Countdown on Health and Climate Change, which is able to follow climate sensitive health indicators around the world and highlight differential exposure by both region and income level [100]. A paper by Weinhold et. al. [102] used Köppen climate zones as a type of framework to study disease prevalence and mortality, leading to significant associations being made between health outcomes of children and climate types.

Despite these exceptions, the majority of international health indices are still static and lack the dynamic ability to simulate how real-life climate events would change health trajectories. Furthermore, most datasets are focused on other age groups, with adults being the largest focus, and do not analyze the obscure child-specific risks. This study will aim to incorporate climatic variability and geographic diversity into a health model that is dynamic and age-stratified. This model will then be applied to a diverse set of regions around the world that span climates and socioeconomic situations.

2.2.2 Model build considerations

This study will devise a novel health model incorporating the various factors considered in this section, in order to compare populations in six different countries. The countries selected for this work together with their populations and climate details are listed in Table 2.1.

Country	Population	Climate
Canada	40,100,000	Humid continental
U.S.	341,800,000	Humid subtropical
Brazil	217,600,000	Tropical savanna
U.K.	68,300,000	Temperate maritime
Australia	26,400,000	Oceanic
Japan	125,800,000	Humid subtropical

Table 2.1: Countries with their Population and Climate.

There were three main factors in deciding upon the six countries that are included in

the model simulations: population, climate, and geographical location. The selection aimed to reflect a range of climate zones, population structures, and geographic regions to ensure global representation.

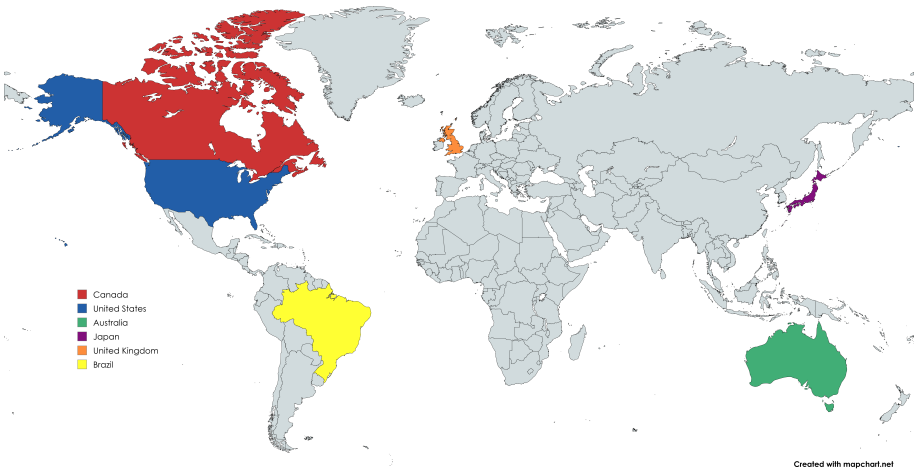


Figure 2.1: Map of Modeled Countries.

Figure 2.1 allows visualization of where the countries are located around the globe. To capture the influence of environmental variation, country selection aimed to include as many of the seven major climate types as possible. Each climatic zone is associated with distinct environmental conditions that may affect physical health in different ways. Including a diverse set of climate regions enables a comparative analysis of how climate-related factors contribute to health outcomes across global settings. The differences and definitions of the seven climatic zones are as follows:

1. Humid continental: Large temperature difference with cold winters and hot summer and precipitation occurs throughout the entire year [49].
2. Humid subtropical: Hot and humid summers with mild winters, lots of precipitation with thunderstorms in the summer and tropical cyclones [49].
3. Tropical savanna: Has a wet and dry season with consistently high temperatures with heavy rainfall in wet season and very little during dry season [48].
4. Temperature maritime: Mild temperatures that are fairly consistent between seasons, precipitation is even distributed across the year [3].
5. Hot desert: Extreme heat during the day and cool nights with low humidity, very little rain that occurs sporadically [42].

6. Subtropical Highland: Warm average temperatures that significantly decrease at night, precipitation is variable and depends on height of the location [101].

7. Oceanic: Mild temperatures with high humidity and a small range in temperatures and a moderate to high amount of precipitation throughout the year [87].

Each climate is characterized by various factors such as increased chance of drought, shorter growing season, high temperatures that worsen air quality, and so on. It should be noted that all countries considered are made up of more than one climatic zone. For the purposes of this study, each country has been assigned to its most population-relevant climate zone. Canada, for example, is classified as humid continental, as the majority of its population resides in regions characterized by large seasonal temperature variation, which are hot summers, cold winters, and year-round precipitation [24]. Both the United States and Japan fall under humid subtropical climates due to the fact that their summers are hot and humid, with both countries receiving considerable precipitation in the summer and with the accompanying risk of the impact of tropical cyclones [24, 70]. Brasil is mainly composed of tropical savanna areas; it has both wet and dry seasons that have high temperatures and vastly different amounts of precipitation [70]. The United Kingdom is considered temperate maritime since it has mild temperatures across all seasons with little variance and where precipitation is distributed fairly evenly throughout the year [70]. Australia falls under the category of oceanic since it has mild to high temperatures with high humidity and a moderate to high amount of precipitation yearly [70]. These six countries include most of the climate types and spread over five of the inhabitable continents on Earth; they also have substantially different population sizes. These factors provide this study with a large variety of climates and populations on which we may apply our model.

Discrete time weather and disease events add a layer of real-world unpredictability to the simulation. Just as the COVID-19 pandemic emerged unexpectedly with widespread consequences, the timing and impact of future pandemics and droughts cannot be precisely forecast. They are essential to include because they cause sudden effects that vary by age group depending on the nature and severity of the event. These shock events are implemented stochastically in the model and produce immediate, nonlinear increases in mortality rates when triggered. This reflects the abrupt and often disproportionate burden that real-world crises place on vulnerable subpopulations. Incorporating these disruptions into long-term simulations improves realism and allows for more accurate stress-testing of national health trajectories.

Although this study will be applied to all ages, the main focus will be on the age range of 0-14 years old. The health of children is particularly important due to their heightened physiological vulnerability and the long-term implications of early-life exposures. We propose a model that incorporates all of these factors simultaneously, so that we may better predict, quantify, and analyze possible outcomes that may arise from a variety of circumstances. This technique of mathematical modeling will give us a method to quantify the

different possible outcomes, as well as analyze each of them. Modeling will allow predictions to be made about future health trends, and knowing them ahead of time will give us the opportunity to prevent them if the results are not positive.

Population Compartments and Governing Equations

For each country under consideration, the population has been compartmentalized into four groups. Each population is defined as a function of time, where t is the time in years. Any individual of the population belongs to exactly one of the following:

Symbol	**Description**
$P_k(t)$	Children (0–14 years)
$P_y(t)$	Youth (15–29 years)
$P_a(t)$	Adults (30–64 years)
$P_s(t)$	Seniors (65+ years)

Table 2.2: Age-Stratified Population Compartments.

The governing equations each represent the evolution of the population between compartments, and describe the changes of population over time in each compartment due to birth, aging, death, and environmental factors that are linked to health:

$$\frac{dP_k}{dt} = b_y P_y(t) + b_a P_a(t) - \Big(d_k + f_k F + A_k A\Big) P_k(t) - \frac{P_k(t)}{15}, \tag{2.1}$$

$$\frac{dP_y}{dt} = \frac{P_k(t)}{15} - \Big(d_y + f_y F + A_y A\Big) P_y(t) - \frac{P_y(t)}{15}, \tag{2.2}$$

$$\frac{dP_a}{dt} = \frac{P_y(t)}{15} - \Big(d_a + f_a F + A_a A\Big) P_a(t) - \frac{P_a(t)}{35}, \tag{2.3}$$

$$\frac{dP_s}{dt} = \frac{P_a(t)}{35} - \Big(d_s + f_s F + A_s A\Big) P_s(t). \tag{2.4}$$

Each of the included parameters is associated with specific demographic or environmental factors that influence the modeled system:

Parameter(s)	Description
b_y, b_a	Age-specific fertility rates
d_k, d_y, d_a, d_s	Baseline mortality rates for each age group
a_k, a_y, a_s	Aging-out transition rates between compartments
f_i	Proportion of compartment with food-related disease
A_i	Proportion of compartment with air-related disease
F, A	Mortality rate of food-related and air-related diseases

Table 2.3: Model Parameters and Their Descriptions.

Model Assumptions

1. Consistent diabetes and asthma mortality rate: Due to lack of data, each country will have identical mortality rates for diabetes and asthma.

2. Uniform aging: Each age stratified compartment will have a uniform amount of the population age out at each time step, represented by $\frac{1}{age_range}$.

3. Static birth and death rate: Each of these rates will remain constant for the duration of the 65 year simulation.

4. Drought and disease are modelled as independent Bernoulli events: The occurrence of a drought or disease shock event is determined independently using fixed probabilities at each time step.

5. Uniform age group impacts: All individuals within the same age compartment experience the same risk from environmental and health stressors.

6. No births from seniors or children: All births will come from the youth and adult compartments since the number of births from the other two compartments are negligible.

7. Multiplicative impact from discrete events: When a discrete event occurs (e.g., a drought), age-specific multipliers are applied directly to the mortality rate terms in the differential equations, proportionally increasing mortality for each age group. These multipliers are designed to capture short-term shocks to the population dynamics and will be described in detail in the following section.

8. Realistic randomized disease rates: Age distributed mortality patterns are randomly selected from the considered historical pandemics.

Environmental Feedbacks

Environmental stressors such as food insecurity and air pollution are integrated into the model as a constant, modifying the baseline mortality rate. These chronic pressures differ from discrete shock events such as droughts or epidemics, as they accumulate gradually and

persistently impact public health.

Chronic food exposure is modeled using a hazard multiplier F, which is applied to all age compartments through distinct sensitivity coefficients f_k, f_y, f_a, and f_s. These coefficients are associated with how vulnerable each age group is to poor diet quality, food insecurity, or undernutrition. For example, seniors and adults, who are more susceptible to complications from diabetes and malnutrition, have larger f values [66]. This model component captures increased death rates due to long-term food-related health burdens such as type 2 diabetes, cardiovascular disease, and micronutrient deficiencies.

Air quality effects are modeled using a hazard multiplier A, which scales death rates multiplicatively and interacts with age-specific vulnerability coefficients A_k, A_y, A_a, and A_s. These coefficients account for differing sensitivity to chronic air pollutant exposure (e.g., $PM_{2.5}$, NO_x). Children and the elderly, due to less developed or declining respiratory and cardiovascular function, are especially at risk, and therefore A_k and $A_s > A_y$ and A_a. This modeling approach reflects empirical findings linking air pollution to elevated risks of asthma exacerbation, chronic bronchitis, and cardiovascular events.

Together, F and A serve as multiplicative stressors, scaling death rates in each compartment proportionally to their age-specific sensitivity.

Drought Shock Events

Drought is modeled as a discrete stochastic shock that temporarily increases mortality rates across all age groups. It represents acute disruptions in food systems, water access, sanitation, and healthcare services, each of which may contribute to spikes in mortality.

For the purposes of this study, a drought is defined as an event classified at severity level D3 or D4 on the standard drought intensity scale. These levels correspond to "Extreme Drought" and "Exceptional Drought," respectively, which are associated with severe impacts on agriculture, water supply, and public health. Lesser drought categories (D0–D2), while more common, were excluded from the model due to their relatively minor or short-term health effects. This threshold aligns with classifications used by national drought monitoring systems such as the Canadian Drought Monitor and the U.S. Drought Monitor, both of which define D3 and D4 as conditions that involve widespread crop failure, prolonged water shortages, and significant socioeconomic disruption [2] [29].

Each year, there is a small fixed probability that a drought will occur. Once active, drought effects persist for the current and following year, reflecting lagging impacts on health infrastructure and food systems. During these periods, mortality rates d_i for each age group are scaled by a country specific multiplier.

To account for variation in climate resilience, infrastructure, and historical drought impacts, the multipliers were estimated separately for each country based on reported impacts of past droughts.

Country	**0–14 (Children)**	**15–29 (Youth)**	**30–64 (Adults)**	**65+ (Seniors)**
Canada	×1.03	×1.01	×1.01	×1.04
United States	×1.02	×1.015	×1.02	×1.03
Australia	×1.03	×1.02	×1.03	×1.05
Brazil	×1.04	×1.03	×1.04	×1.06
United Kingdom	×1.015	×1.01	×1.015	×1.03
Japan	×1.02	×1.015	×1.02	×1.04

Table 2.4: Drought mortality multipliers by age group and country. Multipliers represent proportional increases in baseline age–specific death rates when a drought event is active. Values are *constructed* from country hazard exposure and age–specific vulnerability evidence (see Methods below), triangulating national/agency reports and climate–health studies rather than taken directly from one dataset [2, 4, 9, 16, 29, 30, 54, 64, 88, 95, 96, 112].

These values are not copied from a single statistical table; rather, they are derived by combining (i) country–level drought hazard frequency/severity indicators and outlooks [2, 29, 64], with (ii) evidence on age-specific vulnerability to drought-linked outcomes [4, 16, 95, 96, 112], and (iii) country-specific public-health context reports [9, 30, 54, 88].

For Canada, the multipliers reflect relatively mild mortality effects, grounded in findings from the Canadian Institute for Climate Choices, which reports a moderate but growing burden of climate-related health stressors, particularly among vulnerable populations [12]. In the United States, similar moderate values were chosen, supported by studies from the CDC and the U.S. Global Change Research Program indicating regional variability in heat and drought related mortality but generally manageable public health infrastructure [95].

Australia received slightly elevated multipliers, especially for the 65+ age group, based on recurring severe droughts and bushfire linked respiratory burden. Sources from the Australian Institute of Health and Welfare link drought and extreme heat to spikes in cardiovascular and respiratory deaths, especially in the elderly [4]. For the United Kingdom and Japan, the selected multipliers are modest, reflecting limited drought exposure but acknowledging indirect health consequences (e.g., heat waves in Japan during droughts have strained hospital resources) [54, 88].

Brazil received the highest drought multipliers among the countries studied, justified by both the frequency and severity of droughts in regions including the Northeast and Amazon. According to Brazil's Ministry of Health and regional public health assessments, droughts have been linked to spikes in child malnutrition, increased respiratory disease, and higher

elderly mortality, particularly in rural and underserved areas [9, 30]. These values were selected to represent realistic stress amplification while preserving model stability.

Table 2.4 summarizes the selected values for each country and age group. Drought events are activated independently from disease events but can occur simultaneously, compounding their effects on population dynamics. This structure allows for realistic modeling of compounding crises, such as when disease outbreaks follow periods of resource scarcity or population displacement.

Disease Shock Events

In this simulation framework, discrete disease events represent temporary but impactful epidemics that occur infrequently. When active, they increase age specific mortality rates to reflect the elevated risk during major outbreaks. Unlike continuous health stressors such as diabetes or air quality, disease events are treated as transient shocks that last for a defined period –in this case, 3 years– and occur randomly with a fixed annual probability (5%). This 5% was decided upon directly from the number of major epidemics between the years 1920 - 2020. To determine a major epidemic, the criteria was as follows: the viral pandemic had to have effects lasting at least 1 year in duration and having the total death toll exceed 150,000 people worldwide.

Each time a disease event is triggered in the simulation, one of five historically significant viral pandemics is randomly selected to determine the age-specific mortality impact. These include the Spanish Flu (1918), the Asian Flu (1957), the Hong Kong Flu (1968), the Swine Flu (2009), and COVID-19 (2019). Each of these pandemics is known for its distinct mortality distribution across age groups, with some, such as the Spanish Flu, having disproportionate impacts on young adults, and others, such as COVID-19, affecting the elderly most severely. Their inclusion allows the model to incorporate diverse epidemiological patterns grounded in real-world historical events.

Each of these pandemics had distinct age-related mortality patterns. Based on historical mortality data and available age-specific death distributions, we define age-specific mortality multipliers for each event as follows:

Pandemic	**0–14**	**15–29**	**30–64**	**65+**
Spanish Flu	1.03	1.15	1.10	1.05
Asian Flu	1.02	1.10	1.08	1.04
Hong Kong Flu	1.015	1.08	1.06	1.03
Swine Flu	1.01	1.05	1.04	1.02
COVID-19	1.005	1.02	1.03	1.10

Table 2.5: smth

The multipliers presented in table 2.5 were derived from available mortality estimates and age-specific death distributions reported by the WHO and the U.S. Centers for Disease Control and Prevention, along with historical analyses of major pandemic events [15, 17, 19, 83, 110, 111]. The total number of deaths globally was combined with available data or scientific estimates for the age distribution of those deaths to generate relative mortality burdens. These were then scaled into reasonable multiplicative modifiers to age-specific death rates in the model.

The Spanish Flu received the highest multipliers overall due to its extraordinarily high global death toll and its unusual age distribution, where young adults suffered disproportionately [83]. COVID-19, in contrast, produced a lower total death burden but had a sharply skewed impact on elderly populations, which is reflected in the strong multiplier for the 65+ group [19, 110]. The Swine Flu, while widely spread, resulted in relatively fewer deaths, with a modest increase in mortality among younger populations [15]. The Asian Flu and Hong Kong Flu exhibited moderate overall mortality and less dramatic age-specific variation, resulting in multipliers that reflect intermediate severity [17].

These values are not direct death rates but relative adjustments to each compartment's baseline mortality during an active epidemic. Their magnitudes were chosen to reflect historical severity while maintaining plausible population dynamics in simulation. This allows the model to explore the effect of stochastic pandemic events on long term demographic trends with age specific realism.

These multipliers are applied directly to the base mortality rates (d_k, d_y, d_a, d_s) for each age group during the active duration of the disease event, effectively redefining these parameters to reflect the increased rates. The structure ensures that disease events not only vary in timing and duration but also in severity and age specificity — providing a more nuanced, realistic simulation of population health dynamics.

This approach is implemented numerically by modifying the death rates dynamically when a disease is active, randomly selecting one pandemic, and applying its associated multipliers. The randomness reflects the unpredictability of outbreak timing and nature, while the calibrated multipliers are grounded in epidemiological evidence.

In the code, a disease shock is implemented as follows: for each year of the simulation, a random number is drawn from a uniform distribution. If this number is less than the fixed disease probability (0.05), a disease event is triggered. When a disease occurs, a random choice is made among the five historical pandemics, and its associated age specific multipliers are applied to the baseline death rates for the current and next two years (3 years total). During these years, the modified death rates are passed to the ODE solver, altering the trajectory of the age compartments accordingly. If multiple events occur close together, their impacts are not stacked; instead, the most recent shock determines the active multipliers.

This assumption prevents unrealistic compounding of mortality effects when stressors overlap in short succession, which could otherwise exaggerate losses beyond empirically observed ranges. Conceptually, it reflects that health systems and population mortality respond to the dominant crisis at a given time step rather than summing hazards linearly—an intentional simplification adopted for model stability and interpretability.

Parameter values and Calibration

The model used in this study requires 17 parameters to simulate population dynamics under environmental stress. These include birth rates, baseline death rates, age transition (aging) rates, and sensitivity coefficients related to food and air quality. While the model structure is consistent across all countries, each country has its own unique parameter set calibrated to reflect its demographic and environmental conditions.

Table 2.6 presents the parameters used for Canada, which were determined using demographic data and estimates from Canadian health economics literature[11, 12, 73]. In particular, food- and air-related mortality coefficients were inferred from national reports and epidemiological studies due to the absence of direct risk quantification.

Parameter Type	0-14	15-29	30-64	65+
Birth rates (b_i)	-	0.032866	0.02397	-
Death rates (d_i)	0.001225	0.0006	0.0031	0.06873
Aging rates (a_i)	0.0667	0.0667	0.0286	-
Food hazard rates (f_i)	0.00240	0.00784	0.0764	0.2686
Air hazard rates (A_i)	0.13335	0.1463	0.0849	0.0996

Table 2.6: Parameter Values for Canada.

The values in table 2.6 were determined through Canadian demographic statistics and health economics literature in order to estimate the food and air related risks which are not exactly known [11, 12, 36, 73]. For Canada the initial population values are as follows:

Initial Condition	**Population (millions)**
$P_k(0)$	6.4
$P_y(0)$	8.0
$P_a(0)$	18.7
$P_s(0)$	7.1

Table 2.7: Initial age-specific population values for Canada [92].

The parameter set for the **United States** was derived from publicly available demographic data, government health statistics, and published literature on chronic disease prevalence [16, 26, 95, 103]. As with all countries in this study, the birth rates, death rates, and aging rates reflect country-specific dynamics, while the food and air hazard coefficients were selected to reflect estimated age-specific vulnerabilities based on diabetes and asthma prevalence, respectively.

Parameter Type	**0–14**	**15–29**	**30–64**	**65+**
Birth rates (b_i)	–	0.01100	0.01100	–
Death rates (d_i)	0.000147	0.001125	0.00445	0.05000
Aging rates (a_i)	0.06670	0.06670	0.02860	–
Food hazard rates (f_i)	0.00900	0.03600	0.17700	0.27300
Air hazard rates (A_i)	0.12000	0.09000	0.08500	0.07500

Table 2.8: Parameter values used for the United States.

These parameters were selected to reflect both the chronic burden of disease and the demographic transitions expected over a 65-year horizon. In particular, the elevated food hazard rates in the adult and senior groups reflect the high prevalence of diet-related illnesses such as obesity, diabetes, and cardiovascular conditions in the United States.

The initial population sizes for each age group were taken from recent census data:

Initial Condition	**Population (millions)**
$P_k(0)$	60.4
$P_y(0)$	88.1
$P_a(0)$	123.2
$P_s(0)$	54.4

Table 2.9: Initial age-specific population values for the United States[92].

The parameter set for **Australia** was derived using national demographic statistics, public health datasets, and global epidemiological estimates [4, 26, 103]. The aging and birth rates reflect national averages, while the food and air hazard coefficients were calibrated using age-specific prevalence of diabetes and asthma reported in Australian health surveys and international datasets [4, 26, 52, 66].

Parameter Type	0–14	15–29	30–64	65+
Birth rates (b_i)	–	0.01140	0.01140	–
Death rates (d_i)	0.000154	0.000432	0.002611	0.044351
Aging rates (a_i)	0.06670	0.06670	0.02860	–
Food hazard rates (f_i)	0.00280	0.03050	0.09500	0.21000
Air hazard rates (A_i)	0.08800	0.10200	0.09500	0.08600

Table 2.10: Parameter values used for Australia.

These parameter values include moderate mortality rates and slightly elevated asthma prevalence among younger populations in Australia [4, 26, 52]. The food hazard coefficients reflect growing concerns around diet-related diseases, particularly among adults and the elderly [47, 66, 79].

The initial population sizes for each age group were based on recent demographic estimates:

Initial Condition	Population (millions)
$P_k(0)$	4.9
$P_y(0)$	5.3
$P_a(0)$	12.4
$P_s(0)$	4.7

Table 2.11: Initial age-specific population values for Australia[92].

The parameter set for the **United Kingdom** was compiled from national demographic data, NHS health records, and international health statistics[26, 88, 103]. The birth and death rates reflect current demographic patterns, while the food and air hazard coefficients were selected based on age-specific estimates of diabetes and asthma prevalence reported by Public Health England and the Global Burden of Disease database[26, 52, 66].

Parameter Type	0–14	15–29	30–64	65+
Birth rates (b_i)	–	0.01120	0.01120	–
Death rates (d_i)	0.000187	0.000675	0.004215	0.047891
Aging rates (a_i)	0.06670	0.06670	0.02860	–
Food hazard rates (f_i)	0.00310	0.03340	0.10900	0.22900
Air hazard rates (A_i)	0.07000	0.09200	0.08800	0.07500

Table 2.12: Parameter values used for the United Kingdom.

These parameter values reflect a moderate but growing burden of lifestyle-related disease in the U.K., particularly among working-age adults and seniors [26, 52, 66]. Asthma prevalence remains elevated in children and young adults, contributing to slightly higher air hazard coefficients in those age groups.

The initial population sizes for each age group were taken from national census estimates:

Initial Condition	**Population (millions)**
$P_k(0)$	12.0
$P_y(0)$	13.3
$P_a(0)$	31.0
$P_s(0)$	12.6

Table 2.13: Initial age-specific population values for the United Kingdom[92].

The parameter set for **Japan** was developed using official demographic statistics, Ministry of Health reports, and published global datasets on noncommunicable diseases [26, 54, 103]. Japan's low birth rate and rapidly aging population are reflected in the selected parameters, along with relatively low prevalence of diabetes and asthma, especially among children and younger adults [26, 52, 66].

Parameter Type	**0–14**	**15–29**	**30–64**	**65+**
Birth rates (b_i)	–	0.00580	0.00580	–
Death rates (d_i)	0.001500	0.000802	0.004500	0.061849
Aging rates (a_i)	0.06670	0.06670	0.02860	–
Food hazard rates (f_i)	0.00050	0.00100	0.01100	0.05200
Air hazard rates (A_i)	0.05000	0.06000	0.04100	0.03200

Table 2.14: Parameter values used for Japan.

These parameters highlight Japan's low chronic disease burden in early and middle age, as well as the disproportionately high health vulnerability among its elderly population [26, 52, 66]. The extremely low food hazard coefficients in younger age groups reflect the country's comparatively healthy diet and low rates of childhood obesity and diabetes [66, 77, 91].

The initial population sizes for each age group were estimated from national census data:

Initial Condition	Population (millions)
$P_k(0)$	14.7
$P_y(0)$	20.6
$P_a(0)$	48.8
$P_s(0)$	36.4

Table 2.15: Initial age-specific population values for Japan[92].

The parameter set for **Brazil** was constructed using demographic data from the Brazilian Institute of Geography and Statistics (IBGE), regional health surveys, and international databases such as WHO and IHME [9, 26, 30, 103]. The food and air hazard coefficients reflect both a rising burden of chronic diseases and substantial environmental risk exposures, especially in rural and underserved regions [26, 52, 66, 105].

Parameter Type	**0–14**	**15–29**	**30–64**	**65+**
Birth rates (b_i)	–	0.01258	0.01258	–
Death rates (d_i)	0.000310	0.000830	0.004120	0.045600
Aging rates (a_i)	0.06670	0.06670	0.02860	–
Food hazard rates (f_i)	0.00350	0.01200	0.14100	0.20670
Air hazard rates (A_i)	0.12500	0.11500	0.08500	0.06200

Table 2.16: Parameter values used for Brazil.

These parameters reflect the dual burden of malnutrition and rising diet-related disease in Brazil, particularly in younger and middle-aged populations. Air hazard rates are elevated across all groups due to persistent exposure to pollution, especially in urban centers and deforestation-prone regions.

The initial population sizes for each age group were estimated using the most recent national demographic projections:

Initial Condition	Population (millions)
$P_k(0)$	46.8
$P_y(0)$	42.7
$P_a(0)$	101.2
$P_s(0)$	27.0

Table 2.17: Initial age-specific population values for Brazil[92].

Together, these country specific parameter sets provide the foundation for age structured simulation of population dynamics under environmental stress. By tailoring birth, death,

and hazard coefficients to each country, the model captures geographic variation in health outcomes and enables meaningful cross-national comparisons.

Relation to Existing Models

This model builds conceptually on previous models such as the SIR model. Similarly to this classic compartmental epidemiological model, it uses transitions between the discrete population states. However, instead of modeling infectious transmissions, it shows demographic flow and allows for analysis of environmental stress. This model also draws some inspiration from other previous models including the PREVENT model for chronic disease burden and the GBD project which aggregates DALYs across multiple populations [26, 57]. This new model builds on these by going beyond and introducing droughts and disease events in the form of stochastic shocks, giving climate or environmental feedback instead of health system interventions only, and it incorporates age stratification with realistic demographic turnover through birth, death, and aging.

2.3 Simulation Techniques

In order to numerically evaluate the system of differential equations that is outlined in the model framework, this study employs a Python-based simulation approach that combines continuous population dynamics with discrete stochastic climate and disease events. The objective is to simulate how the population health evolves over the next 65 years, aiming to capture age-specific effects of environmental stressors.

Numerical Integration with *solve_ivp*

The system of differential equations is solved by using the solve_ivp function from SciPy's integrate module. This is an appropriate function to use, as it uses adaptive stepsize control to solve ODEs. At each time step $t_i \in [0, T]$ (with t_0 the initial time and $t_i = t_{i-1} + \Delta t$ for $i = 1, \ldots, N$), the system of equations which controls the four populated compartments is evaluated on the interval $[t_{i-1}, t_i]$. Because the drought and disease status may change from one year to the next, the model must be re-evaluated at each time step independently to accommodate conditional parameter updates, which are the increased mortality rates during both of the stochastic events. The Runge–Kutta method of order 5(4) (RK45) is used as the default integration scheme, which is ideal since it offers a balance between computational speed and numerical precision.

Time Horizon and Granularity

This simulation covers a range of 65 years, discretized into 66 points. This allows for population trends to emerge and analyzed yearly or by the decade while keeping the computation and visualizations manageable. 65 was chosen to specifically allow the population of the children's compartment to entirely reach the seniors compartment. The time step being 1 year is a logical choice since it aligns with the natural scale of demographic processes such as aging and mortality.

Stochastic Drought Implementation

To introduce environmental uncertainty, a random binary drought vector is generated for each simulation run using a uniform random number generator. The probability of drought in a given year is country specific and reflects historical frequencies of severe drought events based on regional climate data, which can be seen in table 2.18.

A drought is triggered in a given year if a random number between 0 and 1 falls below the country's specified probability threshold. The drought probabilities used in the model are summarized in table 2.18.

Country	**Annual Drought Probability (%)**
Canada	7.7
United States	6.0
Australia	7.0
United Kingdom	10.0
Japan	4.0
Brazil	10.0

Table 2.18: Annual probability of drought by country [2, 29, 64].

When a drought is triggered, its effects persist for two consecutive time steps: the year it begins and the following year. This structure reflects the real world persistence of drought related impacts, such as reduced agricultural output, strained water resources, and delayed recovery of public health systems.

During active drought periods, age specific mortality rates are increased multiplicatively based on country and age calibrated drought multipliers. These modified death rates are passed into the system of ODEs as conditional arguments at each affected time step.

Data Storage and Initialization

Before simulation begins, the model initializes with country specific population distributions across four age compartments: children (0–14), youth (15–29), adults (30–64), and seniors

(65+). These initial values were compiled from national census data and reflect the approximate demographic structure at the model's starting point[11, 92, 103].

Country	**0–14 (Children)**	**15–29 (Youth)**	**30–64 (Adults)**	**65+ (Seniors)**
Canada	6.4	8.0	18.7	7.1
United States	60.4	88.1	123.2	54.4
Australia	4.9	5.3	12.4	4.7
United Kingdom	12.0	13.3	31.0	12.6
Japan	14.7	20.6	48.8	36.4
Brazil	46.8	42.7	101.2	27.0

Table 2.19: Initial population sizes (in millions) by age group and country.

The initial state for each country is stored as a vector of the form `initial_state = [Pk(0), Py(0), Pa(0), Ps(0)]`, corresponding to the four age compartments. An output matrix named `solution` with dimension $66, 5$ is pre-allocated to store simulation results. The first column holds the time values (from 0 to 65 years), while the remaining columns store the evolving population values for each age group.

During simulation, the main loop updates each time step independently, allowing for real time adjustments based on stochastic events such as drought and disease. This design enables conditional parameter switching while maintaining a consistent output format for analysis and visualization.

Reproducibility and Flexibility

This simulation was built to allow for easy parameter modification. A user could change the demographic or environment parameters by editing the parameters list, adjusting the drought frequency to test different scenarios regarding climate change, and/or swapping out the hazard coefficients in order to simulate other countries or scenarios.

Chapter 3

Results and Discussion

3.1 Cross-Country Comparison of Baseline Simulations

This section presents the baseline simulation results for all six modeled countries under standard environmental and demographic conditions. Although simulations were performed for each nation, only the results for Canada are displayed here in detail, as the patterns observed were qualitatively similar across countries. All additional figures have been included in the Appendix to avoid redundancy and visual clutter.

To ensure the stochastic components of the model (drought and disease shocks) produce stable and representative population trajectories, each country's simulation was iterated 25 times. The stability of these runs was quantitatively evaluated using the *Sum of Squared Errors* (SSE) of the cumulative mean change in each compartment relative to its previous checkpoint (Figure 3.1).

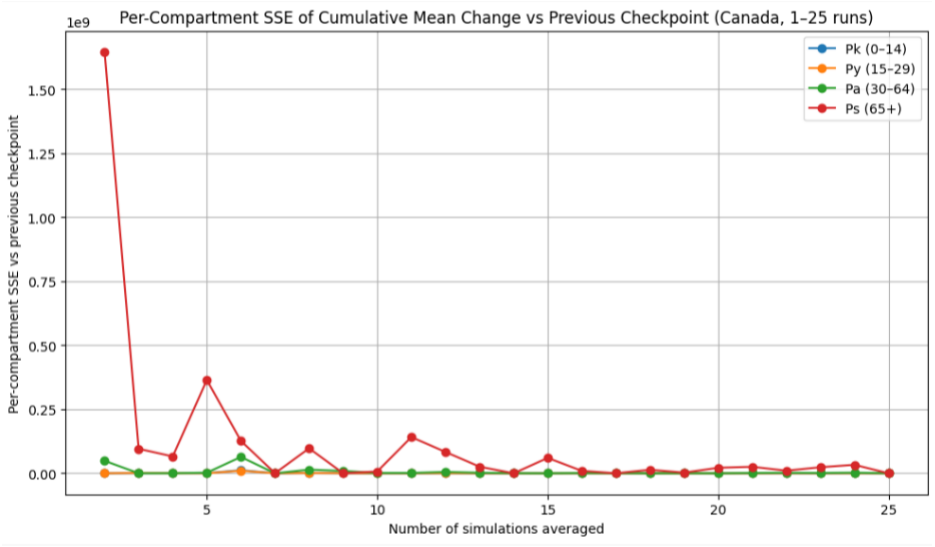


Figure 3.1: Per-compartment SSE of cumulative mean change across iterations.

Mathematically, for each age compartment $i \in \{k, y, a, s\}$, the cumulative mean population after n iterations was compared to the cumulative mean at iteration $n-1$ via:

$$\mathrm{SSE}_i(n) = \sum_{t=1}^{T} \left(\bar{P}_i^{(n)}(t) - \bar{P}_i^{(n-1)}(t) \right)^2,$$

where $\bar{P}_i^{(n)}(t)$ represents the mean population trajectory for compartment i at time t averaged over n iterations. The results (Figure 3.1) show that the *seniors* compartment (P_s) displays the largest initial variability, which is expected given that older populations are the most sensitive to stochastic mortality fluctuations. However, after approximately $n = 20$ iterations, $\mathrm{SSE}_s(n)$ converges to a negligible value, and the fluctuations across all compartments fall below a visually and numerically meaningful threshold. Formally, for $n \geq 20$, $\max(\mathrm{SSE}_i(n)) < 10^{-2}$ relative to initial variability, satisfying practical convergence criteria. Thus, all subsequent analyses are based on the averaged output from 20 independent stochastic simulations per country.

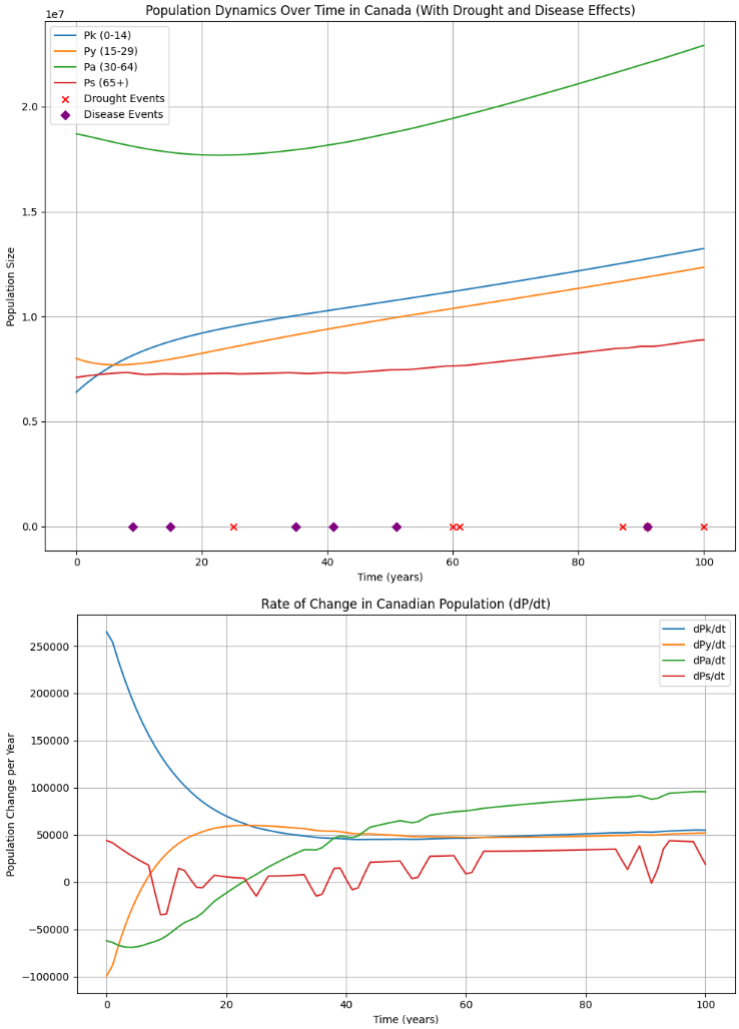


Figure 3.2: Canada Plots of Population Dynamics and Rate of Change.

Figure 3.2 shows one representative simulation run for Canada, illustrating the temporal evolution of the four age compartments and the timing of discrete shock events. The upper panel depicts the overall population trajectories, while the lower panel shows the corresponding rates of change (dP_i/dt) over time. The red crosses and purple diamonds indicate the occurrence of droughts and disease events, respectively. This visualization highlights how each shock event produces transient dips in the population rate of change, most notably in the adult and senior compartments, followed by gradual recovery. Such transient deviations are consistent with the model's stochastic design and reflect realistic short-term demographic perturbations.

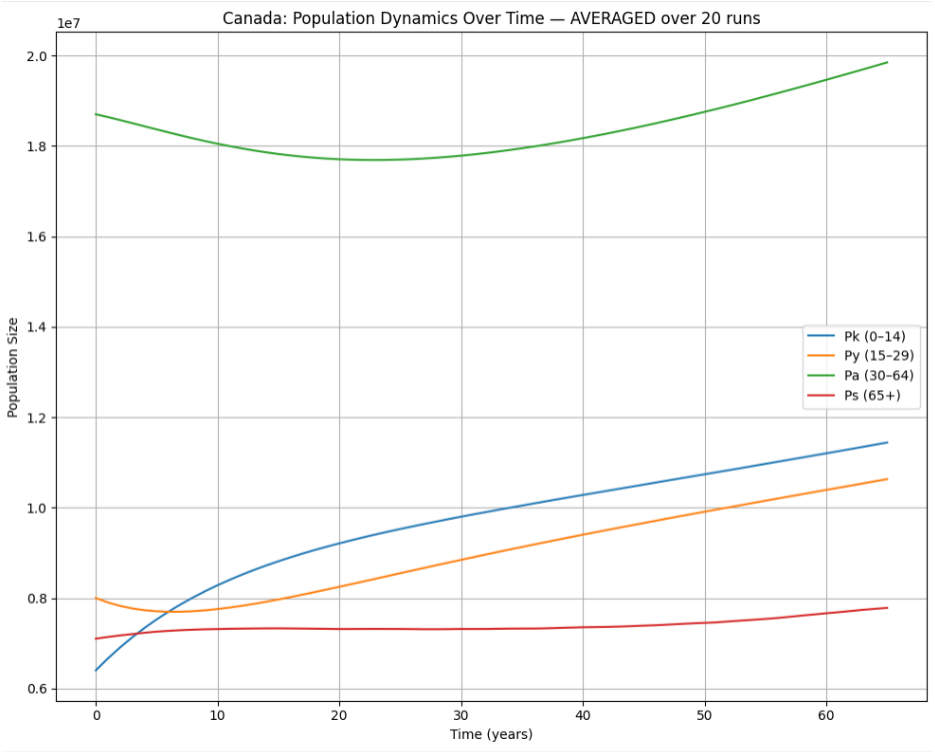


Figure 3.3: Population Dynamics in Canada, Averaged Over 20 Iterations.

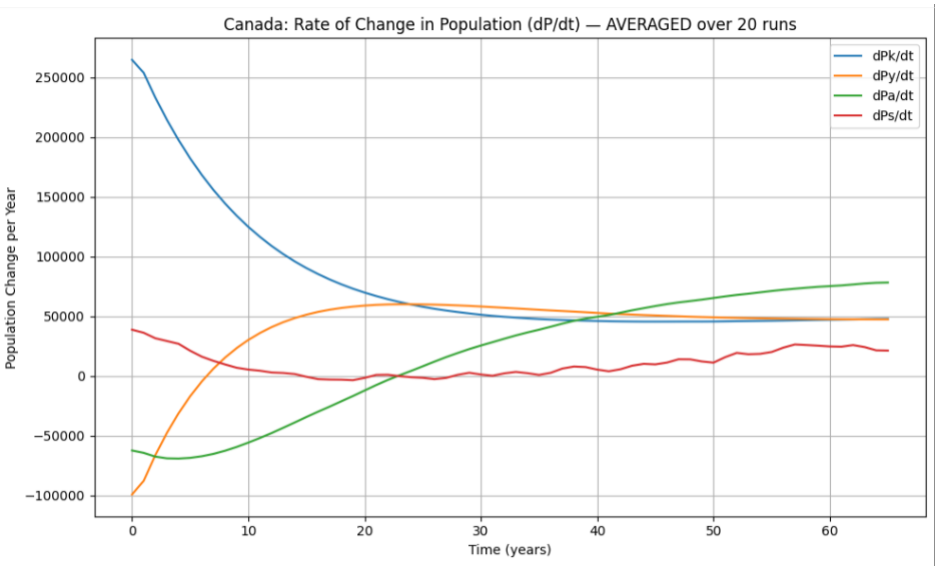


Figure 3.4: Rate of Change in Canada's Population Averaged Over 20 Iterations.

When results are averaged across 20 independent runs (Figure 3.3), the population trajectories become smooth and the stochastic noise is eliminated. Under these baseline conditions, the child (P_k) and youth (P_y) populations display steady long-term growth, the adult (P_a) group stabilizes after an initial decline, and the senior (P_s) compartment shows slow but persistent increases. The corresponding rate-of-change plots (Figure 3.4 confirm that all four groups converge toward near-equilibrium behavior by the later simulation years, indicating demographic stability under current stress levels.

To illustrate the stochastic variability across runs, Figure 3.5 presents the distribution of discrete events (droughts and disease outbreaks) over all 20 iterations.

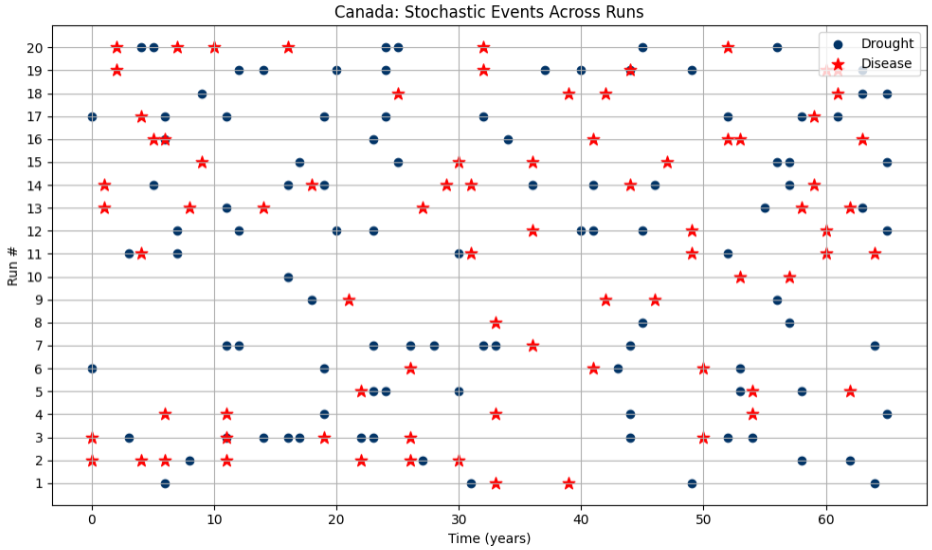


Figure 3.5: Stochastic Shock Events in Canada for 20 Iterations.

Each horizontal line represents one run, and each marker indicates the occurrence of an event in time. The scatter pattern confirms that the drought and disease processes are independent and uniformly distributed across the simulation period, with no systematic bias toward particular years. This visualization provides further evidence that the model's random event generator behaves as expected.

To complement the ratio-based comparisons, Figures 3.6 and 3.7 present each age group's proportion of the total population at the start and end of the simulation period. These normalized comparisons highlight demographic composition rather than absolute growth, offering insight into population aging and intergenerational balance under prolonged environmental and epidemiological pressures.

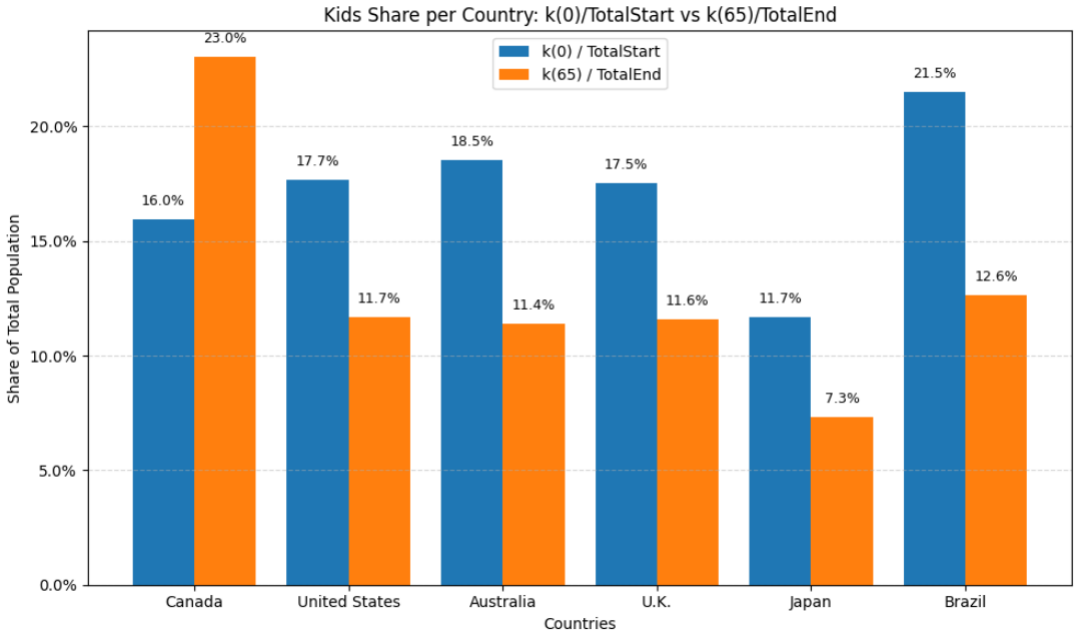


Figure 3.6: Ratio of Kids Compartment from the Start to End of the Averaged Simulations.

Figure 3.6 illustrates the share of children (P_k) as a fraction of each country's total population at the beginning and end of the 65-year baseline simulation. Across most countries, the share of children declines substantially over time, consistent with low fertility and gradual demographic aging. Canada again stands out as an exception, with the child share increasing from roughly 16% to nearly 24% of the total population. This contrasts sharply with the United States, Australia, and the United Kingdom, where the child share falls from approximately 17–18% to about 10–11%. Japan exhibits the lowest child share overall, declining to below 8% by the end of the simulation, reflecting its well-documented population aging trend. Brazil maintains a moderately high child share relative to other nations but still experiences a proportional decline from roughly 22% to 13%. Together, these trends reveal that most modeled populations are projected to undergo a demographic contraction in their younger cohorts, with Canada representing the only clear case of youth rejuvenation.

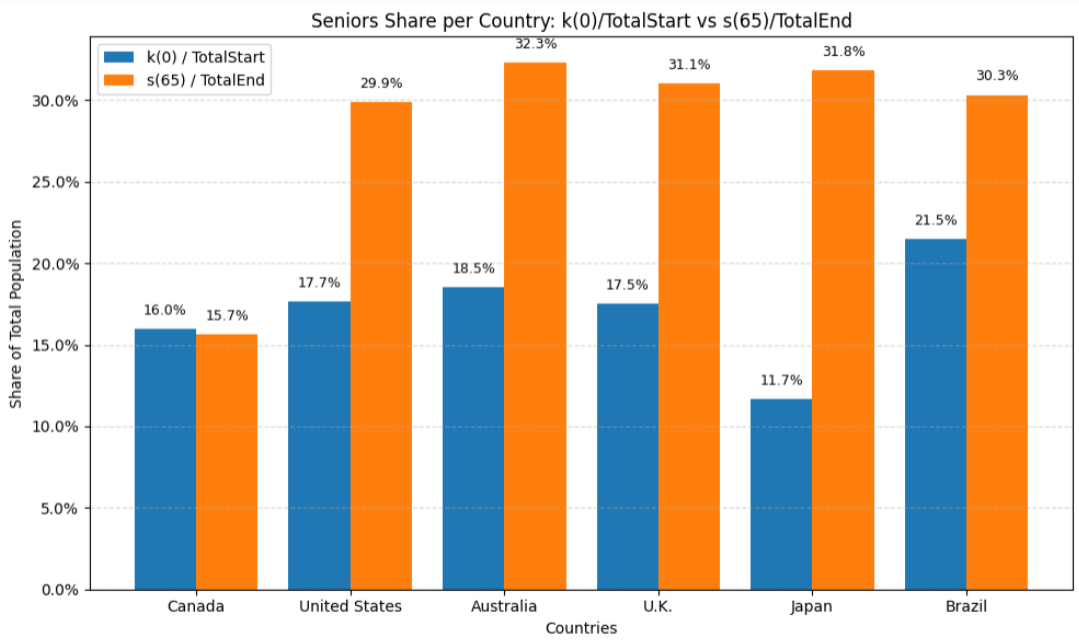


Figure 3.7: Ratio of Kids Compartment from the Start to End of Seniors Compartment for the Averaged Simulations.

Figure 3.7 shows the corresponding change in the *seniors* compartment (P_s) as a fraction of each country's total population. Here, the opposite pattern is observed: nearly every country exhibits a marked increase in the share of seniors over time, consistent with longer life expectancies and declining birth rates. The United States, Australia, the United Kingdom, Japan, and Brazil all reach senior shares exceeding 30% of the total population by the end of the simulation. Canada remains the only country with an approximately balanced age structure, with its senior share remaining stable near 16%, indicating that the modeled stressors—drought and disease—do not lead to excessive aging in its demographic composition. Japan again stands out for having the smallest youth fraction and one of the highest senior fractions, capturing its accelerated aging trajectory.

Taken together, the share-based and ratio-based analyses provide a coherent demographic picture: Canada maintains both absolute and relative growth in its younger population while keeping a stable proportion of older individuals, suggesting resilience against long-term population aging. By contrast, other countries exhibit simultaneous youth decline and senior expansion, consistent with real-world observations of aging societies. These results reinforce the model's capacity to capture structural demographic shifts and highlight how modest variations in fertility, mortality, and environmental vulnerability can lead to divergent long-term population outcomes.

Overall, the baseline simulations reveal a consistent pattern: Canada maintains long-term demographic stability and growth; Australia, the United Kingdom, and Brazil exhibit

moderate declines; and Japan and the United States show more substantial age-structural contraction. Because the iteration behavior and convergence trends were nearly identical across all countries, only Canada's results are shown in the main text for clarity. Equivalent plots for the remaining countries are provided in Appendix A

3.2 Projected Scenario Simulations

3.2.1 Projected Scenario 1: Time-Varying Birth and Death Rates

While the baseline simulations assumed constant birth and death rates, this first projected scenario introduces time-varying demographic parameters that reflect expected trends through the year 2090. The objective is to assess how gradual changes in fertility and mortality, consistent with long-term global projections, influence overall population structure and intergenerational balance.

Obtaining age-specific projections for birth and death rates proved challenging, as most international datasets only provide aggregated total rates extending into the future. This presents a difficulty, since each compartment in the model (P_k, P_y, P_a, P_s) is governed by its own birth and death parameters. To approximate realistic demographic shifts while preserving internal consistency, the following method was applied. First, the ratio of projected to current national birth and death rates was computed using data extending to the year 2090, the same duration as the simulation [92]. For example, if a country's total birth rate is projected to decrease by 5% over 65 years, each age-specific birth parameter was proportionally scaled by this same ratio, maintaining the relative structure between compartments while ensuring the aggregate change matched the observed projection. This approach allows each compartment to evolve in proportion to the national trend, preserving demographic relationships while incorporating realistic long-term changes in fertility and mortality.

Figure 3.8 shows the resulting average population dynamics for Canada under these time-varying rates, averaged across 20 stochastic runs. Compared to the baseline (constant-rate) simulation, the main difference lies in the more moderate growth of the child (P_k) and youth (P_y) compartments, and a slight reduction in the long-term adult (P_a) and senior (P_s) populations. While the overall shape of the trajectories remains similar, the magnitude of growth has decreased, suggesting that declining fertility and slightly increasing mortality temper the expansion observed in the baseline scenario. The adult population still exhibits a recovery after an initial decline, but the slope is noticeably shallower, reflecting slower replacement dynamics under lower fertility.

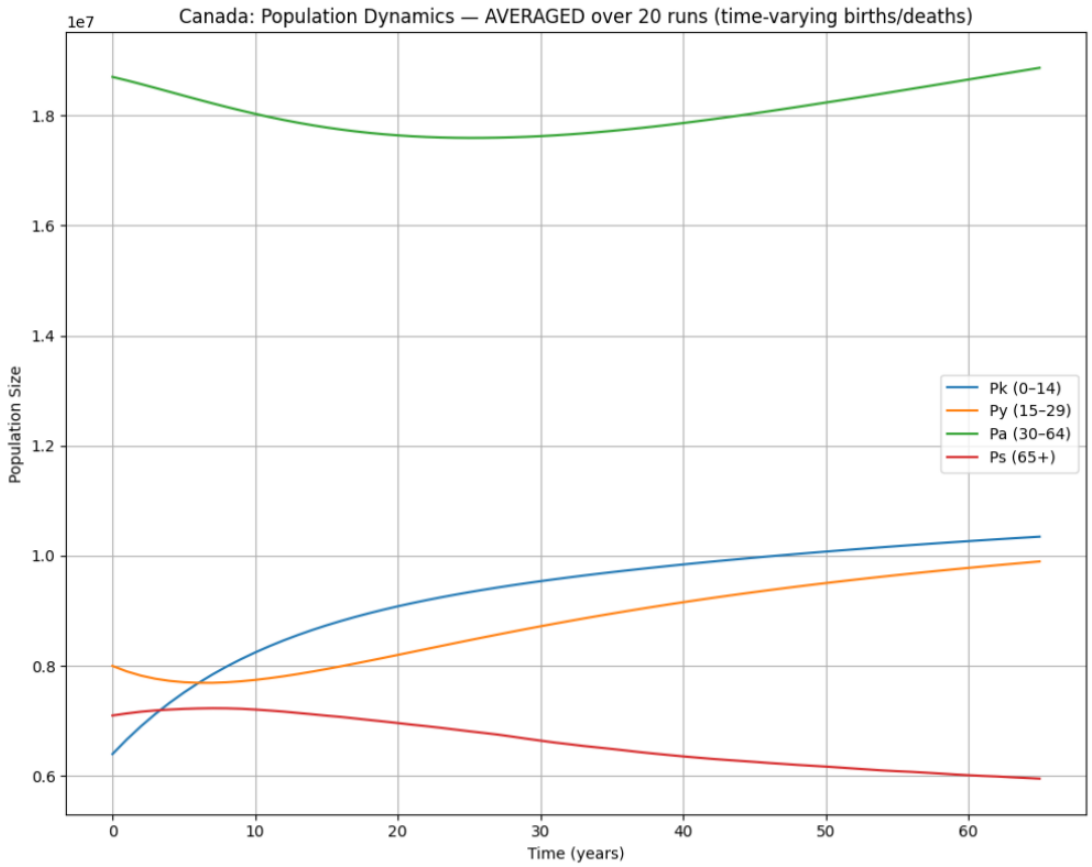


Figure 3.8: Canadian population dynamics averaged over 20 runs with time-varying birth and death rates.

The corresponding rates of change (Figure 3.9) confirm these observations. The derivatives dP_k/dt and dP_y/dt remain positive throughout the simulation, but the peak growth rates are lower and occur earlier than in the baseline case. The adult compartment (dP_a/dt) stabilizes more quickly and reaches equilibrium around year 40, while the senior compartment (dP_s/dt) trends slightly negative after year 30, indicating a gradual decline in elderly population size toward the end of the projection. This overall flattening of the growth curves demonstrates the demographic dampening effect of declining fertility combined with modestly rising mortality rates.

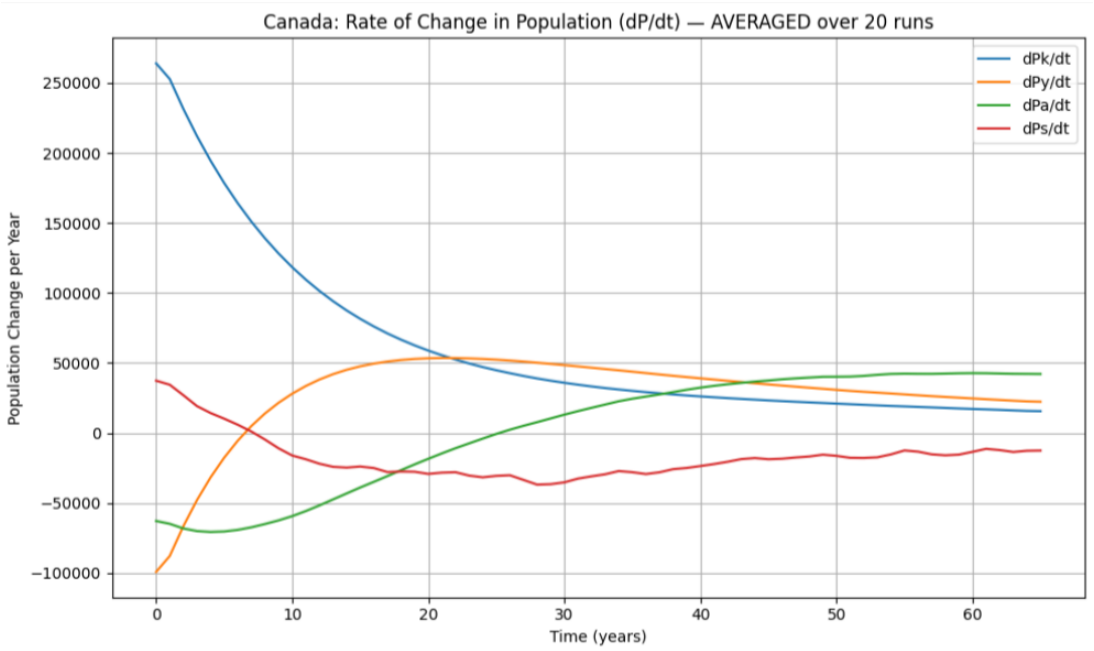


Figure 3.9: Rates of change in Canadian population compartments (dP_i/dt) under time-varying birth and death rates.

At the cross-country level, Figures 3.10 and 3.11 summarize the population shares of children and seniors under projected demographic trends. Compared with the baseline scenario, the differences are more pronounced. In Figure 3.10, the child share (P_k/Total) decreases across nearly all countries, though Canada still retains the highest proportion at approximately 23% by simulation end. For the United States, Australia, and the United Kingdom, child shares fall to around 11%, while Japan drops below 8%, reinforcing its continued trajectory of demographic contraction. Brazil remains intermediate, maintaining a child share near 12%. Overall, this figure indicates a global tendency toward youth decline under reduced fertility, though Canada remains an exception.

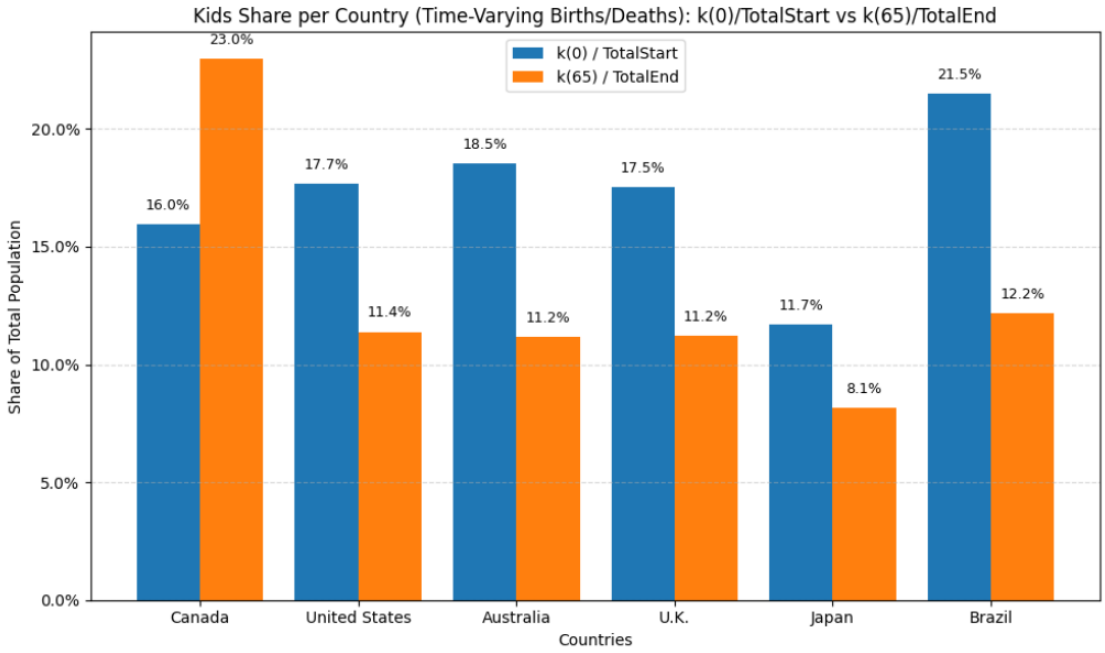


Figure 3.10: Share of children in total population at start and end of simulation under time-varying birth and death rates.

In contrast, Figure 3.11 highlights the corresponding increase in the share of seniors across most countries. The United States, Australia, the United Kingdom, and Japan all experience substantial growth in the elderly share, exceeding 27%–31% of total population by 2090. Canada's senior proportion, however, decreases slightly to about 13%, maintaining a balanced age structure despite the nationwide fertility decline. This result emphasizes that while most modeled nations exhibit strong demographic aging when exposed to time-varying rates, Canada's projected pattern remains relatively stable and sustainable. The contrast between Figures 3.10 and 3.11 thus illustrates how even small proportional changes in fertility and mortality can meaningfully alter the long-term population balance.

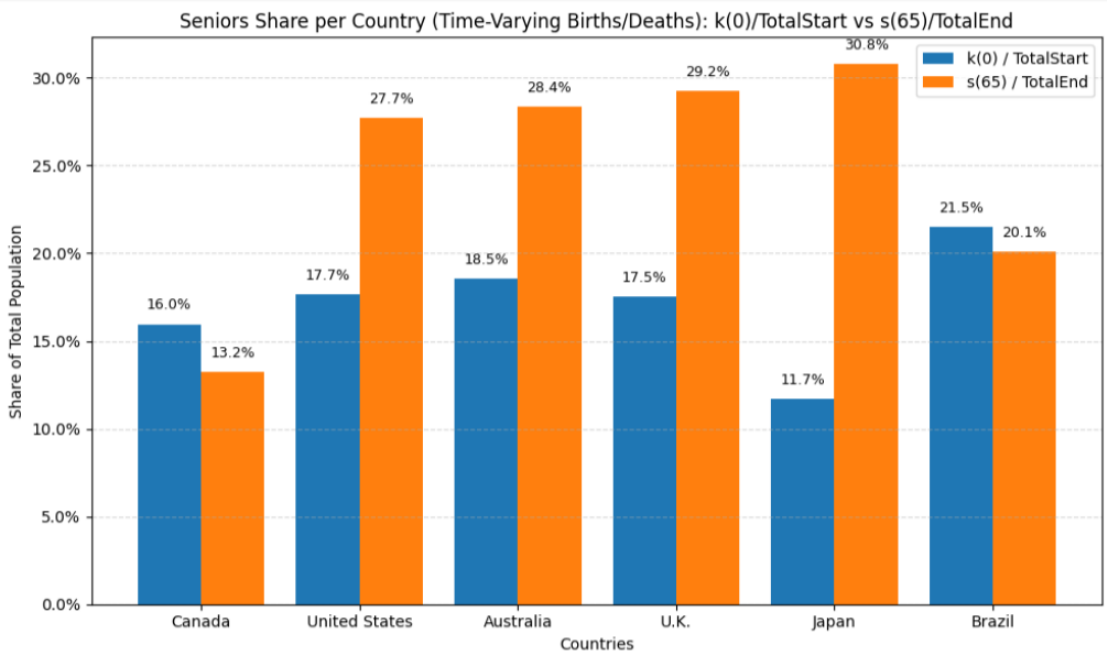


Figure 3.11: Share of seniors in total population at start and end of simulation under time-varying birth and death rates.

Overall, the inclusion of time-varying birth and death rates yields a more conservative growth outlook relative to the baseline scenario. While the general trajectories and inter-compartment relationships remain consistent, the system converges toward slower population growth and higher demographic stability. These results suggest that the model's long-term projections are sensitive to modest demographic shifts, and that sustained fertility decline—if realized—could substantially limit the pace of population recovery following environmental and epidemiological stress events.

3.2.2 Projected Scenario 2: Combined Disease Burden

This scenario evaluates the long-term demographic consequences of simultaneously increasing mortality risk from two major chronic health stressors: diabetes and asthma. Rather than increasing disease prevalence directly, this approach models a future in which the *severity* and *lethality* of each condition rises over time due to deteriorating environmental conditions and worsening global health determinants.

To capture these effects quantitatively, the disease intensity coefficients were systematically scaled relative to the baseline. For diabetes, represented by the food insecurity factor F, the parameter was increased by a factor of 2.3. This adjustment is consistent with projections from the Global Burden of Disease (GBD) study, which estimate that global deaths from type 2 diabetes will more than double between 2021 and 2050, driven primarily by

population aging, poor dietary transitions, and a growing burden of metabolic comorbidities [66]. For asthma, represented by the air-quality factor A, a more modest 10% increase was applied, aligning with forecasts linking climate change, wildfire activity, and rising particulate exposure ($PM_{2.5}$) to increasing asthma-related hospitalizations and mortality [52, 84]. Together, these two adjustments represent a compounding disease scenario in which chronic illness severity, rather than prevalence, steadily worsens through the simulation horizon.

Two alternative trajectories for disease progression were modeled: a **linear ramp** and an **exponential ramp**. In the linear case, disease severity increases at a constant rate over time, while in the exponential case, the growth rate of disease intensity accelerates toward the end of the century, mimicking nonlinear environmental feedbacks. Both approaches were designed to assess how the rate of health deterioration influences long-term demographic stability.

Figure 3.12 presents the Canadian population dynamics averaged over 20 stochastic runs for the **linear ramp** condition. Compared to the time-varying birth–death scenario, overall population growth remains positive but noticeably slower. The child (P_k) and youth (P_y) groups continue to increase but at a reduced pace, while the adult (P_a) and senior (P_s) compartments exhibit mild contraction. The shape of the curves indicates that the elevated disease burden dampens growth without inducing outright population collapse. This outcome suggests that moderate health deterioration primarily delays population recovery rather than fundamentally altering demographic equilibrium.

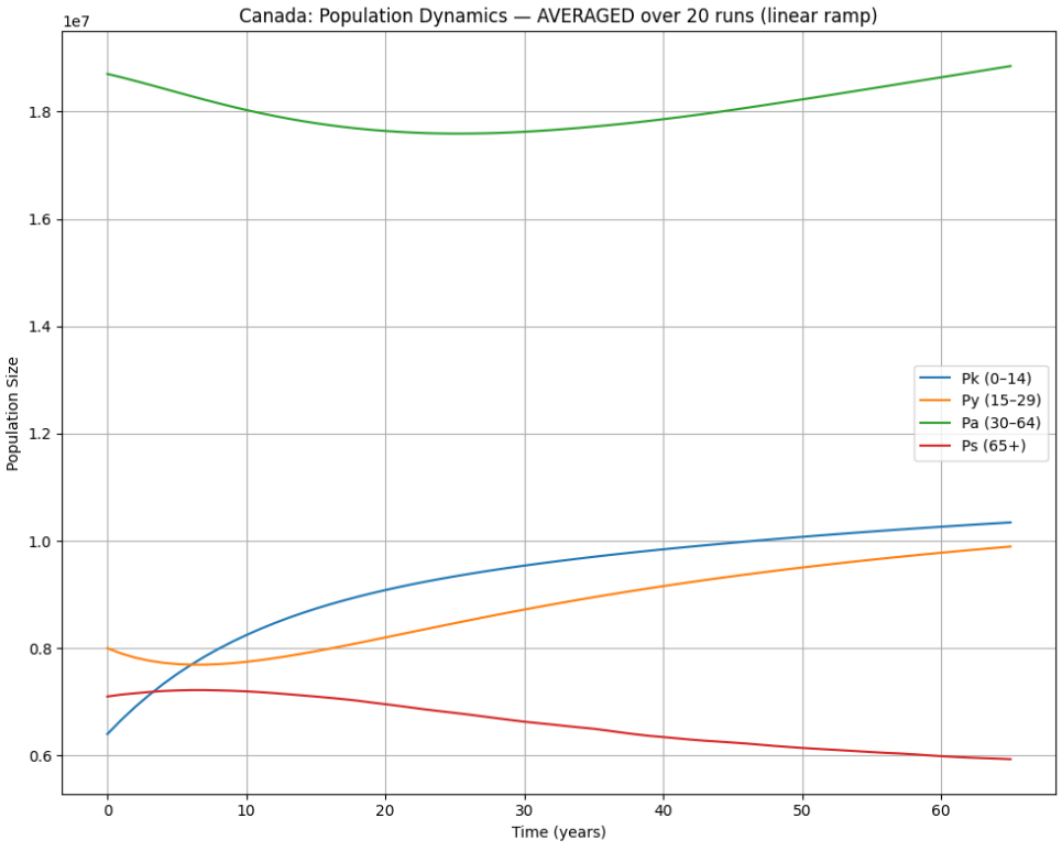


Figure 3.12: Canadian population dynamics under combined disease burden with linearly increasing chronic disease severity.

The corresponding rate-of-change curves (Figure 3.13) reinforce this interpretation. The derivatives dP_k/dt and dP_y/dt remain positive, though slightly attenuated relative to the time-varying baseline. Meanwhile, dP_a/dt and dP_s/dt settle closer to zero, indicating slower adult regeneration and a gradual decline in the senior compartment. This reflects the anticipated demographic compression that accompanies higher chronic disease lethality—particularly in older age groups.

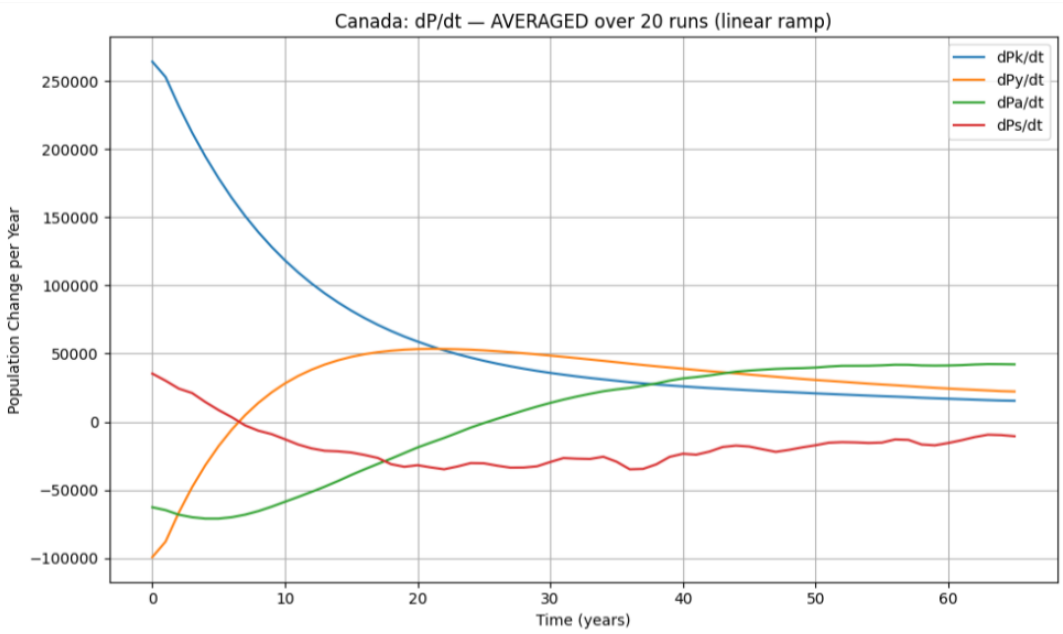


Figure 3.13: Rates of change (dP_i/dt) for Canadian compartments under the linear disease ramp scenario.

Figures 3.14 and 3.15 display the corresponding results for the **exponential disease ramp**. Under this assumption, disease impacts intensify more rapidly over time, yielding a more pronounced reduction in both the adult and senior compartments. The overall shapes of the trajectories remain qualitatively similar to the linear case, but the magnitude of suppression is larger in the latter half of the simulation. By year 65, the senior population (P_s) is approximately 5% lower than under the linear scenario, while child and youth growth remain nearly identical. This suggests that nonlinear acceleration in chronic disease lethality mainly affects the oldest cohorts, amplifying mortality in later decades.

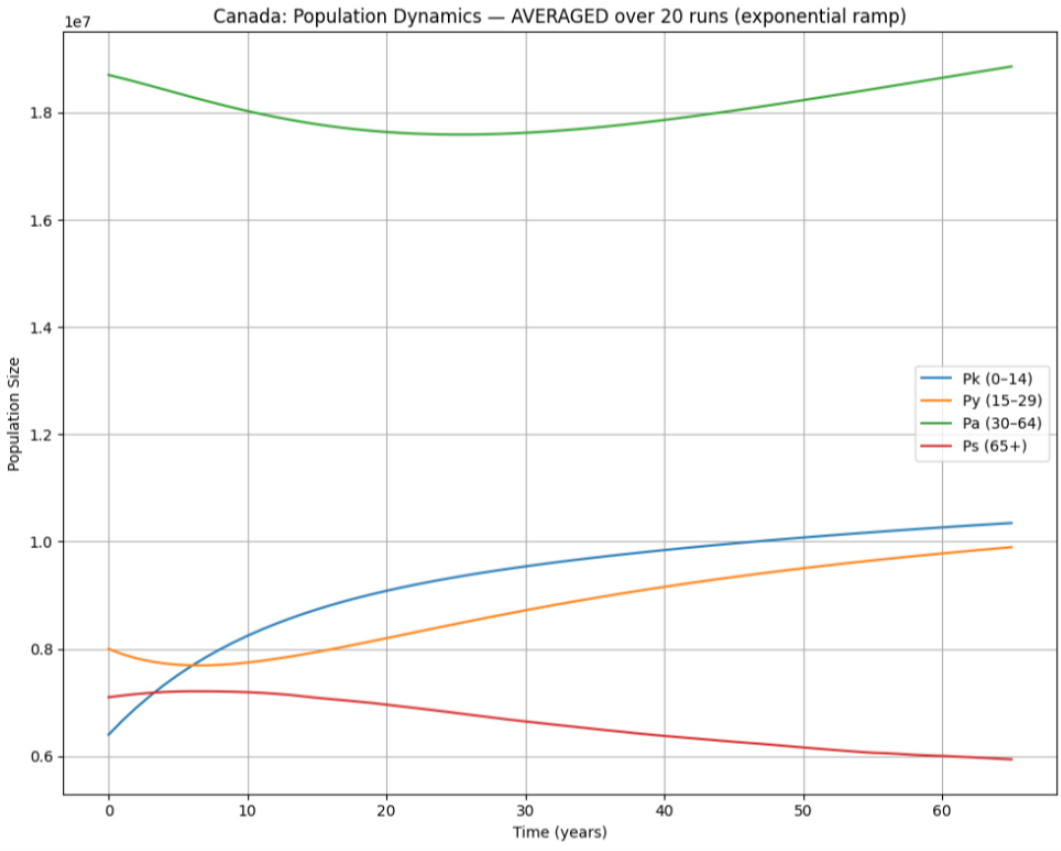


Figure 3.14: Canadian population dynamics under combined disease burden with exponentially increasing chronic disease severity.

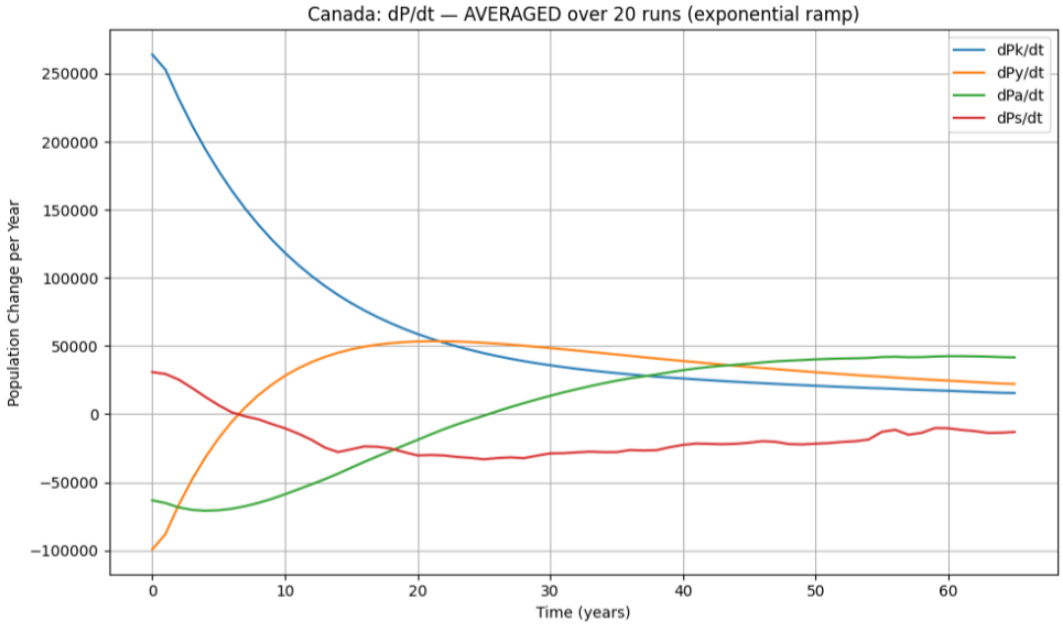


Figure 3.15: Rates of change (dP_i/dt) for Canadian compartments under the exponential disease ramp scenario.

Figures 3.16 and 3.17 provide cross-country comparisons of the age-structural outcomes under both disease ramp assumptions. In Figure 3.16, the share of children relative to the total population decreases slightly in all countries compared to the time-varying birth–death case, though the difference between the linear and exponential ramps is marginal. Canada once again maintains the highest child share (approximately 23%), while other countries cluster around 11%, indicating that even elevated chronic disease burden does not significantly alter youth growth patterns. This resilience reflects the model's assumption that chronic disease mortality primarily affects adults and seniors rather than children.

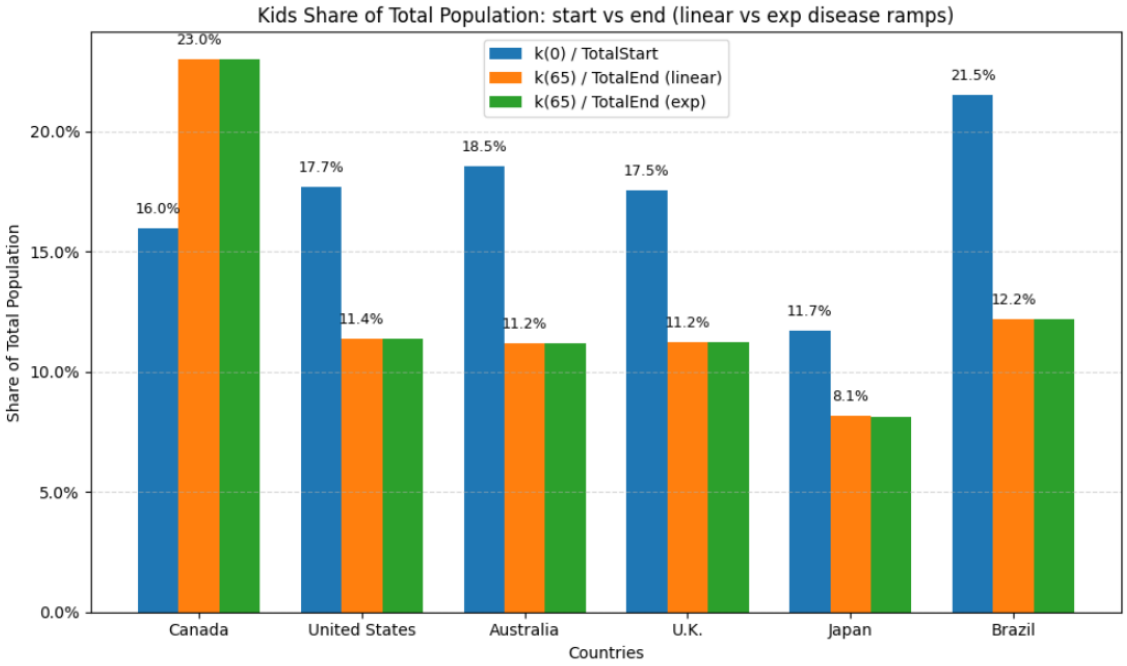


Figure 3.16: Cross-country comparison of child population share under linear and exponential disease severity ramps.

In contrast, Figure 3.17 shows that the *senior share* is more strongly impacted by the nonlinear (exponential) disease progression. For most countries, the senior fraction remains near 28–31% under the linear ramp but increases only marginally (or even stabilizes) under the exponential version, indicating that accelerated chronic disease lethality curbs elderly population growth late in the projection. Canada again stands out for maintaining a stable senior proportion of roughly 13–14%, suggesting robust demographic balance even under compounding health stress. The close alignment between the linear and exponential outcomes for most countries also implies that chronic disease severity, though impactful, does not independently destabilize long-term population equilibrium unless coupled with additional shocks or fertility decline.

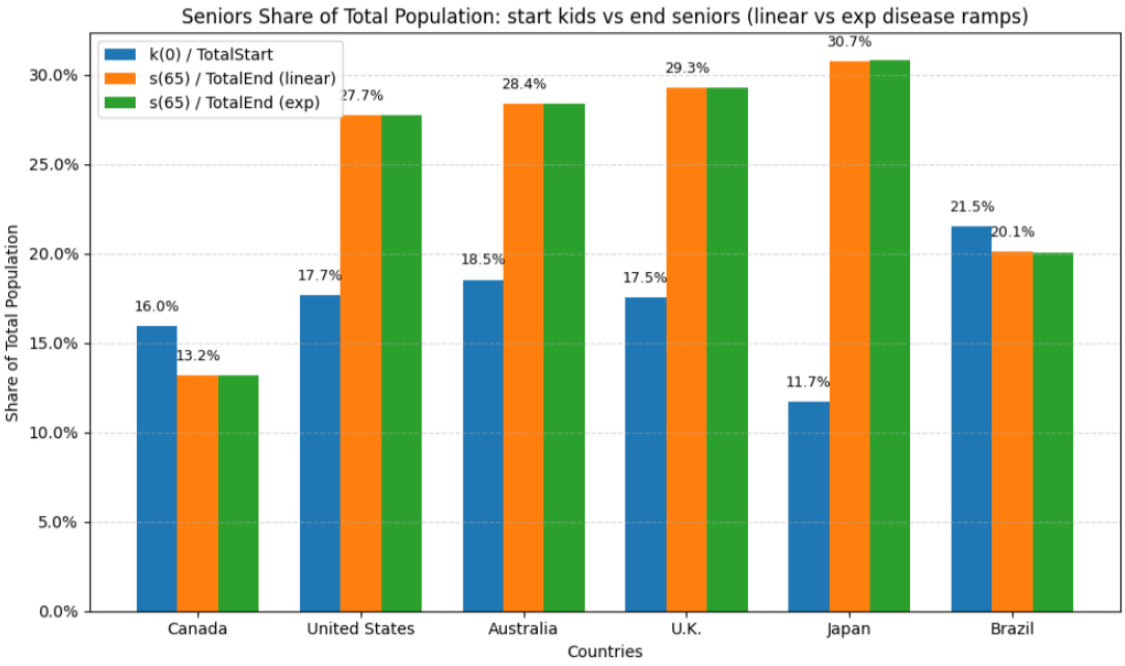


Figure 3.17: Cross-country comparison of senior population share under linear and exponential disease severity ramps.

Overall, Scenario 2 demonstrates that gradual increases in chronic disease mortality—whether linear or exponential—produce a measurable but not catastrophic demographic impact. Compared to the baseline and time-varying birth–death simulations, population growth slows, and aging intensifies modestly, but the system remains dynamically stable. This suggests that even under significant health stress, resilience mechanisms such as fertility-driven replenishment and moderate mortality adaptation can sustain long-term population balance.

3.2.3 Projected Scenario 3: Increased Environmental Shock Events

This final scenario examines the demographic effects of heightened environmental volatility under progressively intensifying chronic disease, and with time-varying birth and death rates. It therefore integrates all prior modifications—non-static demographic parameters and compounding chronic disease severity—while additionally increasing the annual probabilities of drought and pandemic events. This composite scenario represents the most realistic future projection considered in this study, capturing both gradual and acute stressors simultaneously.

Numerous climate studies, including the most recent IPCC assessments, indicate that extreme drought events are expected to become between 1.5 and 3 times more frequent

across populated regions by mid-century under moderate-to-high warming trajectories [41, 64]. To reflect this, each country's baseline drought probability was doubled. In parallel, global pandemic risk assessments estimate an annual pandemic probability of approximately 2.5–3.3%, implying a 22–28% likelihood of another large-scale outbreak within a single decade [14]. Drivers include increasing human–animal contact, ecosystem disruption, and climate-mediated zoonotic spillover [58, 59]. Accordingly, the baseline annual probability of a disease event was also doubled (from 5% to 10%). Together, these changes produce a dynamic environment where both chronic and acute shocks evolve over time, testing the long-term resilience of each population.

Figure 3.18 shows the Canadian population trajectories averaged over 20 stochastic runs under this combined setting of time-varying rates, compounding disease burden, and ramping shock probabilities. Despite the increase in event frequency and mortality pressure, population trajectories remain stable, with only modest reductions compared to Scenario 2. The child (P_k) and youth (P_y) groups continue to rise steadily, while the adult (P_a) and senior (P_s) compartments flatten earlier and at lower levels. Relative to the baseline simulation—which assumed constant parameters and fixed shock probabilities—the total population growth rate is notably reduced, but demographic collapse does not occur. This illustrates strong structural resilience even under high environmental volatility.

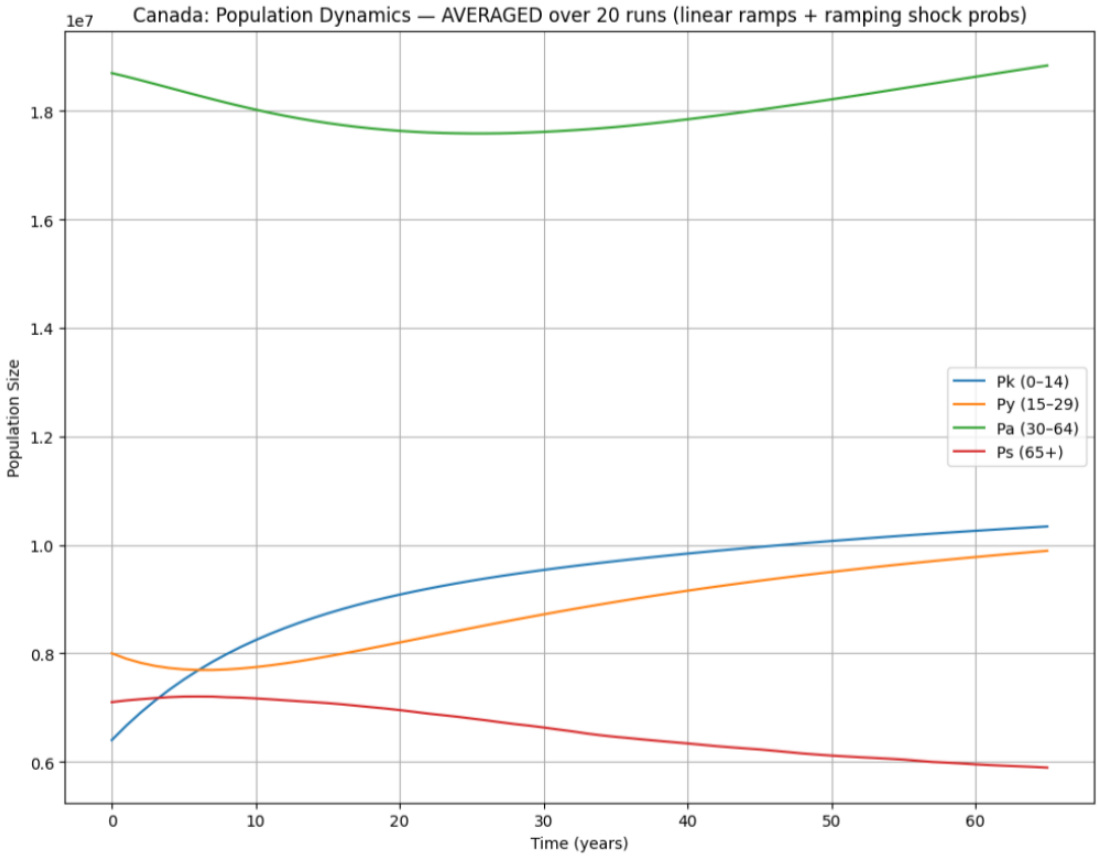


Figure 3.18: Canadian population dynamics averaged over 20 runs with linearly increasing disease severity and ramping drought/pandemic probabilities.

The derivative curves in Figure 3.19 further demonstrate this attenuation effect. Peak values of dP_k/dt and dP_y/dt are lower than in any previous scenario, reflecting the cumulative influence of chronic and acute stress. The adult rate (dP_a/dt) transitions to near-zero earlier, suggesting slowed economic-age growth, while the senior derivative (dP_s/dt) remains slightly negative, indicating a continuing but gradual contraction in the elderly population. Nonetheless, these rates stabilize toward the end of the simulation, implying a new quasi-equilibrium rather than progressive decline. Compared to the baseline, the rate dynamics exhibit a clear downward shift but retain the same qualitative structure, confirming that the system's stability is preserved even under intensified conditions.

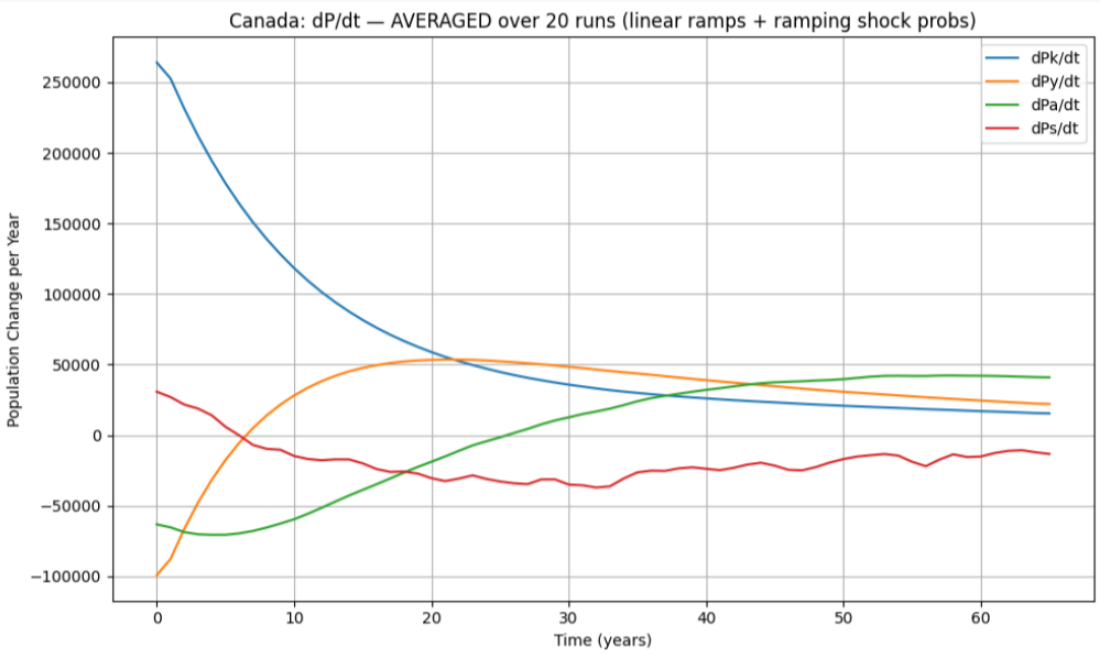


Figure 3.19: Rates of change (dP_i/dt) for Canadian compartments under combined chronic and stochastic stressors.

The increased frequency of discrete shocks is illustrated in Figure 3.20, which maps all drought and pandemic events across 20 runs. Unlike the baseline stochastic distribution, where events were relatively uniform, this scenario displays clear clustering and higher density of markers in later simulation years—reflecting the intentional ramping of event probabilities. Drought (blue) and disease (red) events both become substantially more frequent after year 30, mimicking the compounding risks projected for the latter half of the century. This visualization confirms that the probabilistic ramping mechanism functioned as intended and that the model successfully integrates increasing environmental volatility over time.

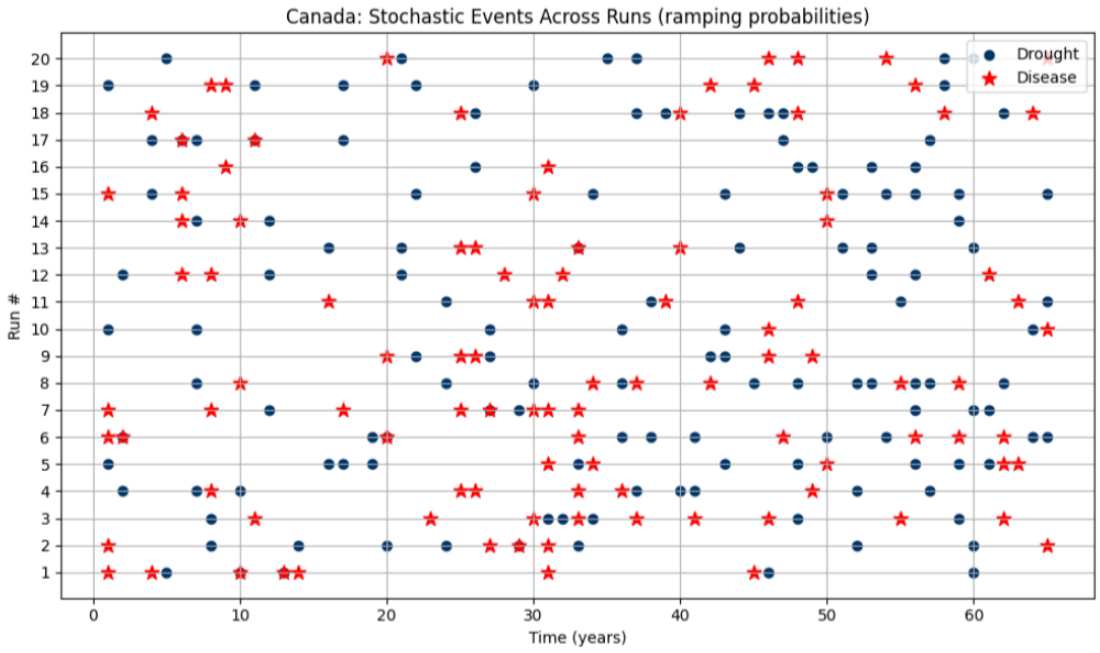


Figure 3.20: Distribution of stochastic drought and disease events across runs under ramping shock probabilities.

At the cross-country level, Figures 3.21 and 3.22 show the proportional changes in children and seniors relative to total population under the final combined scenario. Child shares (P_k/Total) in Figure 3.21 remain highest in Canada at approximately 23%, while all other countries show slight further declines compared to Scenario 2. The difference between this combined-shock scenario and the time-varying demographic case is modest, suggesting that even substantial increases in drought and pandemic frequency exert limited influence on long-term fertility dynamics. Japan remains the most youth-constrained population, while Brazil retains a moderate level of child representation. Overall, these results indicate that acute shocks primarily redistribute short-term demographic losses across age groups rather than permanently suppressing birth-driven growth.

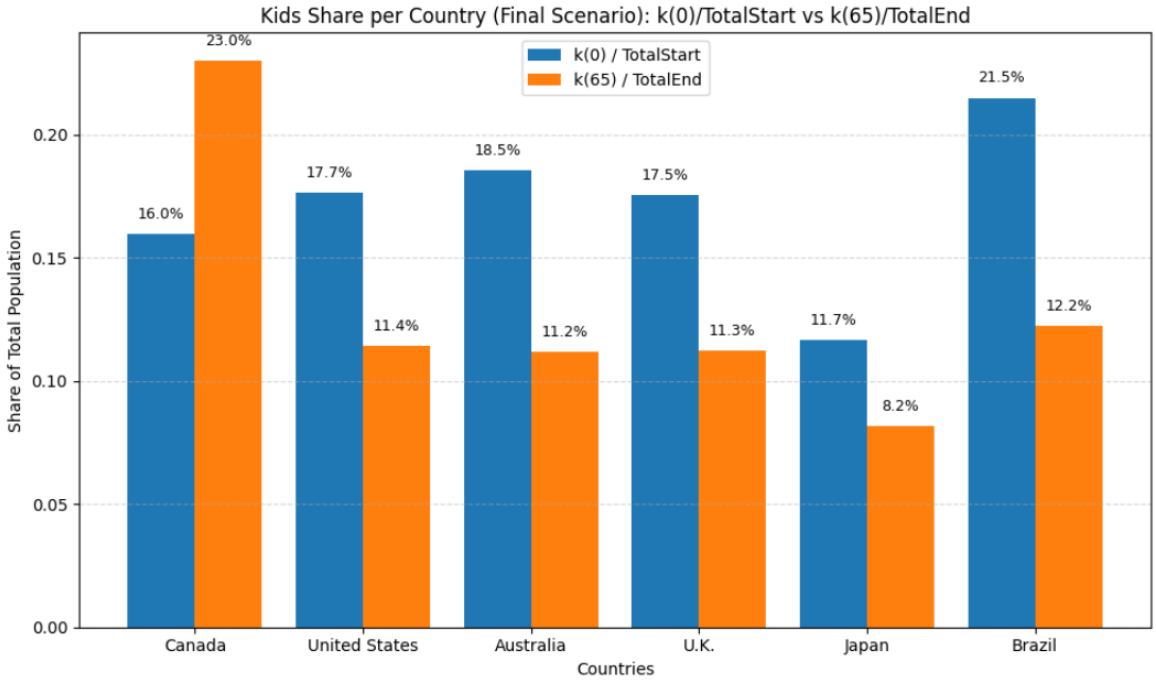


Figure 3.21: Cross-country comparison of child population shares under final combined-shock scenario.

The senior share results (Figure 3.22) complete this picture. For most countries, senior proportions remain elevated but slightly lower than in the pure time-varying scenario, suggesting that increased environmental shocks disproportionately affect older cohorts. Canada again shows exceptional balance, with its senior fraction falling to roughly 13%, nearly identical to the prior disease scenario, reinforcing its relative demographic resilience. In contrast, Japan and Australia maintain the highest senior shares (around 29–30%), continuing their long-term aging trend even under repeated shocks. These outcomes suggest that while external volatility exacerbates mortality among the elderly, it does not fundamentally reverse aging trajectories already driven by fertility decline.

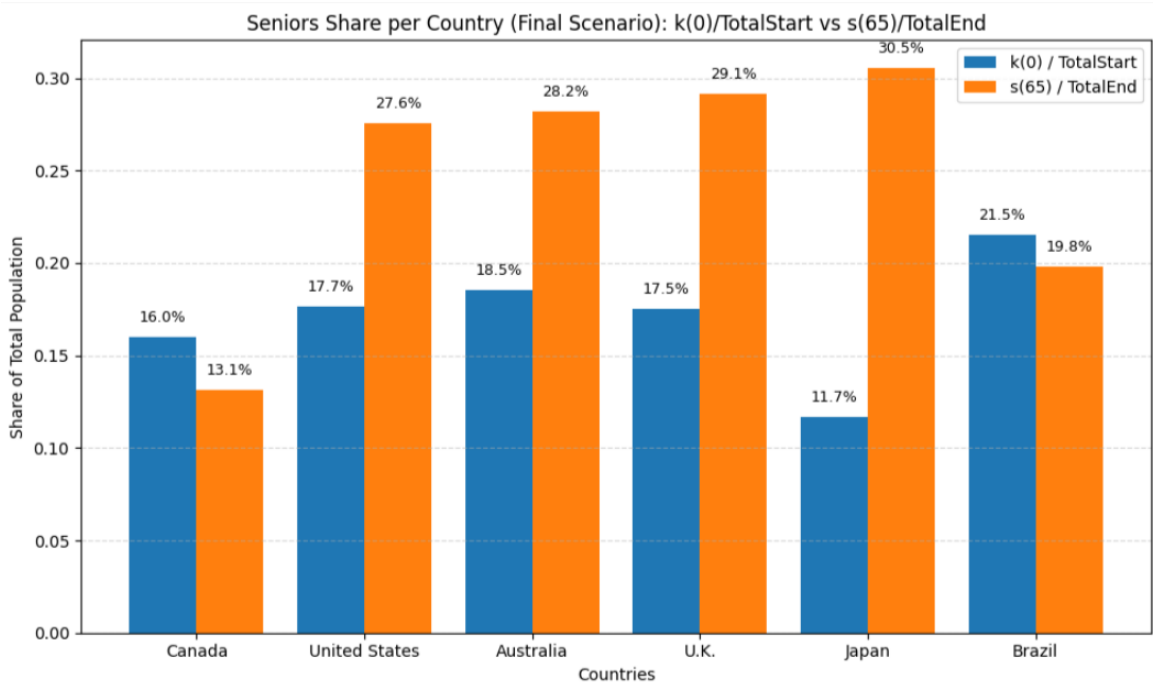


Figure 3.22: Cross-country comparison of senior population shares under final combined-shock scenario.

In summary, this final scenario demonstrates that compounding environmental and health stressors slow—but do not destabilize—population growth across all countries modeled. Relative to the static baseline scenario, total population levels are reduced, and equilibrium is reached earlier, but all systems converge to stable trajectories. This indicates that the inclusion of time-varying demographic rates, progressive disease lethality, and increasing shock probabilities yields a realistic yet demographically resilient future projection. Canada remains the most balanced and robust population under all tested conditions, while countries such as Japan, the United States, and Australia exhibit heightened sensitivity to aging and environmental stress. Together, these findings underscore the importance of fertility maintenance and adaptive capacity in preserving long-term demographic stability under a future characterized by both chronic and acute global health challenges.

Chapter 4

Conclusion and Future Work

4.1 Summary of Key Findings

This thesis developed and implemented a deterministic, age-structured compartmental model designed to simulate long-term population dynamics under the combined influence of demographic trends, environmental stress, and health-related shocks. The model tracked four population compartments—children, youth, adults, and seniors—across a 65-year horizon, integrating both continuous demographic processes and discrete stochastic events such as droughts and disease outbreaks. By linking demographic structure with environmental and health stressors, the model provides a dynamic framework for evaluating population resilience and age-specific vulnerability in an evolving global context.

The baseline scenario assumed static birth and death rates with fixed probabilities of environmental shocks. Under these conditions, all six modeled countries exhibited stable long-term behavior, though the degree of growth or contraction varied. Canada demonstrated sustained population expansion and intergenerational balance, while countries such as Japan and the United Kingdom displayed gradual demographic contraction consistent with aging populations and lower fertility. Cross-country comparisons revealed that baseline birth–death parameters and sensitivity coefficients largely determined each nation's trajectory, confirming that inherent demographic structure plays a dominant role in shaping resilience under static conditions.

In **Scenario 1**, time-varying birth and death rates were introduced to represent projected demographic transitions through the year 2090. This adjustment moderated growth across all countries, yielding smoother trajectories and lower long-term equilibrium populations relative to the baseline. While fertility decline and modest increases in mortality slowed overall growth, population stability was maintained, demonstrating the model's capacity to capture gradual demographic adaptation. Canada continued to show positive growth with balanced age structure, while other countries exhibited varying degrees of aging, with Japan and the United States showing the strongest senior dominance.

Scenario 2 incorporated a compounding chronic disease burden, increasing the lethality

of diabetes and asthma to reflect projected global health deterioration. Two progression profiles—linear and exponential—were simulated. In both cases, population growth slowed further compared to Scenario 1, primarily due to rising mortality among adults and seniors. The exponential ramp produced slightly stronger late-century declines, though overall demographic stability persisted. This finding underscores that chronic health deterioration primarily compresses growth rather than destabilizing population structure.

Scenario 3, the final and most realistic simulation, combined all prior modifications while introducing ramping probabilities for environmental shocks such as droughts and pandemics. Despite compounding stressors and intensified volatility, population trajectories remained remarkably stable across all countries. Growth rates decreased and equilibrium was reached earlier, but collapse or runaway decline did not occur. Canada again maintained the most balanced age structure, while aging trends in Japan, Australia, and the United Kingdom persisted. Collectively, these results demonstrate that the model captures both the dampening effects of demographic slowdown and the adaptive resilience of population systems even under extreme combined pressures.

Across all scenarios, seniors consistently emerged as the most vulnerable compartment, showing the largest sensitivity to both chronic and acute stressors. Children and youth demonstrated robust growth and recovery capacity, reflecting lower baseline mortality and sustained fertility influence. These findings highlight the importance of intergenerational balance: when fertility and survival rates among younger cohorts remain sufficient, populations can withstand substantial environmental and health volatility without losing structural integrity.

4.2 Implications for Population Health Modeling

The modeling framework presented in this thesis represents an important step toward integrating demographic, environmental, and health dynamics within a unified quantitative system. Unlike traditional compartmental models focused on infectious disease transmission, this model captures long-range interactions between non-communicable diseases, fertility trends, and stochastic environmental shocks. This approach aligns with emerging priorities in population health research, which emphasize systemic, climate-informed forecasting over short-term epidemiological modeling.

One of the key contributions of this work is demonstrating how external stressors—both chronic and acute—interact with age-specific vulnerability to determine long-term population outcomes. By parameterizing hazard coefficients, fertility patterns, and event probabilities separately for each country, the model captures both structural and contextual differences in population resilience. This makes it a flexible tool for comparative demographic forecasting, allowing policymakers and researchers to evaluate how shifts in chronic disease severity, air quality, or environmental event frequency may differentially affect nations with distinct demographic profiles.

The results also emphasize the importance of maintaining fertility stability as a buffer against long-term population decline. Across scenarios, populations with stable or slowly declining fertility (such as Canada and Brazil) demonstrated consistent recovery and age balance, even under heightened stress. In contrast, countries already experiencing demographic contraction, such as Japan, remained vulnerable despite moderate environmental and health improvements. This suggests that future population resilience depends not only on mitigating health and environmental risks but also on preserving reproductive and generational continuity.

From a methodological perspective, the model's hybrid structure—combining deterministic ODE dynamics with stochastic shock events—illustrates how long-term projections can incorporate both predictable trends and probabilistic disturbances. This hybridization bridges traditional demographic modeling and systems-based public health approaches, enabling scenario testing under multiple layers of uncertainty. As such, the framework contributes to the growing field of integrated health–environment modeling and supports the design of forward-looking policy strategies in a rapidly changing global context.

4.3 Limitations

While the model yields meaningful insights into long-term population dynamics, several limitations warrant consideration.

First, although stochastic drought and disease events are included, the underlying compartmental structure is deterministic. This design facilitates interpretability and reproducibility but may underestimate the full range of real-world demographic variability—particularly for smaller or more volatile populations where random deviations have outsized effects. Future work could extend this framework using stochastic differential equations or agent-based models to incorporate random variation directly into compartmental flows.

Second, the chronic disease parameters F and A were derived primarily from Canadian epidemiological scaling factors and then generalized across all countries due to limited international mortality-by-disease data. While this allowed consistent comparative modeling, it introduces potential bias in countries with substantially different healthcare systems, pollution exposure, or baseline health profiles. More granular data on regional disease burden would enable improved calibration and cross-country accuracy.

Third, migration, socioeconomic inequality, and healthcare access were not included in the model. Each of these factors can meaningfully alter population resilience, especially under compounding crises. Similarly, economic feedbacks—such as reduced fertility following recessions or migration shifts due to environmental displacement—remain outside the current model scope but represent promising areas for future integration.

Lastly, while the model incorporates time-varying birth and death rates, these are based on linear interpolations of aggregate national projections rather than endogenous demo-

graphic responses. In reality, fertility and mortality trends interact dynamically with policy, technology, and behavior. Incorporating such feedback mechanisms could improve the predictive realism of future versions of this framework.

4.4 Future Work

Several extensions can build upon this foundation to enhance realism, complexity, and interdisciplinary relevance.

First, integrating more detailed health and environmental data would allow for improved parameterization of chronic disease and hazard coefficients. Future versions could incorporate time-resolved estimates of diabetes, asthma, cardiovascular disease, and climate-exposure mortality from global datasets such as GBD and IHME, allowing for refined calibration of F and A across multiple health dimensions.

Second, extending the model's stochastic structure to include demographic variability—rather than only discrete shock events—would better capture uncertainty in long-term projections. Hybrid stochastic–deterministic or agent-based approaches could simulate how local fluctuations aggregate into macro-level population trends.

Third, incorporating spatial and migratory dynamics would enable modeling of interregional dependency, displacement, and adaptive migration flows under shared global stressors. Such an extension could assess how drought-induced migration or regional fertility differentials affect continental demographic balance over time.

Finally, linking this demographic model with complementary systems such as climate, food security, or economic simulations could support holistic scenario planning. These integrations would facilitate evaluation of cross-sectoral trade-offs—for example, how agricultural drought impacts not only population health but also economic productivity and food availability. Such interdisciplinary coupling would extend the model's utility from theoretical exploration to applied policy forecasting.

Overall, this thesis establishes a flexible, extensible framework for modeling long-term population dynamics under environmental and health stress. By systematically layering static, dynamic, and stochastic processes, it bridges traditional demographic theory with modern systems modeling, providing a foundation for future quantitative research on population resilience in the Anthropocene.

Appendix A

Additional Figures and Graphs

A.1 Scenario 1: Baseline Simulation with Constant Parameters

This section presents the full time–series trajectories for each country under the baseline simulation, where all parameters remain fixed throughout the 65-year horizon. The figures illustrate the evolution of the four population compartments—children, youth, adults, and seniors—in the absence of any external shocks or demographic transitions. These baseline outputs provide the reference against which all subsequent scenarios are compared.

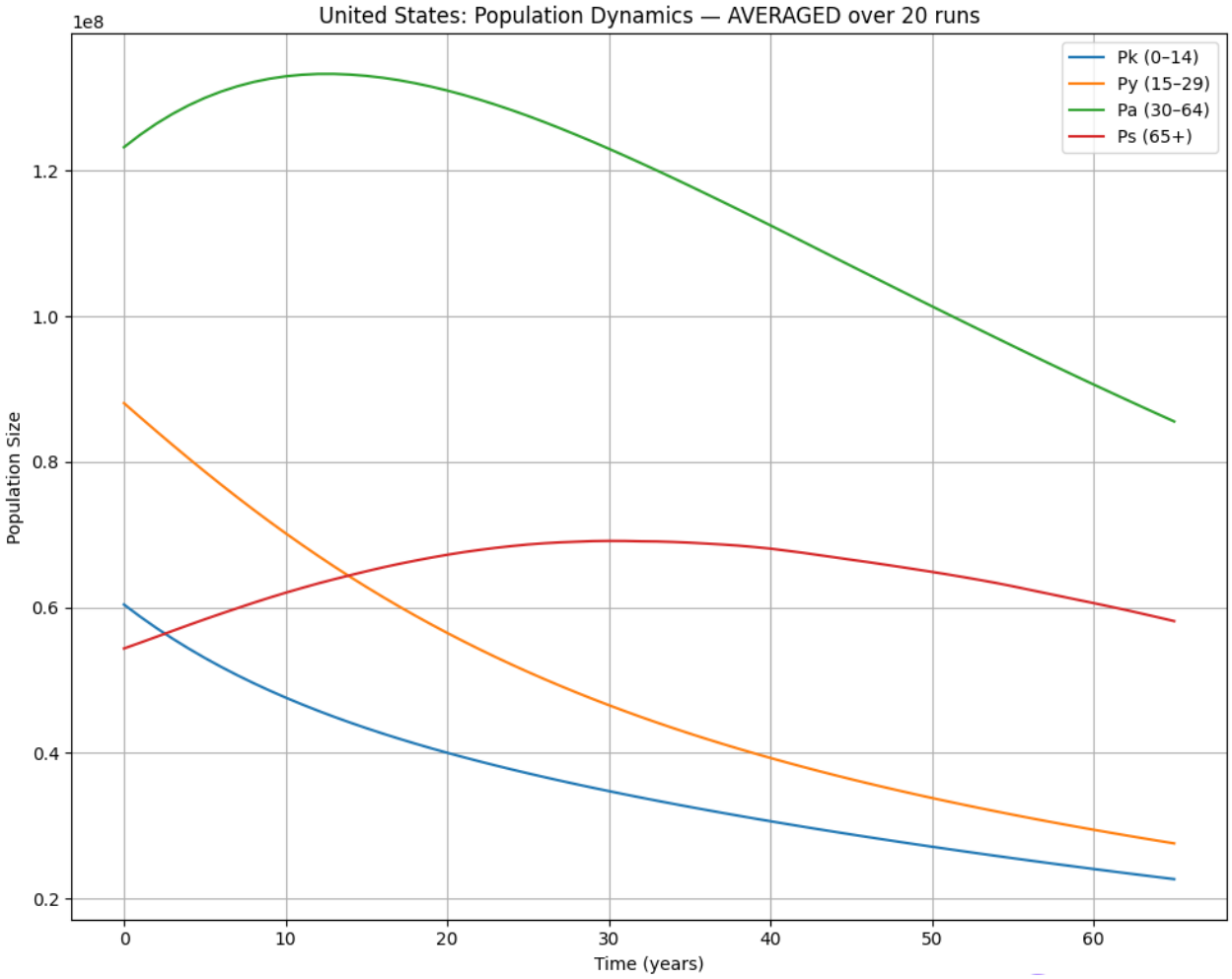


Figure A.1: Baseline population trajectories for the United States (all four compartments).

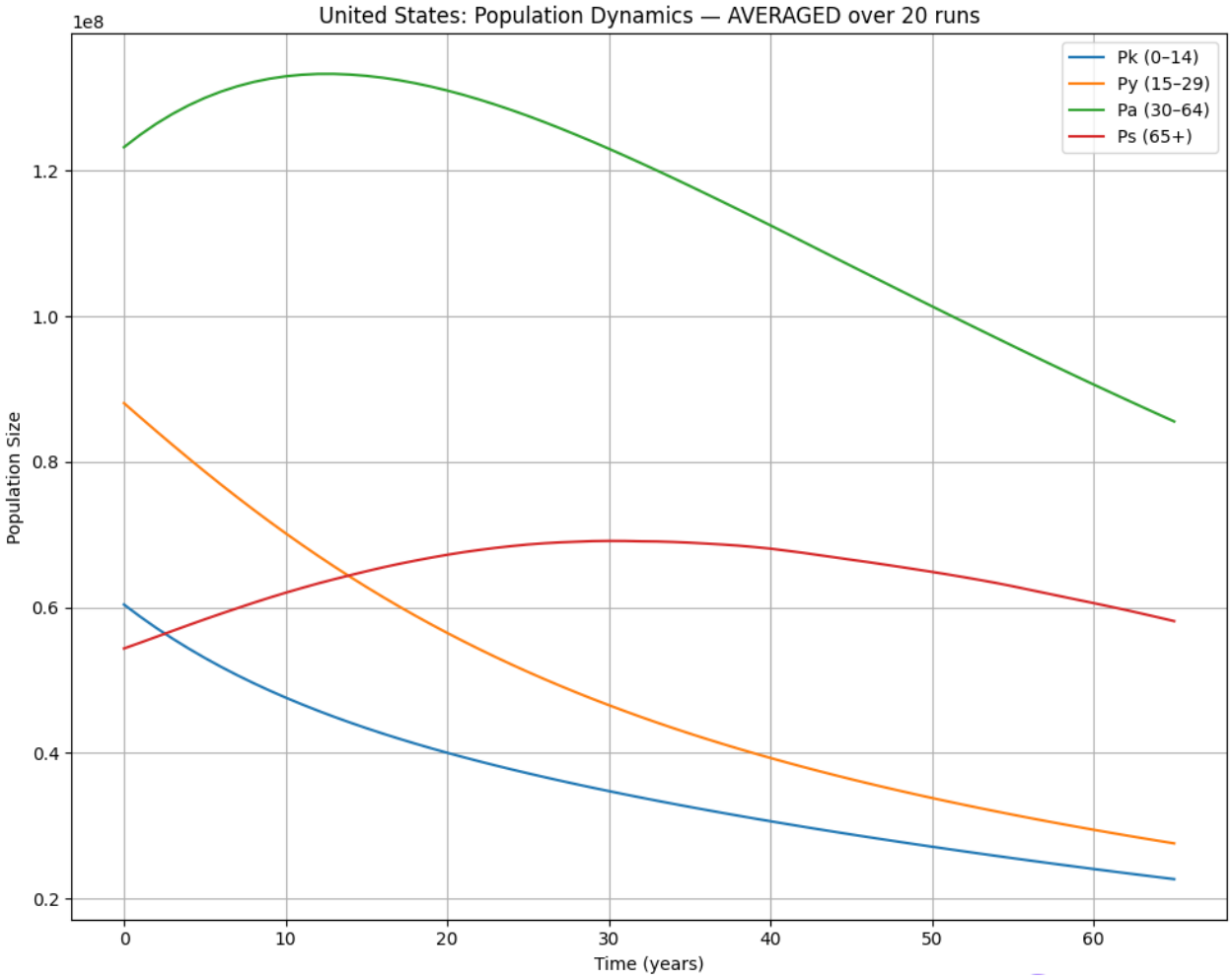


Figure A.2: Baseline population trajectories for Australia (all four compartments).

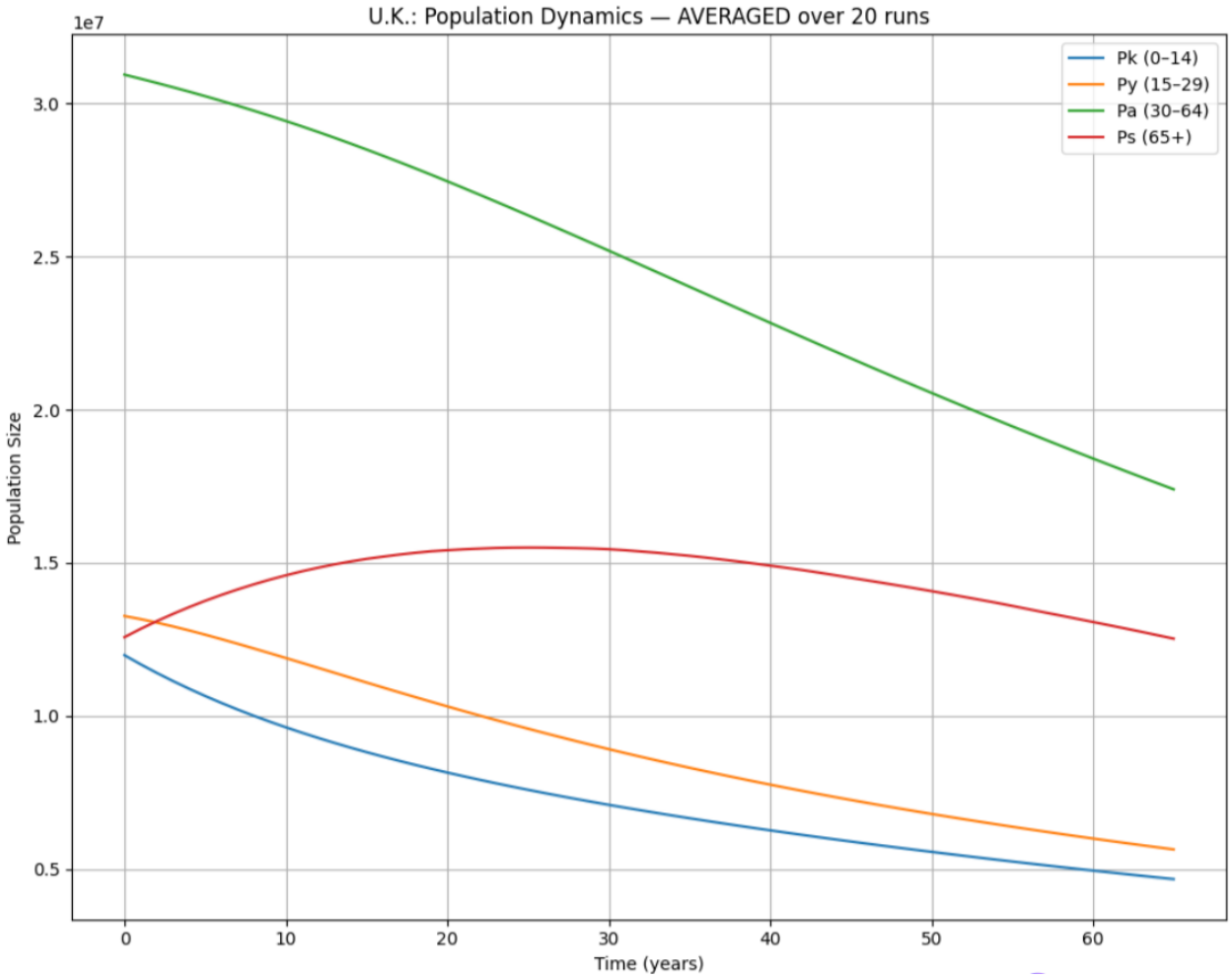


Figure A.3: Baseline population trajectories for the United Kingdom (all four compartments).

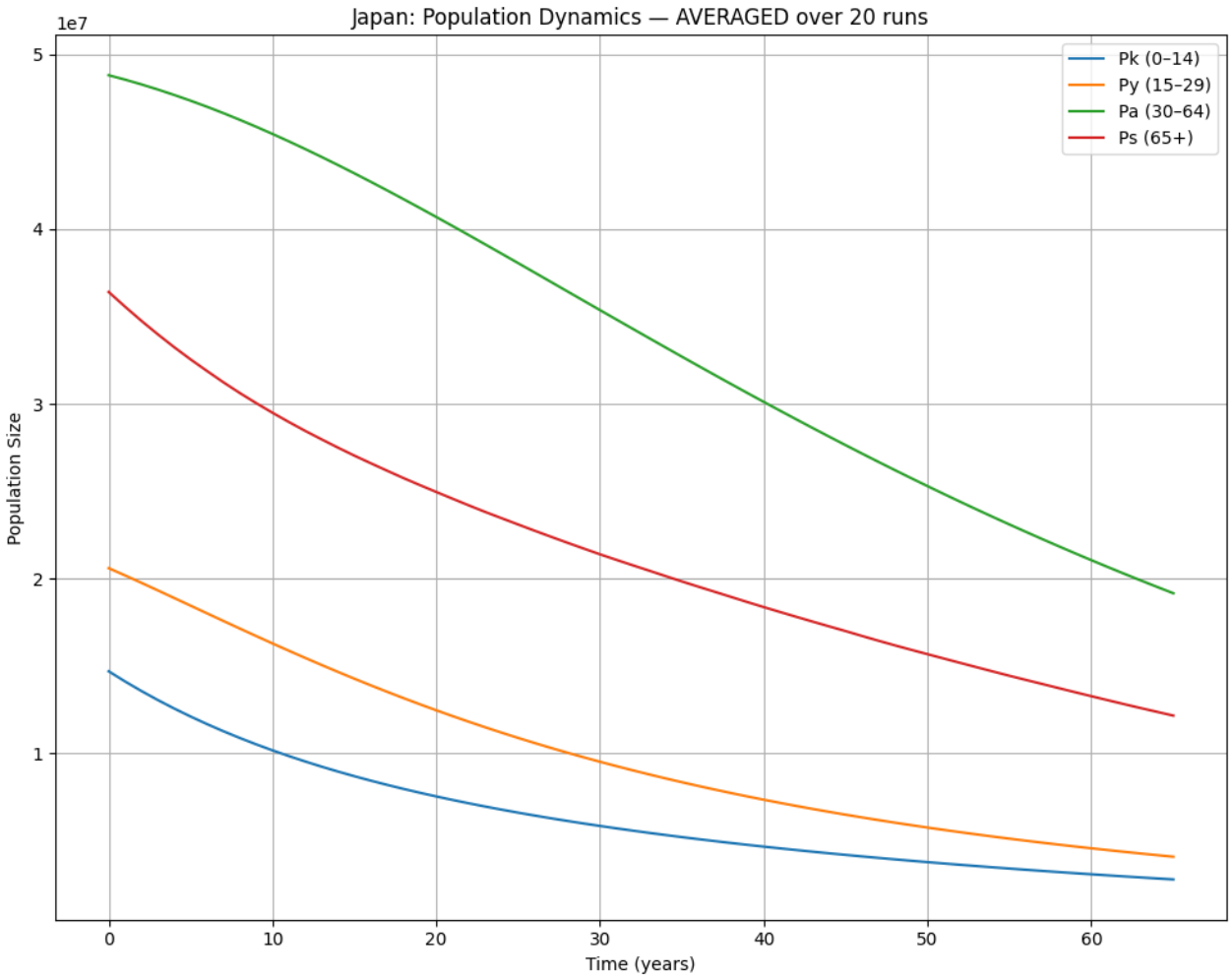


Figure A.4: Baseline population trajectories for Japan (all four compartments).

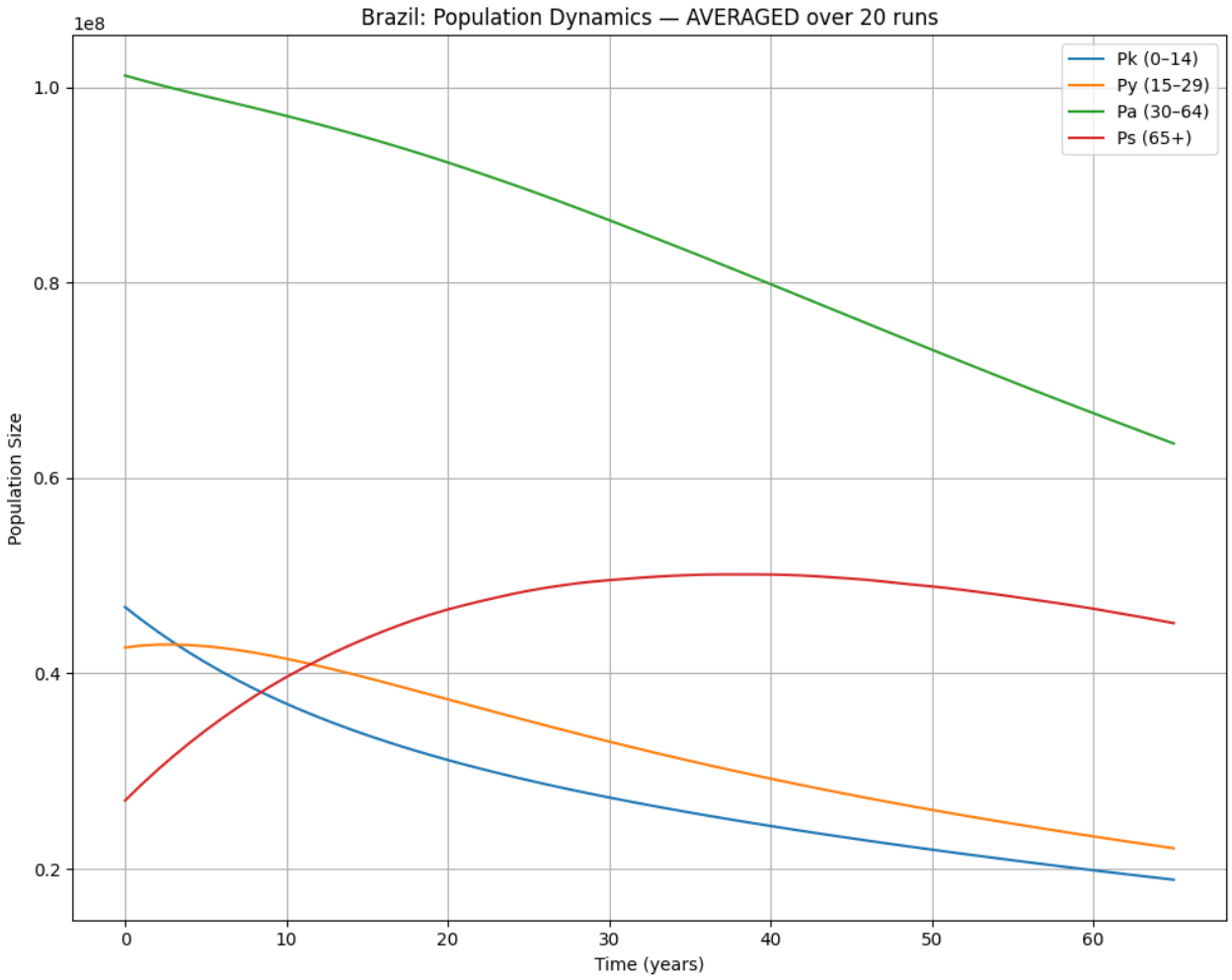


Figure A.5: Baseline population trajectories for Brazil (all four compartments).

A.2 Scenario 2: Simulation with Time-Varying Birth and Death Rates

The following figures show the long-term population dynamics when country-specific birth and death rates are allowed to vary over time, reflecting projected demographic trends through 2090. This scenario captures gradual structural change due to fertility decline and population aging, independent of health or environmental shocks, and serves to isolate the effect of endogenous demographic transition on overall population resilience.

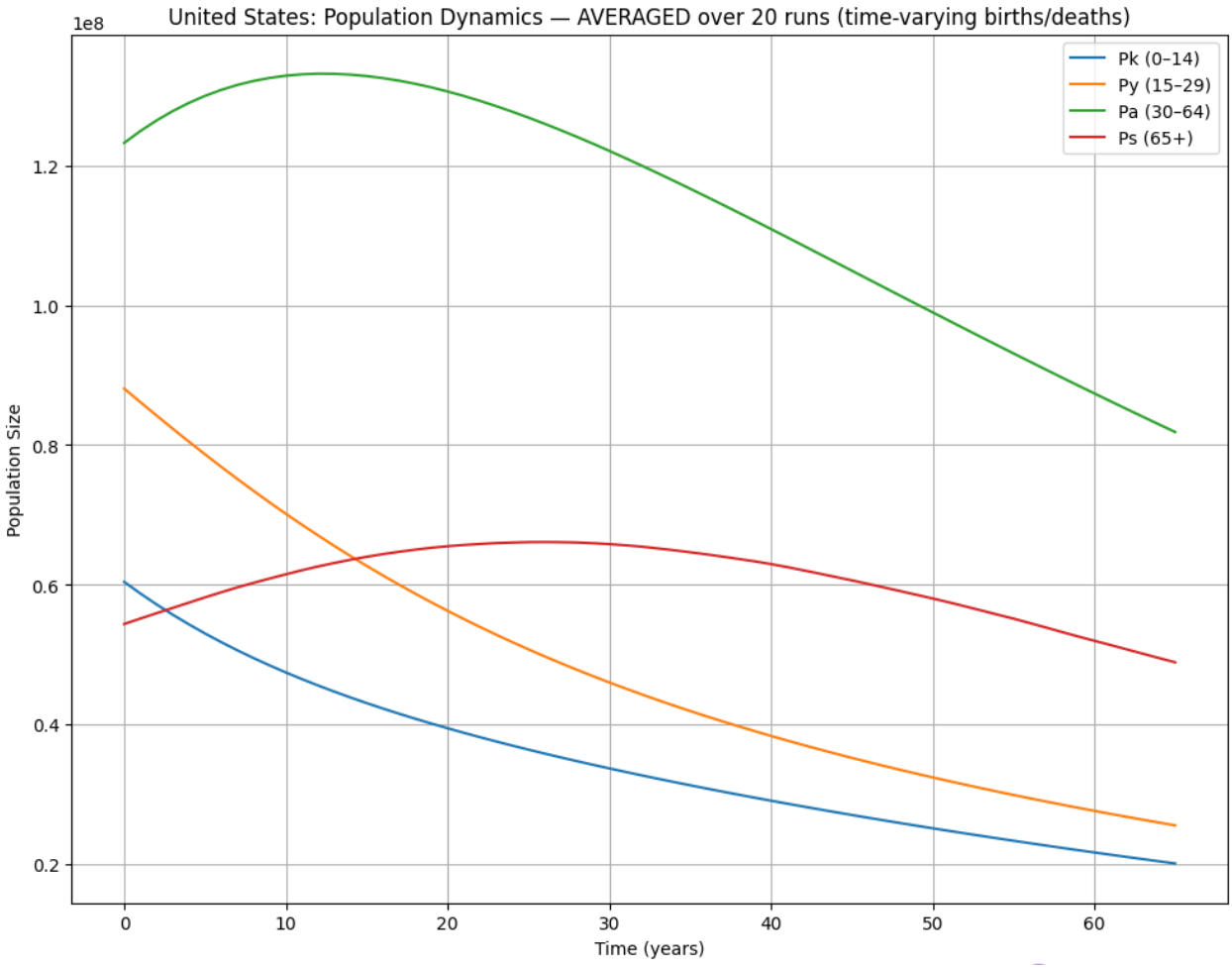


Figure A.6: Population projections under time-varying birth and death rates for the United States.

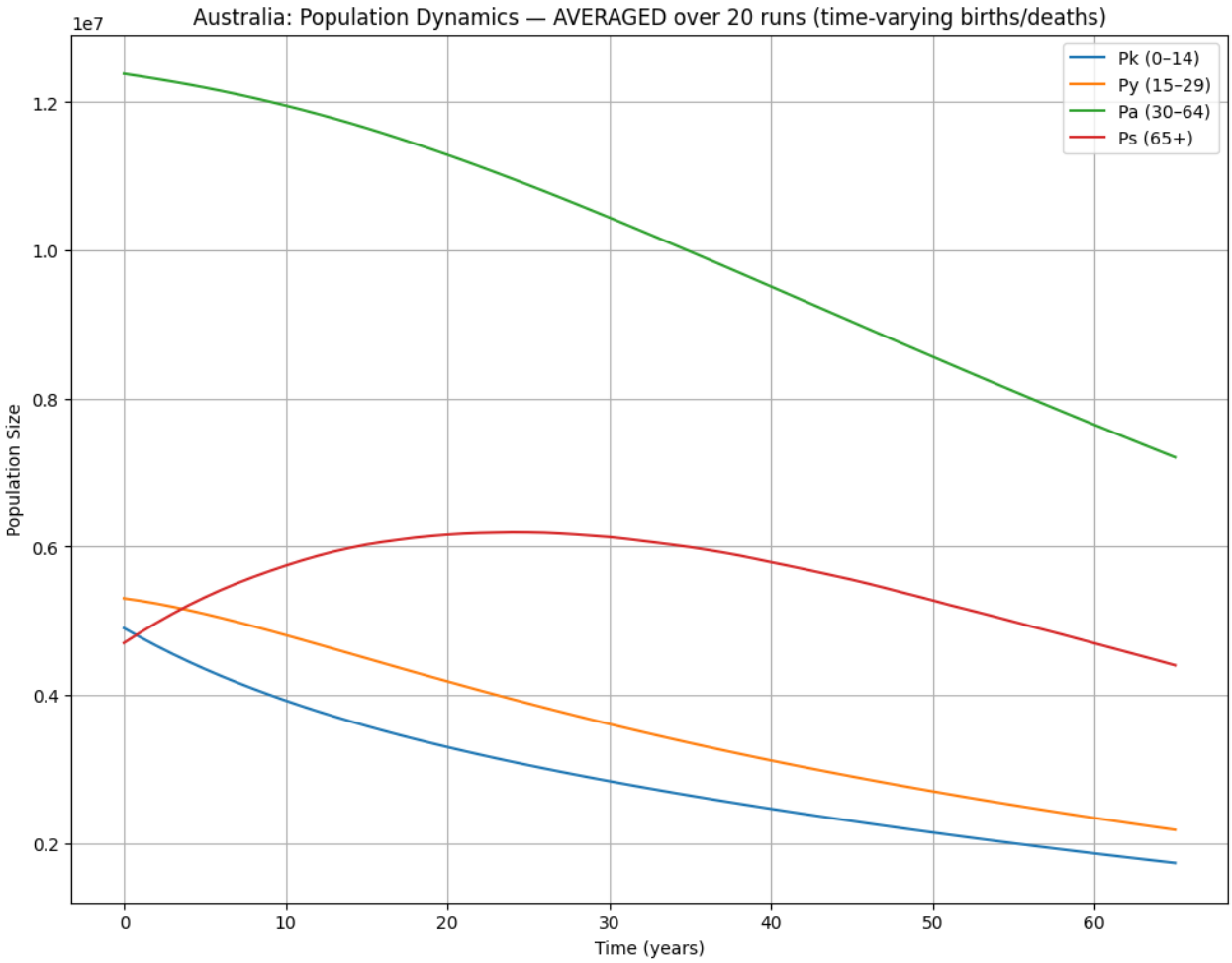


Figure A.7: Population projections under time-varying birth and death rates for Australia.

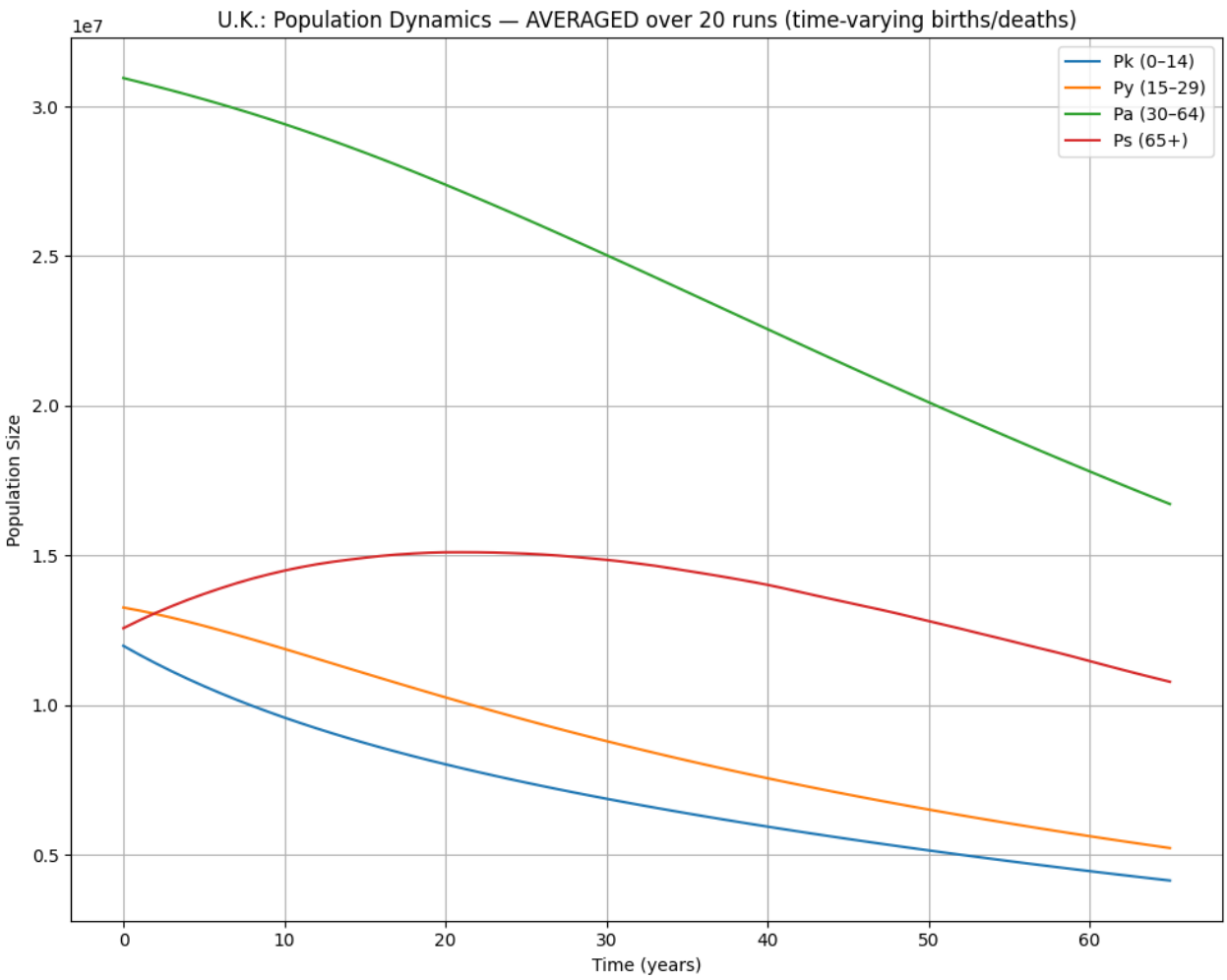


Figure A.8: Population projections under time-varying birth and death rates for the United Kingdom.

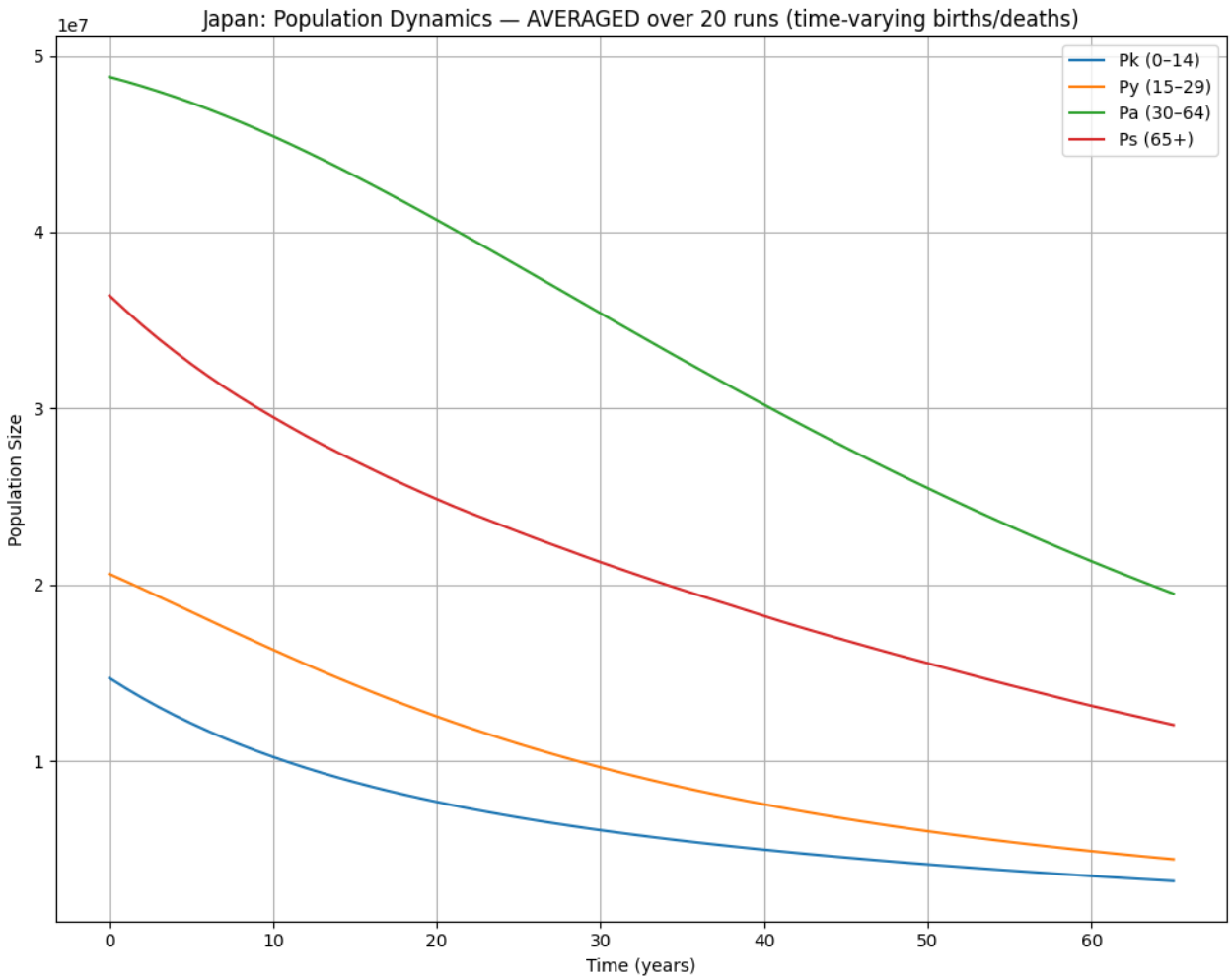


Figure A.9: Population projections under time-varying birth and death rates for Japan.

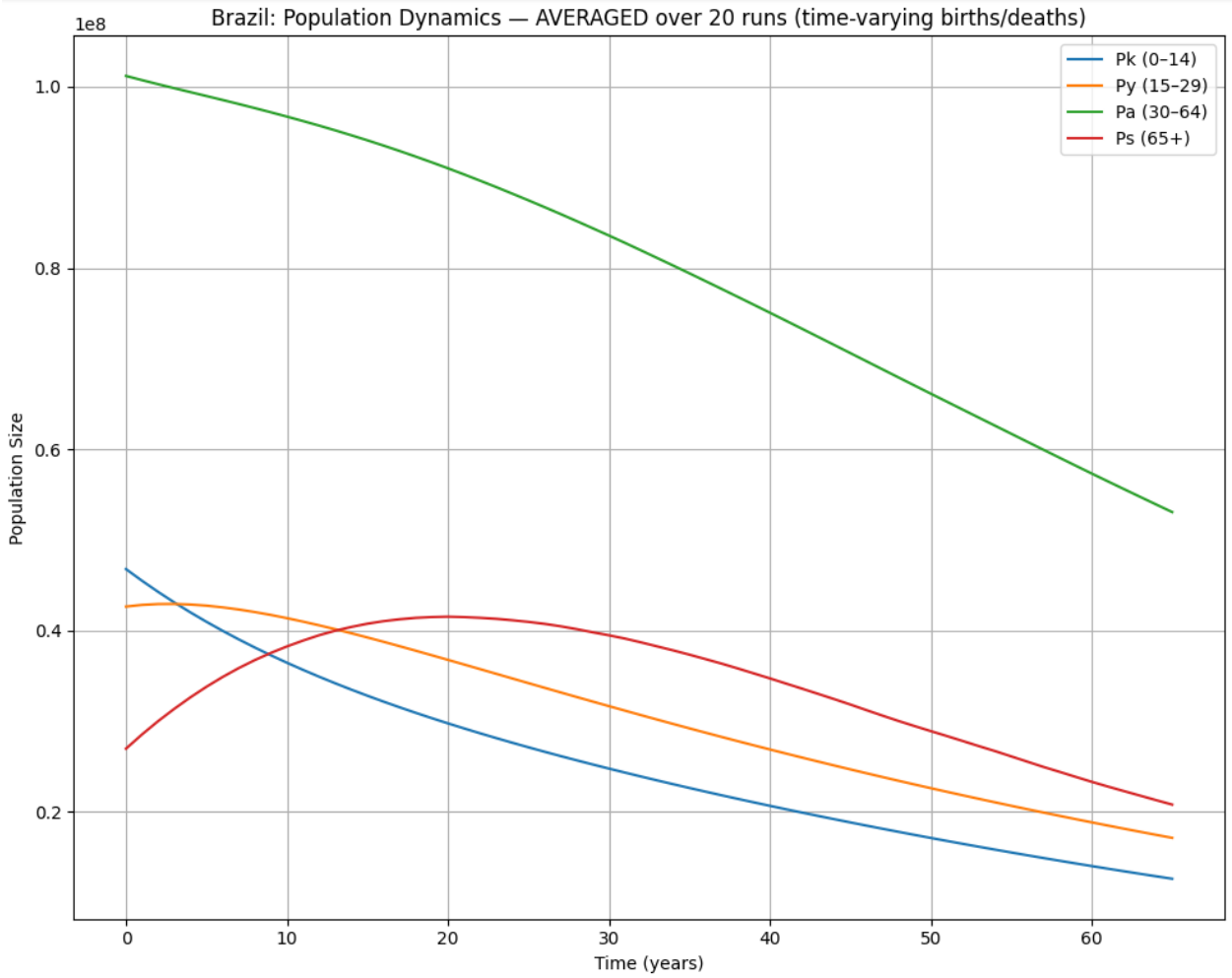


Figure A.10: Population projections under time-varying birth and death rates for Brazil.

A.3 Scenario 3: Elevated Chronic Disease Mortality

This section illustrates population trajectories under increased mortality from chronic conditions such as diabetes and asthma. All other parameters remain constant to isolate the health-related effects of worsened chronic disease burden. Comparisons with earlier scenarios reveal how sustained health deterioration modifies long-term population structure and age-specific vulnerability.

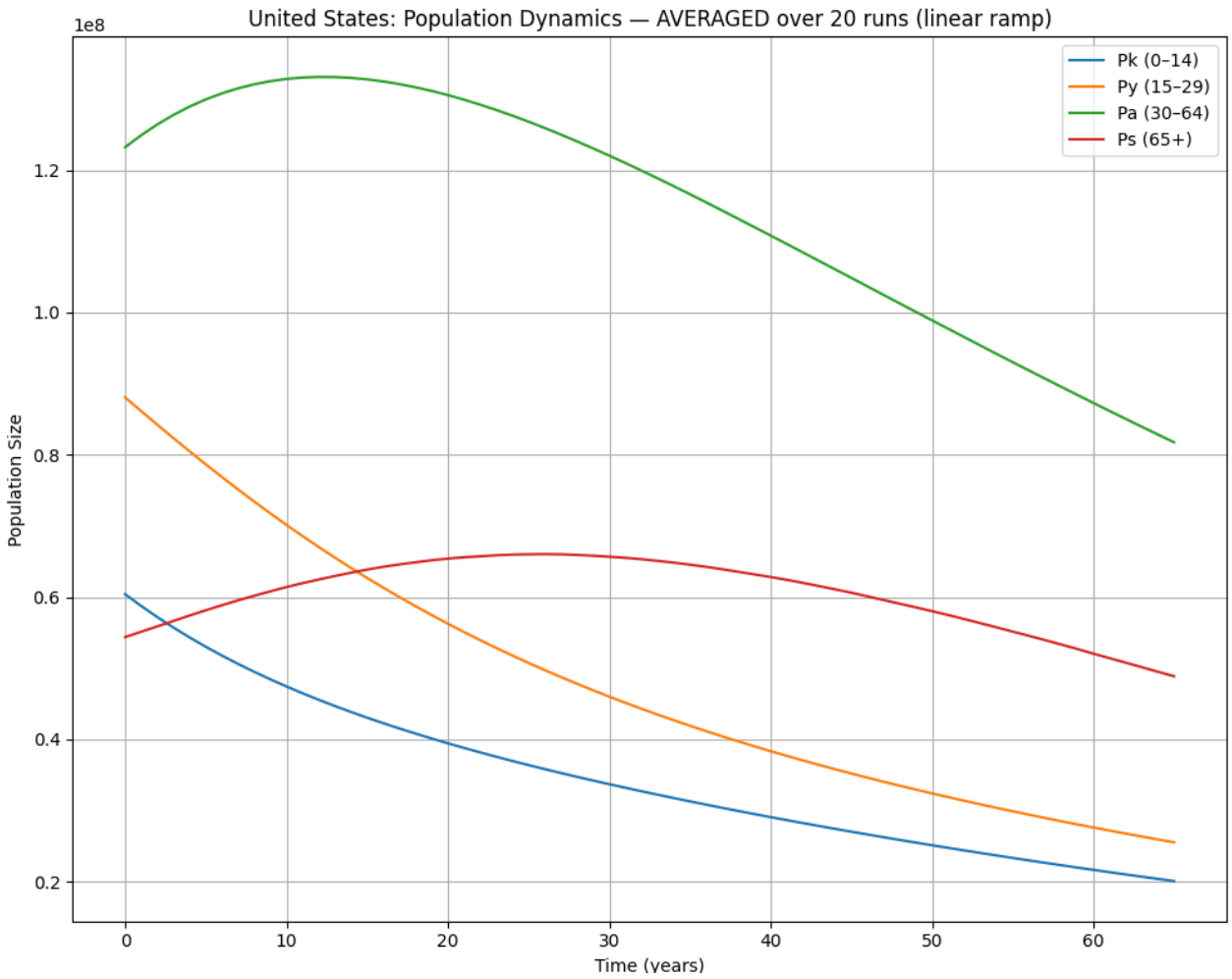


Figure A.11: Impact of elevated chronic disease mortality on the United States age structure.

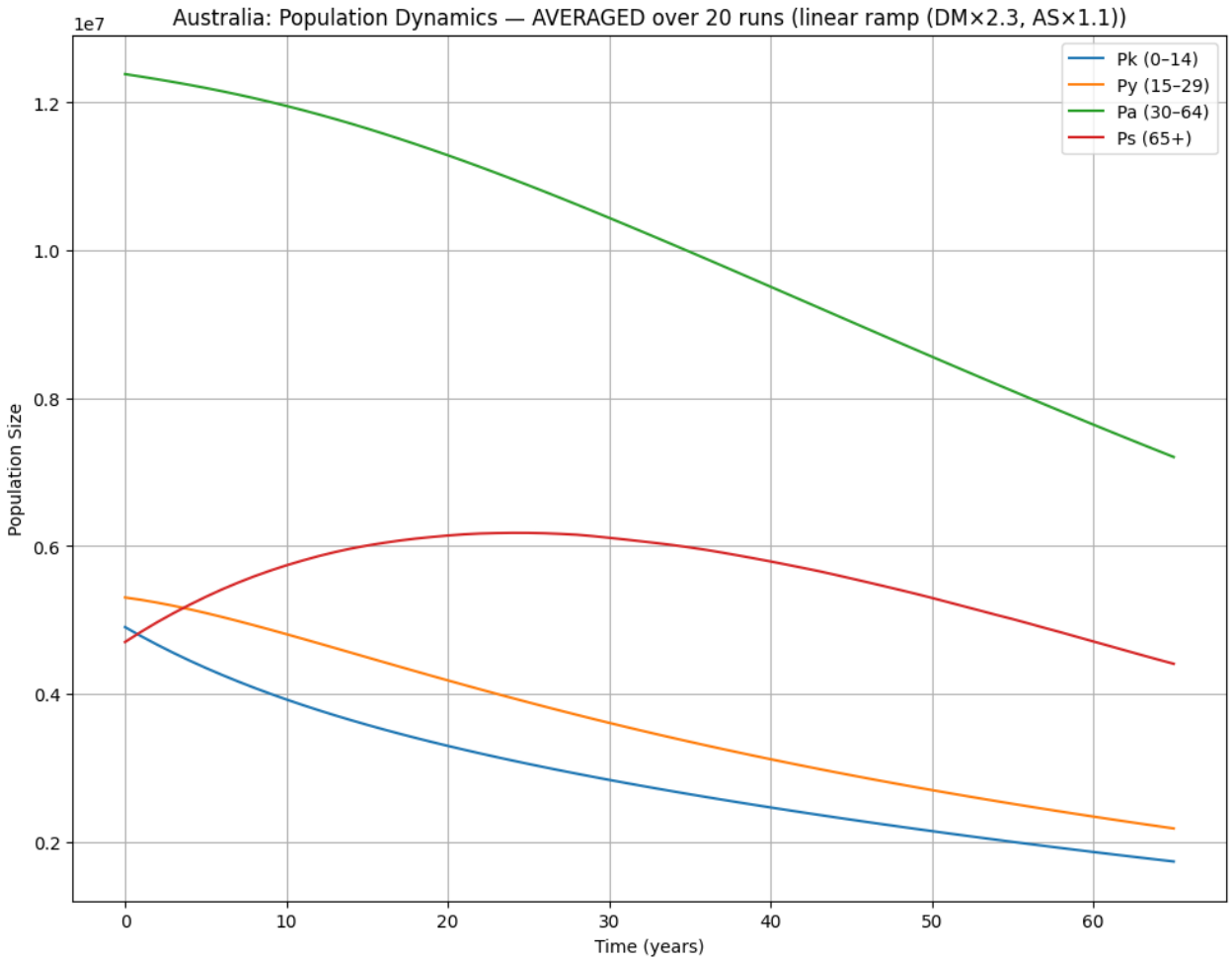


Figure A.12: Impact of elevated chronic disease mortality on Australias age structure.

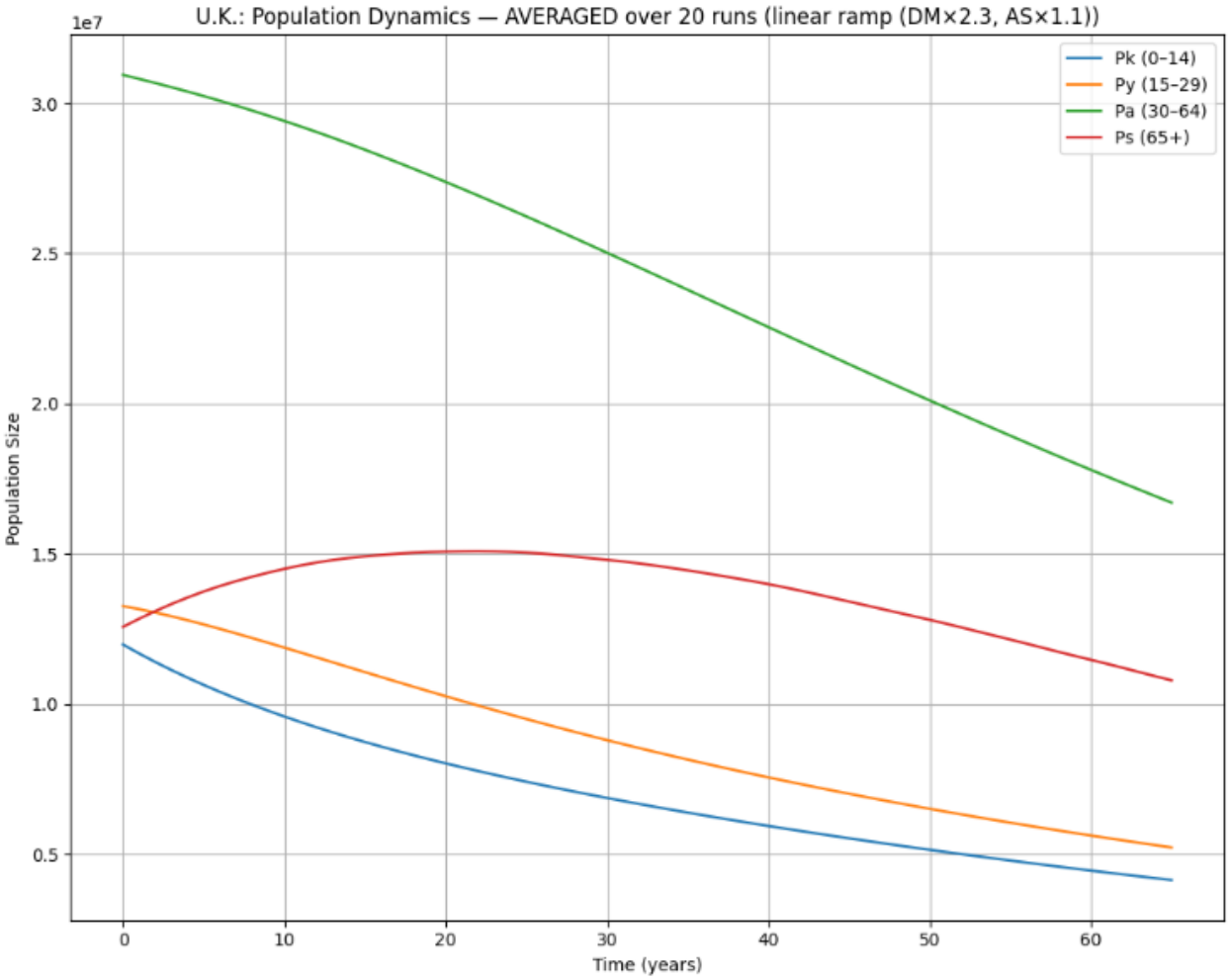


Figure A.13: Impact of elevated chronic disease mortality on the United Kingdoms age structure.

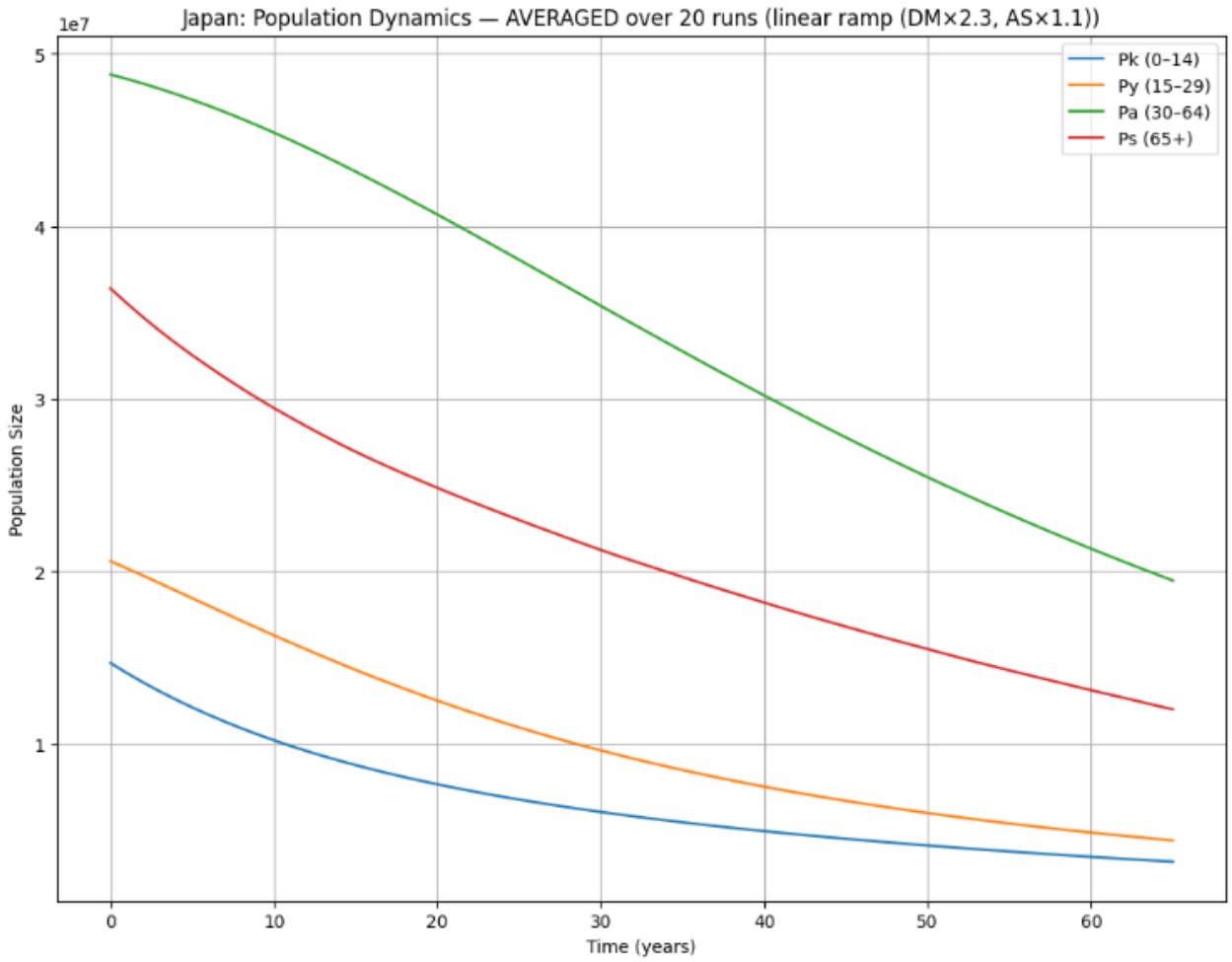
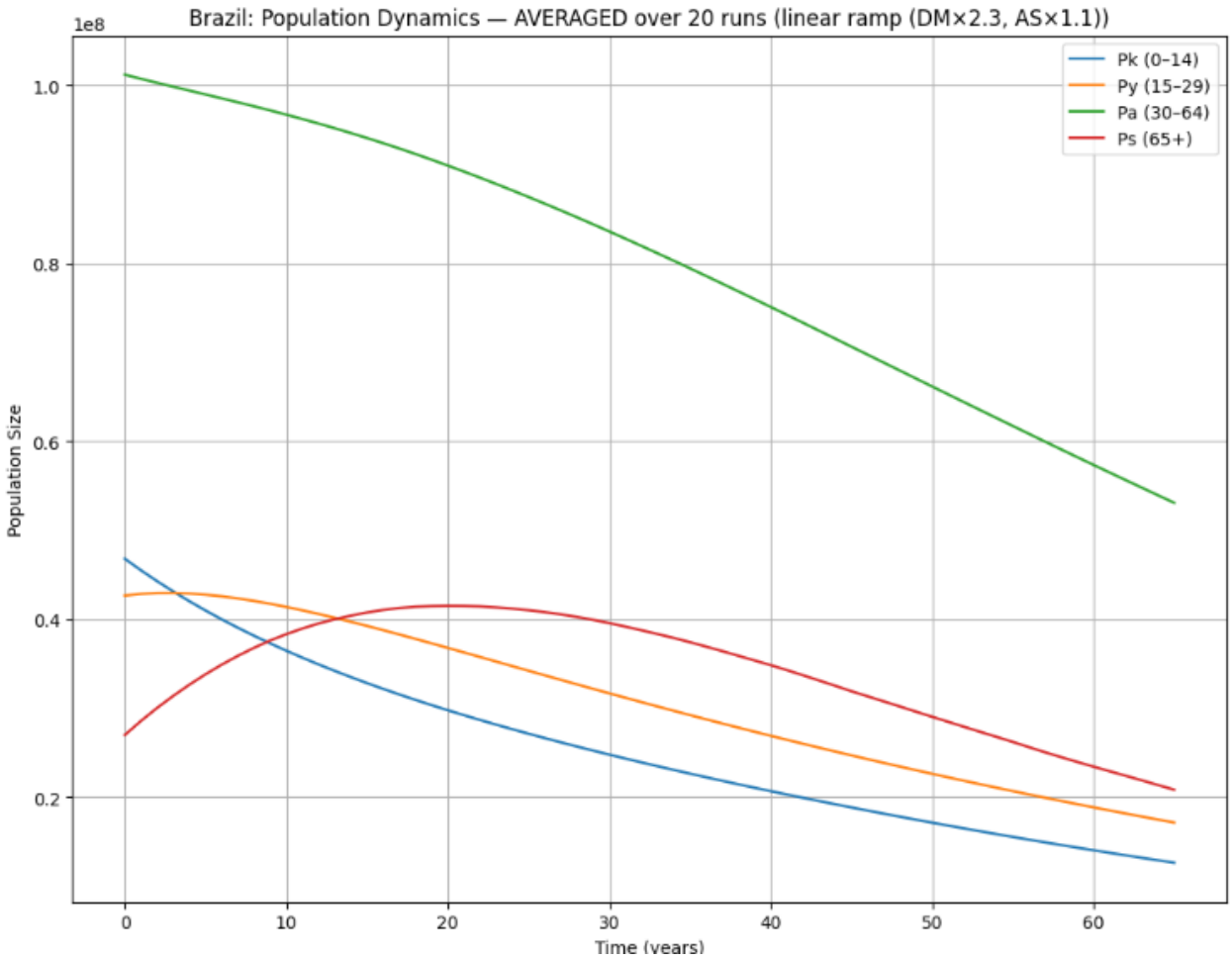


Figure A.14: Impact of elevated chronic disease mortality on Japans age structure.



¡– add filename

Figure A.15: Impact of elevated chronic disease mortality on Brazils age structure.

A.4 Scenario 4: Stochastic Environmental Shock Events

The figures below present results from simulations that incorporate discrete stochastic shocks, including droughts and pandemics. Each country was simulated across twenty independent runs, and the plotted envelopes or distributions represent the resulting variability in total and age-specific population trajectories. These outputs emphasize the compounded influence of random environmental disturbances on demographic stability and recovery potential.

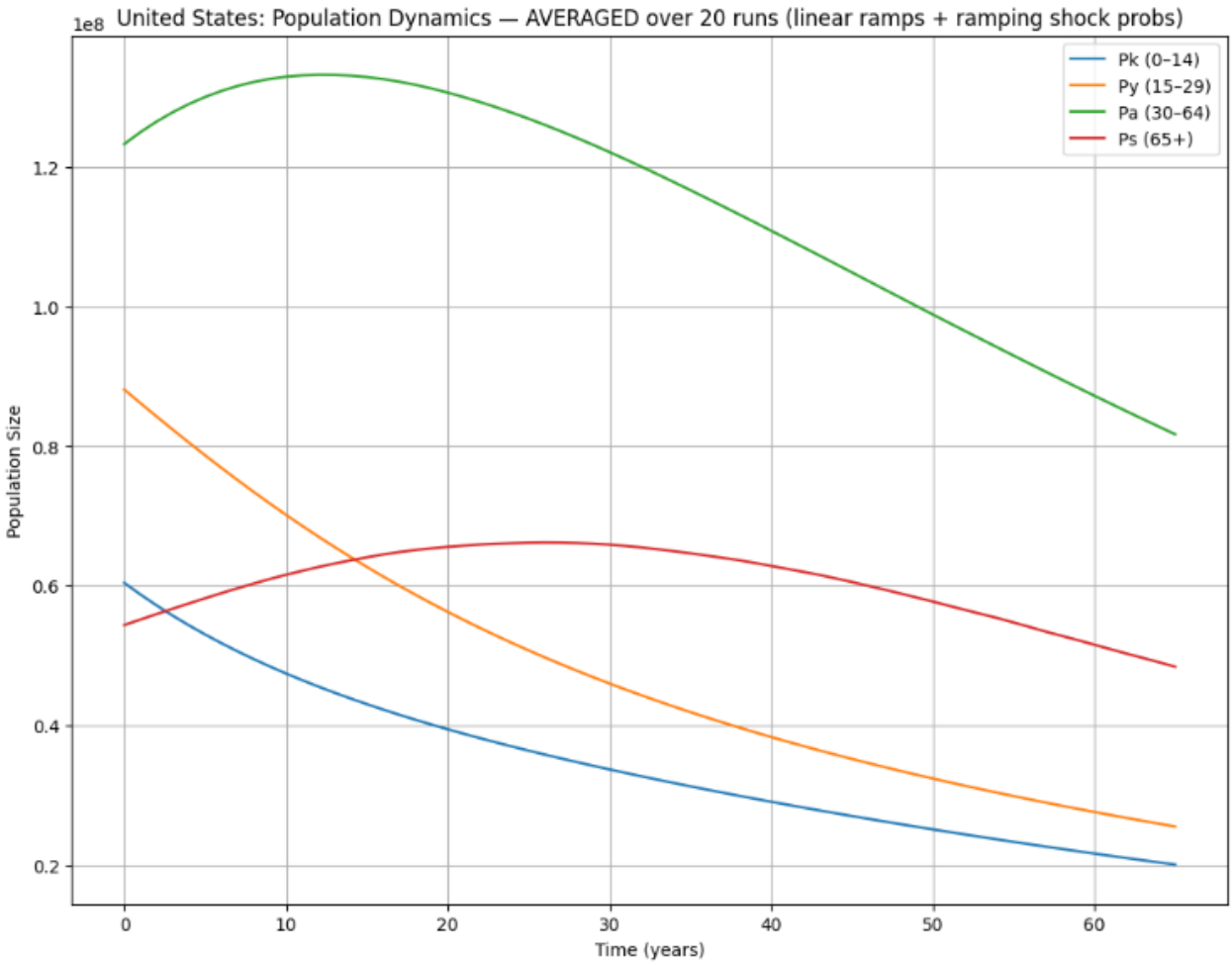


Figure A.16: Distribution of stochastic drought and disease events across runs under ramping shock probabilities.

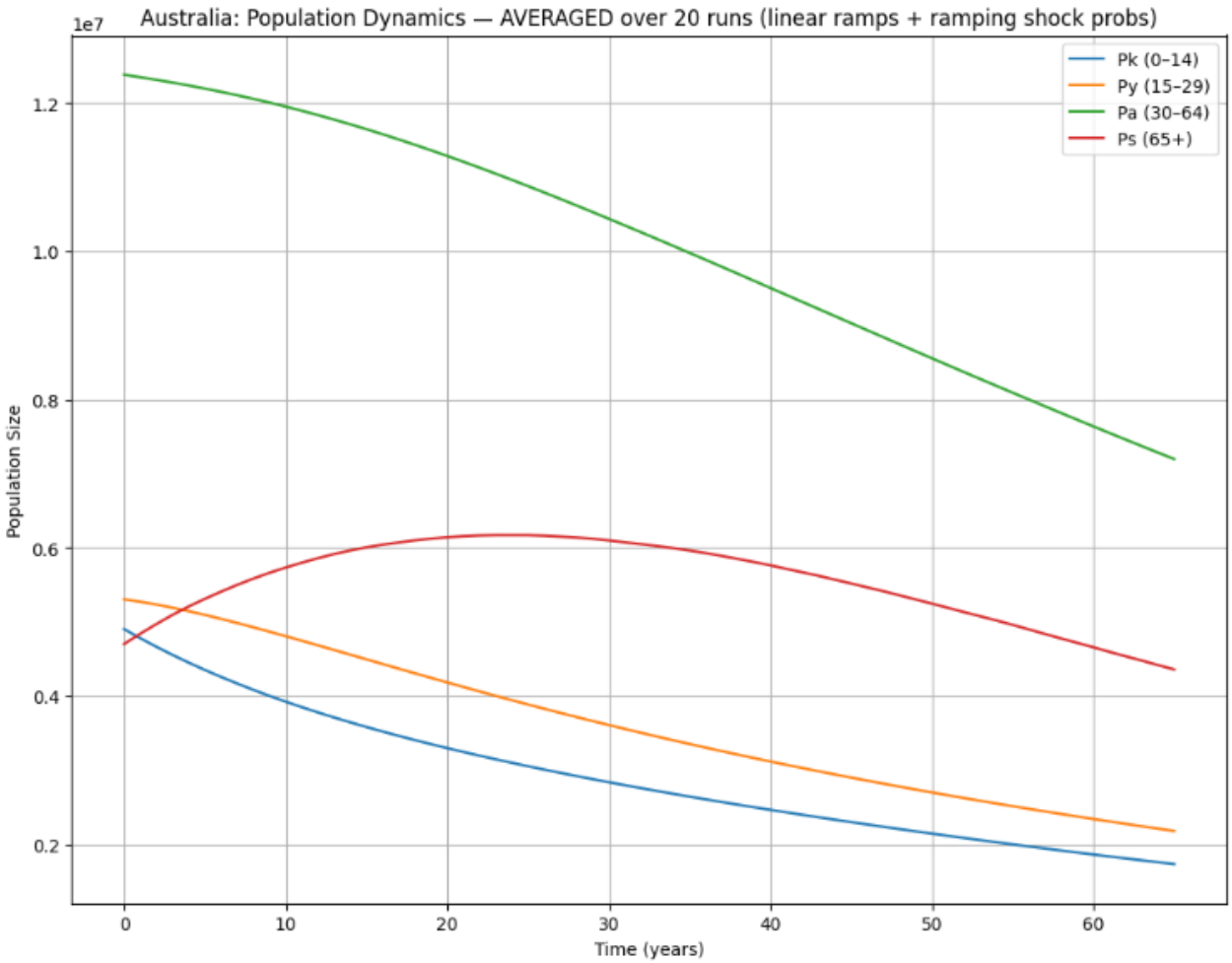


Figure A.17: Distribution of stochastic drought and disease events across runs under ramping shock probabilities.

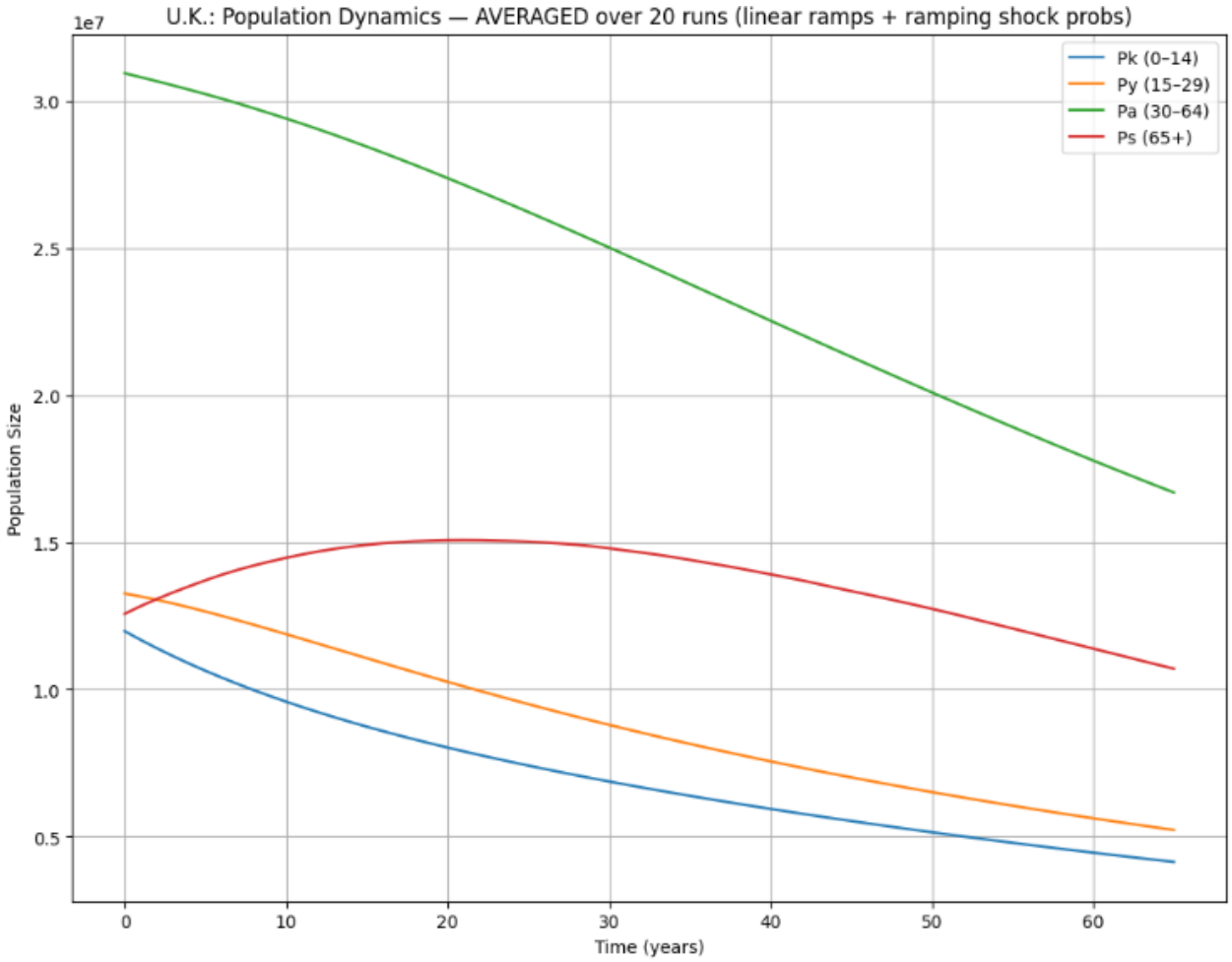


Figure A.18: Distribution of stochastic drought and disease events across runs under ramping shock probabilities.

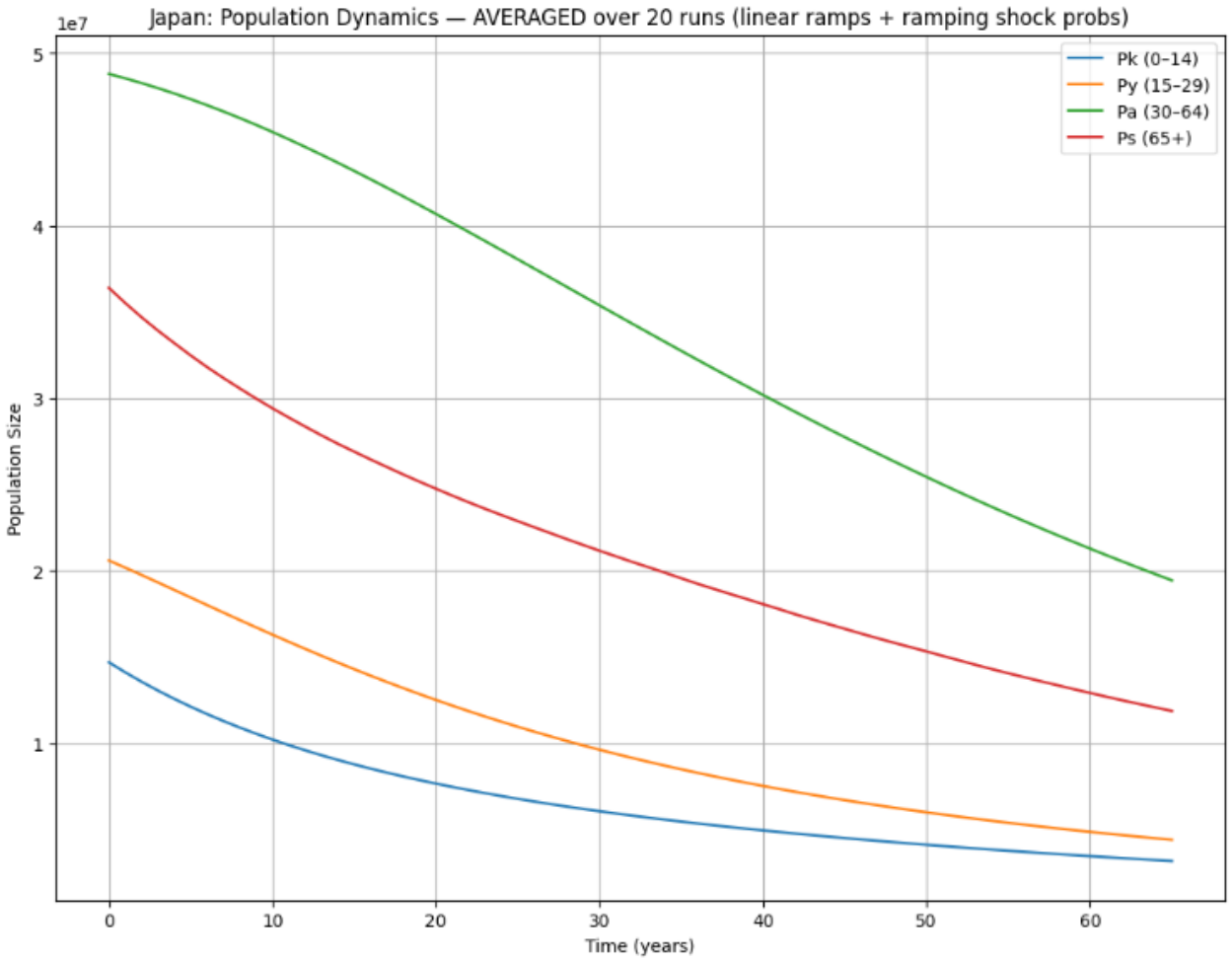


Figure A.19: Distribution of stochastic drought and disease events across runs under ramping shock probabilities.

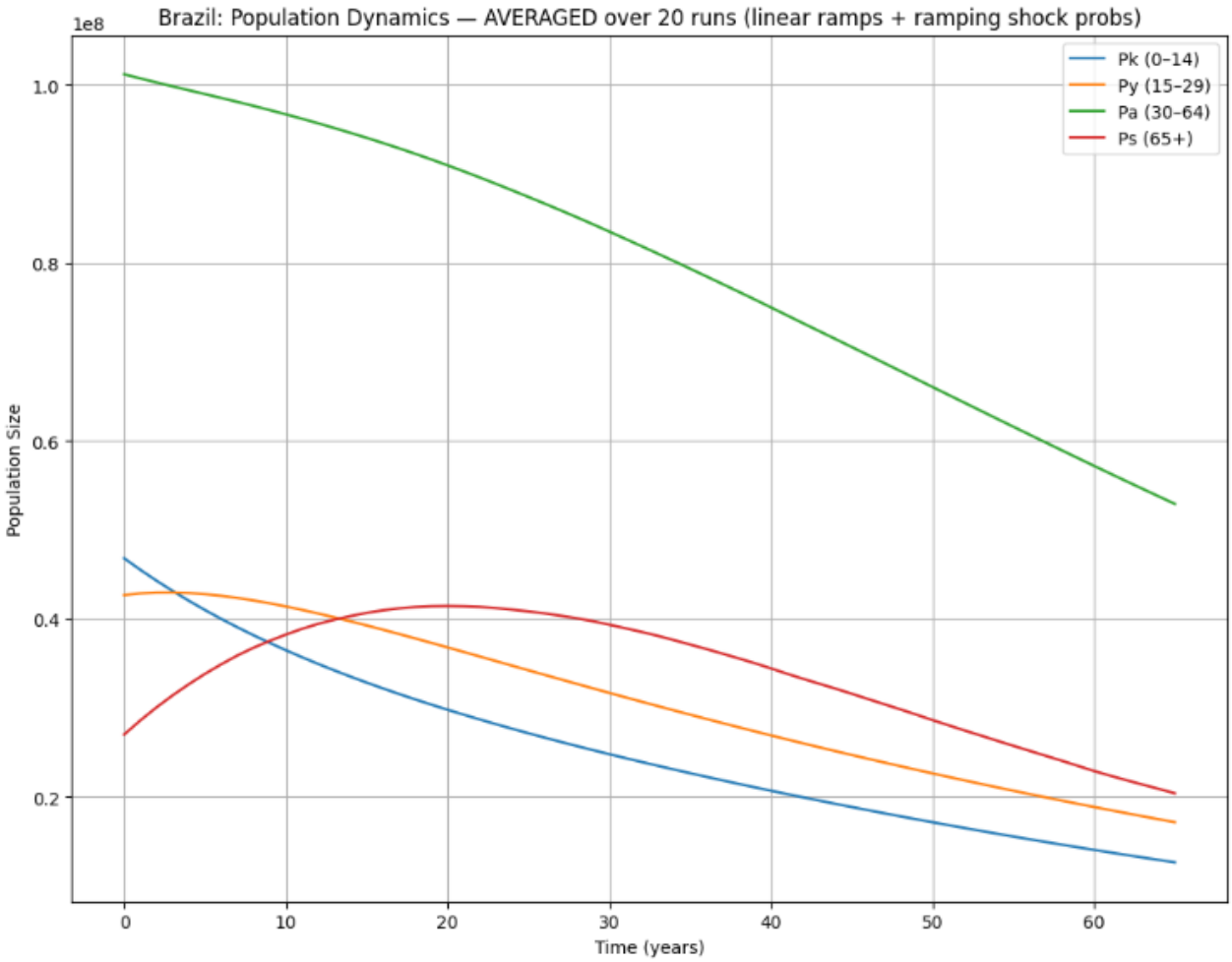


Figure A.20: Distribution of stochastic drought and disease events across runs under ramping shock probabilities.

Bibliography

[1] I Abdulrashid, H. Friji, H. Topuz, K. amd Ghazzai, D. Delen, and Y. Massoud. Analytical approach to evaluate the impact of age demographics. *Stochastic Environmental Research and Risk Assessment*, 2023.

[2] Agriculture and Agri-Food Canada. Canadian drought monitor: Current drought conditions. https://agriculture.canada.ca/en/agricultural-production/weather/canadian-drought-monitor/current-drought-conditions#a8, 2024. Accessed July 2025.

[3] AllTheScience. What is marine climate? https://www.allthescience.org/what-is-marine-climate.htm, 2024. Accessed July 2025.

[4] Australian Institute of Health and Welfare. Impact of heatwaves and drought on health. https://www.aihw.gov.au/reports/australias-health/impact-of-extreme-heat-and-heatwaves-on-health, 2022. Accessed July 2025.

[5] D.W. Bahaye, T. Marijani, and G. Mlay. An age-structured differential equations model for transmission dynamics of pneumonia with treatment and nutrition intervention. *Healthcare Analytics*, 4, 2023.

[6] R. E. Black et al. Maternal and child undernutrition: Global and regional exposures and health consequences. *The Lancet*, 2008.

[7] F. Brauer, C. Castillo-Chavez, and Z. Feng. *Mathematical Models in Epidemiology*. Springer, 2019.

[8] M. Brauer, G. Freedman, J. Frostad, et al. Ambient air pollution exposure estimation for the global burden of disease 2013. *Environmental Science & Technology*, 2016.

[9] Brazilian Ministry of Health. Drought and health in the northeast. https://www.gov.br/saude/pt-br/assuntos/noticias/2021/junho/seca-aumenta-os-riscos-a-saude-em-regioes-do-nordeste, 2021. Accessed July 2025.

[10] Edward W. Butt, Steven C. Turnock, Rachel G. Rigby, Clare Reddington, M. Yoshioka, Jason S. Johnson, Leighton Regayre, Kirsty Pringle, Graham W. Mann, and

Dominick V. Spracklen. Global and regional trends in particulate air pollution and attributable health burden over the past 50 years. *Environmental Research Letters*, 12(10):104017, 2017.

[11] Statistics Canada. Mortality rates, by age group. https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=1310071001, December 4, 2024.

[12] Canadian Institute for Climate Choices. Health effects of climate change. https://climatechoices.ca/reports/health-effects-of-climate-change/, 2022. Accessed July 2025.

[13] K. F. Cann, D. R. Thomas, and R. L. Salmon. Infectious disease outbreaks in the wake of natural flood disasters. *Journal of Water and Health*, 11(S2):S477–S484, 2013. Accessed July 2025.

[14] Center for Global Development. What's next? predicting the frequency and scale of future pandemics. https://www.cgdev.org/event/whats-next-predicting-frequency-and-scale-future-pandemics, 2023. Accessed July 2025.

[15] Centers for Disease Control and Prevention. 2009 h1n1 pandemic (swine flu) mortality estimates. https://www.cdc.gov/h1n1flu/estimates_2009_h1n1.htm, 2010. Accessed July 2025.

[16] Centers for Disease Control and Prevention. Extreme weather and climate events: Population health effects. https://www.cdc.gov/climateandhealth/effects/health_deaths.htm, 2022. Accessed July 2025.

[17] Centers for Disease Control and Prevention. Historical influenza pandemics (including asian and hong kong flu). https://www.cdc.gov/flu/pandemic-resources/basics/past-pandemics.html, 2022. Accessed July 2025.

[18] Centers for Disease Control and Prevention. Childhood asthma data. https://www.cdc.gov/asthma/children.htm, 2023. Accessed July 2025.

[19] Centers for Disease Control and Prevention. Covid-19 provisional death counts by age. https://www.cdc.gov/nchs/nvss/vsrr/covid_weekly/index.htm, 2024. Accessed July 2025.

[20] G. Chen, S. Li, L. D. Knibbs, N. Hamm, W. Cao, J. Guo, H. Ren, M. J. Abramson, and Y. Guo. Association of wildfire particulate matter exposure with mortality and hospital admissions in the usa and canada: A population-based study. *Nature Communications*, 13:7077, 2022. Accessed July 2025.

[21] CIDRAP. Global infectious disease burden among children and adolescents: More than half of deaths in low-income countries. *CIDRAP Research Highlights*, 2023. Based on analysis reporting that infectious diseases caused more than half of deaths in children and adolescents in low-income countries, compared with 6

[22] Aaron J. Cohen, Michael Brauer, Richard Burnett, H. Ross Anderson, Johan Frostad, Kara Estep, Kalpana Balakrishnan, et al. Estimates and 25-year trends of the global burden of disease attributable to ambient air pollution: An analysis of data from the global burden of diseases study 2015. *The Lancet*, 389(10082):1907–1918, 2017.

[23] Felipe J. Colón-González, M. Sewe, A. Tompkins, Henrik Sjödin, A. Casallas, J. Rocklöv, C. Caminade, and R. Lowe. Projecting the risk of mosquito-borne diseases in a warmer and more populated world: A multi-model, multi-scenario intercomparison modelling study. *The Lancet Planetary Health*, 5(7):e404–e414, 2021.

[24] Commission for Environmental Cooperation. Climate zones in north america. `https://www.cec.org/mapmonday/climate-zones-in-north-america/`, 2023. Accessed July 2025.

[25] D. Devine, J. Gaskell, W. Jennings, and G. Stoker. Trust and the coronavirus pandemic: What are the consequences of and for trust? an early review of the literature. *Political Studies Review*, 19(2):274–285, 2021. Accessed July 2025.

[26] GBD 2019 Diseases and Injuries Collaborators. Global burden of 369 diseases and injuries in 204 countries and territories, 1990–2019. *The Lancet*, 2020.

[27] K. L. Ebi and J. C. Semenza. Community-based adaptation to the health impacts of climate change. *American Journal of Preventive Medicine*, 2008.

[28] Environmental Health Indicators New Zealand (EHINZ). Who is more vulnerable to environmental hazards? `https://www.ehinz.ac.nz/indicators/population-vulnerability/what-are-vulnerable-populations/`, 2025. Accessed July 2025.

[29] Esri ArcGIS. Drought occurrence rate map. `https://www.arcgis.com/home/item.html?id=4866d8ab04434d9d873cd4189a2c8c2f`, 2023. Accessed July 2025.

[30] FUNASA - Brazilian National Health Foundation. Drought and public health: Challenges and strategies. `https://www.funasa.gov.br/site/2020/01/31/seca-e-saude-publica-desafios-e-estrategias-de-enfrentamento/`, 2020. Accessed July 2025.

[31] W. J. Gauderman et al. Childhood asthma and exposure to traffic and air pollution. *Environmental Health Perspectives*, 2005.

[32] P. W. Gething et al. A new world malaria map: Plasmodium falciparum endemicity in 2010. *Malaria Journal*, 2011.

[33] M. R. Gold, J. E. Siegel, L. B. Russell, and M. C. Weinstein. *Cost-Effectiveness in Health and Medicine*. Oxford University Press, 2002.

[34] J. D. Goldhaber-Fiebert and M. L. Brandeau. Decision analysis for health policy and planning. In *Handbook of Health Decision Science*. Springer, 2015.

[35] A. Haines, R. S. Kovats, D. Campbell-Lendrum, and C. Corvalán. Climate change and human health: Impacts, vulnerability, and public health. *Public Health*, 2006.

[36] Health Canada. Health impacts of air pollution in canada: Estimates of premature deaths and nonfatal outcomes - 2021 report. `https://www.canada.ca/en/health-canada/services/publications/healthy-living/health-impacts-air-pollution-2021.html`, 2021. Accessed July 2025.

[37] Health Effects Institute. State of global air 2020. `https://www.stateofglobalair.org`, 2020. Accessed July 2025.

[38] M. F. Holick. Vitamin d deficiency. *New England Journal of Medicine*, 2007.

[39] Barry B. Hughes, Randall Kuhn, Christophe Peterson, Dale Rothman, José Solórzano, Colin Mathers, and Janet R. Dickson. Projections of global health outcomes from 2005 to 2060 using the international futures integrated forecasting model. *Bulletin of the World Health Organization*, 89(7):478–486, 2011.

[40] Made in Canada Olivia Bush. Fast food statistics in canada. `https://madeinca.ca/fast-food-canada-statistics/`, December 31, 2024.

[41] Intergovernmental Panel on Climate Change. Ar6 climate change 2021: The physical science basis — chapter 11. `https://www.ipcc.ch/report/ar6/wg1/chapter/chapter-11/`, 2021. Accessed July 2025.

[42] Internet Geography. What is a hot desert? `https://www.internetgeography.net/topics/what-is-a-hot-desert/`, 2024. Accessed July 2025.

[43] IQAir. Air quality in india. `https://www.iqair.com/ca/india`, 2024. Accessed July 2025.

[44] W. O. Kermack and A. G. McKendrick. A contribution to the mathematical theory of epidemics. *Proceedings of the Royal Society A*, 1927.

[45] E. Koipysheva, V. Lebedinsky, and M. Koipysheva. Physical health (definition, semantic content, study prospects. *Research Paradigms Transformation in Social Sciences*, 50:601–605, 2018.

[46] S. Kovats, S. Hales, D. Campbell-Lendrum, J. Rocklöv, Y. Honda, and S. Lloyd. Global risk assessment of the effect of climate change on selected causes of death in the 2030s and 2050s. *The Lancet*, 386(9995):235–251, 2015.

[47] S. Krishnan, P.F. Coogan, D.A. Boggs, L. Rosenberg, and J.R. Palmer. Consumption of restaurant foods and incidence of type 2 diabetes in african american women. *The American Journal of Clinical Nutrition*, 91(2):465–471, 2009.

[48] LibreTexts Geography. Low latitude climates: Tropical savanna. `https://geo.libretexts.org/Courses/Kansas_State_University/Physical_Geography%3A_our_Beautiful_World/07%3A_Climate_Systems/7.04%3A_Low_Latitude_Climates`, 2023. Accessed July 2025.

[49] LibreTexts Geography. Midlatitude and subtropical climates. `https://geo.libretexts.org/Courses/Kansas_State_University/Physical_Geography%3A_our_Beautiful_World/07%3A_Climate_Systems/7.05%3A_Midlatitude_and_Subtropical_Climates`, 2023. Accessed July 2025.

[50] Christopher S. Malley, Susan Anenberg, and Drew Shindell. Improving consistency in estimating future health burdens from environmental risk factors: Case study for ambient air pollution. *Environment International*, 186:108560, 2024.

[51] M. Mayo, R. Bzeih, L. Potter, et al. Impact of the covid-19 pandemic on cancer screening and diagnosis delays: A systematic review. *Journal of Medical Screening*, 28:240–252, 2021. Accessed July 2025.

[52] The Lancet Respiratory Medicine. Rising global asthma trends: Impact of environmental factors. *The Lancet Respiratory Medicine*, 2025. Accessed July 2025.

[53] David Mills, Russell Jones, Casey Wobus, Julia Ekstrom, L. Jantarasami, Alexis St. Juliana, and Allison Crimmins. Projecting age-stratified risk of exposure to inland flooding and wildfire smoke in the united states under two climate scenarios. *Environmental Health Perspectives*, 126(4):047007, 2018.

[54] Ministry of the Environment, Japan. Heatstroke and hospitalization statistics. `https://www.env.go.jp/en/earth/heatstroke.html`, 2023. Accessed July 2025.

[55] P. Mulenga-Cilundika, J. Ekofo, C. Kabanga, et al. Indirect effects of ebola virus disease epidemics on health systems in the democratic republic of the congo, guinea, sierra leone and liberia. *International Journal of Environmental Research and Public Health*, 19(20):13113, 2022. Accessed July 2025.

[56] C. J. L. Murray et al. Global, regional, and national disability-adjusted life years (dalys) for 315 diseases and injuries: A systematic analysis for the gbd study 2015. *The Lancet*, 2015.

[57] C. J. L. Murray and A. D. Lopez. *The Global Burden of Disease: A Comprehensive Assessment of Mortality and Disability from Diseases, Injuries, and Risk Factors in 1990 and Projected to 2020*. Harvard School of Public Health, 1996.

[58] National Academies of Sciences, Engineering, and Medicine. Global health and the future role of the united states. https://www.ncbi.nlm.nih.gov/books/NBK525302/, 2017. Accessed July 2025.

[59] National Academy of Medicine. Pandemic risk assessment and its intersection with climate change: Needs, opportunities, and design considerations. https://nam.edu/perspectives/pandemic-risk-assessment-and-its-intersection-with-climate-change-needs-opportunit 2022. Accessed July 2025.

[60] National Center for Biotechnology Information. Chapter 17: Vulnerable populations. https://www.ncbi.nlm.nih.gov/books/NBK590046/, 2022. Accessed July 2025.

[61] P. J. Neumann, G. D. Sanders, L. B. Russell, J. E. Siegel, and T. G. Ganiats. *Cost-Effectiveness in Health and Medicine*. Oxford University Press, 2nd edition, 2016.

[62] Numbeo. Quality of life index. https://www.numbeo.com/quality-of-life/, 2024. Accessed July 2025.

[63] W. J. Nusselder and J. P. Mackenbach. Health expectancy research: Concepts, outcomes, and applications. *Journal of Epidemiology and Community Health*, 2000.

[64] OECD. Towards a drier world: Global drought outlook. https://www.oecd.org/en/publications/global-drought-outlook_d492583a-en/full-report/towards-a-drier-world_6f10e2e5.html, 2024. Accessed July 2025.

[65] S. K. Ofori et al. Age-stratified model to assess health outcomes of covid-19 vaccination strategies, ghana. *Emerging Infectious Diseases*, 2023.

[66] The Lancet Commission on Diabetes. The global burden of diabetes: Urgent need for action. *The Lancet*, 2023. Accessed July 2025.

[67] W. A. Orenstein and R. Ahmed. Vaccination and immunity gaps in children. *New England Journal of Medicine*, 2017.

[68] Organisation for Economic Co-operation and Development. Better life index. https://www.oecdbetterlifeindex.org, 2024. Accessed July 2025.

[69] J. A. Patz, D. Campbell-Lendrum, T. Holloway, and J. A. Foley. Impact of regional climate change on human health. *Nature*, 2005.

[70] M. C. Peel, B. L. Finlayson, and T. A. McMahon. Köppen-geiger climate type map of the world. https://www.researchgate.net/figure/Koeppen-Geiger-climate-type-map-of-the-World_fig5_26640584, 2007. Accessed July 2025.

[71] M. Pierce, H. Hope, T. Ford, et al. Mental health before and during the covid-19 pandemic: A longitudinal probability sample survey of the uk population. *The Lancet Psychiatry*, 7:883–892, 2020. Accessed July 2025.

[72] PLOS Computational Biology. Age-structured non-pharmaceutical interventions for optimal control. *PLOS Computational Biology*, 2021.

[73] Public Health Agency of Canada. Key health inequalities in canada: A national portrait – executive summary. https://www.canada.ca/en/public-health/services/publications/science-research-data/key-health-inequalities-canada-national-portrait-executive-summary.html, 2018. Accessed July 2025.

[74] C. E. Reid, M. Brauer, F. H. Johnston, M. Jerrett, J. R. Balmes, and C. T. Elliott. Critical review of health impacts of wildfire smoke exposure. *Environmental Health Perspectives*, 2016.

[75] ReliefWeb / UN Somalia Humanitarian Needs Overview. Drought in the horn of africa – regional analysis. https://reliefweb.int/report/somalia/drought-horn-africa-regional-analysis, 2023. Accessed July 2025.

[76] J. D. Runkle, A. Brock-Martin, W. Karmaus, and E. R. Svendsen. Secondary surge capacity: A framework for understanding long-term access to primary care for medically vulnerable populations in disaster recovery. *American Journal of Public Health*, 102(12):e24–e32, 2012. Accessed July 2025.

[77] K. Sahoo et al. Childhood obesity: Causes and consequences. *Journal of Family Medicine and Primary Care*, 2015.

[78] J. Schwartz, C.-C. King, and M.-Y. Yen. Protecting healthcare workers during the covid-19 coronavirus outbreak: Lessons from taiwan's sars response. *Clinical Infectious Diseases*, 71(15):858–860, 2020. Accessed July 2025.

[79] E. Seale, M. de Groh, and L. Greene-Finestone. Fast food consumption in adults living in canada: alternative measurement methods, consumption choices, and correlates. *Applied Physiology, Nutrition, and Metabolism*, 48(2):163–171, 2023.

[80] Marco Springmann, Daniel Mason-D'Croz, Sherman Robinson, Tara Garnett, H. Charles J. Godfray, Douglas Gollin, Mike Rayner, Paul Ballon, and Peter Scarborough. Global and regional health effects of future food production under climate change: A modelling study. *The Lancet*, 387(10031):1937–1946, 2016.

[81] S. Stockwell, M. Trott, M. Tully, et al. Changes in physical activity and sedentary behaviours from before to during the covid-19 lockdown: A systematic review and meta-analysis. *BMJ Open Sport & Exercise Medicine*, 7, 2021. Accessed July 2025.

[82] M. Taquet, R. Xu, M. Husain, S. Luciano, and P. J. Harrison. 6-month neurologic and psychiatric outcomes in 236,379 survivors of covid-19. *The Lancet Psychiatry*, 8:416–427, 2021. Accessed July 2025.

[83] J. K. Taubenberger and D. M. Morens. 1918 influenza: The mother of all pandemics. *Emerging Infectious Diseases*, 2006. Accessed July 2025.

[84] Time Magazine. How wildfire smoke is harming human health — even indoors. `https://time.com/6978159/wildfire-smoke-health-impact/`, 2024. Accessed July 2025.

[85] L. Trasande and Y. Liu. Reducing the staggering costs of environmental disease in children. *Health Affairs*, 2011.

[86] L. Trasande and G. D. Thurston. The role of air pollution in asthma and other pediatric morbidities. *Journal of Allergy and Clinical Immunology*, 2005.

[87] UC Davis Soil Health. Oceanic climate zone. `https://soilhealth.ucdavis.edu/climatic-zones/oceanic`, 2023. Accessed July 2025.

[88] UK Met Office. Droughts and health impacts in the uk. `https://www.metoffice.gov.uk/weather/learn-about/weather/types-of-weather/drought`, 2023. Accessed July 2025.

[89] UN-Water / ADI International. Rural vs. urban water challenges. `https://www.adi-international.org/rural-vs-urban-water-challenges-bridging-the-gap-for-equitable-resource-distributi` 2023. Accessed July 2025.

[90] UNICEF. The impact of drought on children. `https://www.unicef.org/reports/impact-drought-children`, 2019. Accessed July 2025.

[91] UNICEF. State of the world's children: Nutrition. `https://www.unicef.org/reports/state-of-worlds-children-2021`, 2021. Accessed July 2025.

[92] United Nations, Department of Economic and Social Affairs, Population Division. World population prospects 2024. `https://population.un.org/wpp/downloads`, 2024. Online Edition, Rev. 1. Accessed October 2025.

[93] U.S. Environmental Protection Agency. Climate change and the health of socially vulnerable people. `https://www.epa.gov/climateimpacts/climate-change-and-health-socially-vulnerable-people`, 2025. Accessed July 2025.

[94] U.S. Environmental Protection Agency. Climate change and the health of socially vulnerable people. https://www.epa.gov/climateimpacts/climate-change-and-health-socially-vulnerable-people, 2025. Accessed July 2025.

[95] U.S. Global Change Research Program. Fourth national climate assessment – chapter 14: Human health. https://nca2018.globalchange.gov/chapter/14/, 2018. Accessed July 2025.

[96] A. M. Vicedo-Cabrera, F. Sera, Y. Guo, et al. The burden of heat-related mortality attributable to recent human-induced climate change. *Nature Climate Change*, 11:492–500, 2021. Accessed July 2025.

[97] C. G. Victora et al. Maternal and child undernutrition: Consequences for adult health and human capital. *The Lancet*, 2008.

[98] Stein Emil Vollset, H. Ababneh, Yohannes Abate, C. Abbafati, Rouzbeh Abbasgholizadeh, Mohammadreza Abbasian, H. Abbastabar, et al. Burden of disease scenarios for 204 countries and territories, 2022–2050: A forecasting analysis for the global burden of disease study 2021. *The Lancet*, 403(10424):1803–1838, 2024.

[99] E. Vynnycky and R. White. *An Introduction to Infectious Disease Modelling*. Oxford University Press, 2010.

[100] N. Watts et al. The 2023 report of the lancet countdown on health and climate change: Responding to converging crises. *The Lancet*, 2023.

[101] Weather Months. Climate types – highland climate. https://weathermonths.com/climate-types/, 2024. Accessed July 2025.

[102] D. Weinhold and F. J. Gurtner. Climate change and child health: What do we know and what can we do? *Climatic Change*, 2014.

[103] World Bank. Death rate, crude (per 1,000 people) - world bank data. https://data.worldbank.org/indicator/SP.DYN.CDRT.IN, 2024. Accessed July 2025.

[104] World Health Organization. Protecting health from climate change: Connecting science, policy and people, 2009.

[105] World Health Organization. More than 90% of the world's children breathe toxic air every day. https://www.who.int/news/item/29-10-2018-more-than-90-of-the-worlds-children-breathe-toxic-air-every-day, 2018. Accessed July 2025.

[106] World Health Organization. Global health estimates: Leading causes of death and disability in children, 2020.

[107] World Health Organization. Global health estimates: Life expectancy and leading causes of death and disability. https://www.who.int/data/gho/data/themes/mortality-and-global-health-estimates, 2020. Accessed July 2025.

[108] World Health Organization. The top 10 causes of death. https://www.who.int/news-room/fact-sheets/detail/the-top-10-causes-of-death, 2020. Accessed July 2025.

[109] World Health Organization. Air pollution. https://www.who.int/health-topics/air-pollution, 2021. Accessed July 2025.

[110] World Health Organization. Covid-19 mortality and risk factors: Scientific brief. https://www.who.int/publications/i/item/WHO-2019-nCoV-Sci_Brief-Mortality-2021.1, 2021. Accessed July 2025.

[111] World Health Organization. Global excess deaths associated with covid-19. https://www.who.int/data/stories/global-excess-deaths-associated-with-covid-19, 2022. Accessed July 2025.

[112] World Health Organization. Climate change and health fact sheet. https://www.who.int/news-room/fact-sheets/detail/climate-change-and-health, 2023. Accessed July 2025.

[113] World Health Organization. Informing policy and progress: Air quality, energy and health. https://www.who.int/teams/environment-climate-change-and-health/air-quality-energy-and-health/policy-progress, 2023. Accessed July 2025.

[114] World Health Organization. Outdoor air pollution — urban and rural differences. https://www.who.int/air-pollution-evidence/urban-rural-geography, 2023. Accessed July 2025.

[115] M. Zhou, D. Qin, and X. Chen. Long-term cardiovascular and neurological sequelae of covid-19: A systematic review. *International Journal of Environmental Research and Public Health*, 21(5):2345, 2024. Accessed July 2025.